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Global Technology

ABF Substrates – Upgrading Unimicron and NYPCB to OW

An AI-driven ABF substrate up-cycle will last to the end of the decade, driving growth to accelerate and margins to expand. The stocks have started to move, but this cycle will be much stronger and more sustainable – fueling multiple expansion.

Key Takeaways

- ABF substrates are in the early innings of an up-cycle, driven by a significant shift in demand.
- PC-related end markets will fall to ~15% by 2030 vs. ~70% in 2015 – and AI-related end markets will rise to ~75% by 2030 vs. ~10% in 2015.
- We forecast ABF substrate market value growth to accelerate over 2025-2030 – to a 16.1% CAGR (vs. 9.0% over the past five years).
- We project ABF substrate market undersupply from 2027 onwards...
- ...putting further upward pressure on pricing, even though it is already moving, driven by higher material costs.

Based on the fundamental backdrop above, these are the stock implications – it's not too late.

We upgrade Unimicron to OW (from EW) and NYPCB to OW (from UW) – we're bullish for the first time in nearly four years: This AI-driven cycle is structurally different and much stronger, driving significant earnings growth in 2025-28e – we forecast a 105% CAGR for Unimicron and 113% for NYPCB. Given the strong growth, their 2027e PEGs, based on our price targets, look attractive at only 0.3x and 0.5x, respectively.

- Price targets of NT\$500 (up from NT\$120.75) and NT\$515 (up from NT\$165) – implying 36% and 24% upside:** We change valuation methodology from P/B back to the multi-stage residual income (RI) valuation model.

Shawn Kim reiterates OW on SEMCO (009150.KS); price target of KRW430,000 – up from W309,000: See [SEMCO's fundamentals](#).

Shoji Sato is UW on Ibiden: His [recent downgrade mainly reflects overheated earnings expectations](#). He still expects Ibiden's earnings to grow 26% in F26, 23% in F27, and 41% in F28, although his earnings forecast is below market consensus.

How does our positive view reconcile with Shoji Sato's Underweight on Ibiden?

Since January 2025, Ibiden's stock is up 310%, vs. NYPCB up 238%, Unimicron up 164%, and SEMCO up 197%. From here, though, we see room for Unimicron, NYPCB, and SEMCO to outperform Ibiden, although views on growth prospects are aligned.

MORGAN STANLEY TAIWAN LIMITED+

Howard Kao

Equity Analyst

Howard.Kao@morganstanley.com

+886 2 2730-2989

MORGAN STANLEY & CO. INTERNATIONAL PLC+

Shawn Kim

Equity Analyst

Shawn.Kim@morganstanley.com

+44 20 7677-1018

MORGAN STANLEY MUFG SECURITIES CO., LTD.+

Shoji Sato

Equity Analyst

Shoji.Sato@morganstanleymufg.com

+81 3 6836-8404

Yoshihito Hasegawa

Equity Analyst

Yoshihito.Hasegawa@morganstanleymufg.com

+81 3 6836-8910

Ryoichi Watanabe

Equity Analyst

Ryoichi.Watanabe@morganstanleymufg.com

+81 3 6836-8929

Tomonobu Tsunoyama

Equity Analyst

Tomonobu.Tsunoyama@morganstanleymufg.com

+81 3 6836-8936

MORGAN STANLEY TAIWAN LIMITED+

Charlie Chan

Equity Analyst

Charlie.Chan@morganstanley.com

+886 2 2730-1725

Sharon Shih

Equity Analyst

Sharon.Shih@morganstanley.com

+886 2 2730-2865

Irene Yen

Research Associate

Irene.Yen@morganstanley.com

+886 2 2730-2869

GREATER CHINA TECHNOLOGY HARDWARE

Asia Pacific

Industry View

In-Line

Exhibit 1 : What's changed

Company	Ticker	Rating		PT		Last Close	Upside (%)	EPS change (%)		
		Now	Old	Now	Old			25e	26e	27e
Unimicron	3037.TW	OW	EW	500	120.75	369.0	38%	-3%	6%	60%
Nan Ya PCB	8046.TW	OW	UW	515	165.00	416.0	24%	-10%	11%	71%
SEMCO	009150.KS	OW	OW	430,000	309,000	358,000	20%	NA	7%	4%

Source: FactSet, Morgan Stanley Research estimates. Pricing as of the close on February 11, 2026 for Unimicron and NYPCB; February 19, 2026 for SEMCO.

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Executive Summary

ABF substrate value growth to accelerate over the next five years amid a shift in demand to AI

Based on our bottom-up analysis, we project that ABF substrates will be in an up-cycle until the end of this decade...

...with industry undersupply starting from 2027 onwards.

What was once a PC-driven market – with PCs accounting for ~70% of it in 2015 – is undergoing a significant shift in demand....

...with AI-related applications (GPU, ASIC, CPU, networking, etc.) capturing ~75% of the end market by 2030, per our estimate.

This will drive ABF substrates value to accelerate at a 16.1% CAGR over 2025-2030e (vs. just 9.0% over 2020-2025e), as AI demand proliferates.

T-glass constraint worth monitoring, but should have limited impact on high-end segments:

Our analysis points towards a ~20% supply/demand imbalance for T-glass this year, which is even more severe vs. 2025. However, there should be improvements from 2H26 onwards, as more supply comes online and more alternative solutions come to market.

We believe this will have more impact on lower-end segments, especially BT substrate suppliers, but should have relatively limited impact on higher-end ABF substrate suppliers.

That said, this does mean that there will be limited upside potential in the near term, but should not lead to downside risks.

Are we too late? No

We are still in the early innings of this AI-driven up-cycle: The stocks have started to move higher, but as favorable pricing and margins start to manifest, earnings estimates will continue to rise, as we have seen in past cycles.

This cycle will also be much stronger and more sustainable vs. past cycles because of AI:

In the past, demand was very much consumer-driven (i.e., PCs). The cyclical nature was more pronounced with less visibility.

- *This AI-driven cycle, will be structurally stronger and longer, with greater demand visibility.*

We believe this will also result in multiple expansion compared to past cycles – as we have witnessed in several tech segments since 2023. We believe the stocks have priced in only ASP increases this year, driven mainly by increased material costs, but have not fully reflected the AI-driven demand and potential pricing adjustments that will follow as the supply-demand imbalance starts to manifest from 2027 onwards.

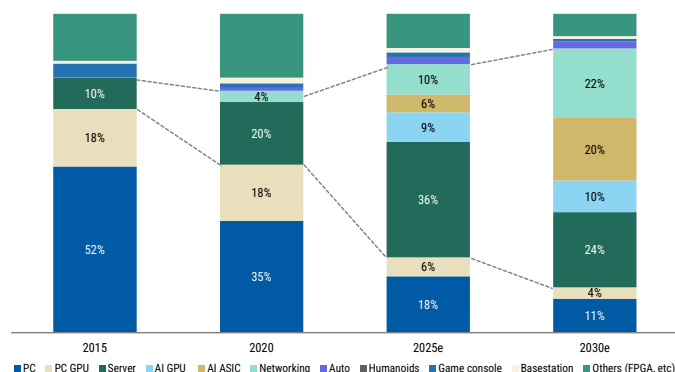
Risks

Risks include:

- AI demand downside
- T-glass being more of a constraint than expected
- More ABF capacity announcements
- Weaker non-AI demand
- CoWoP displacing ABF substrates

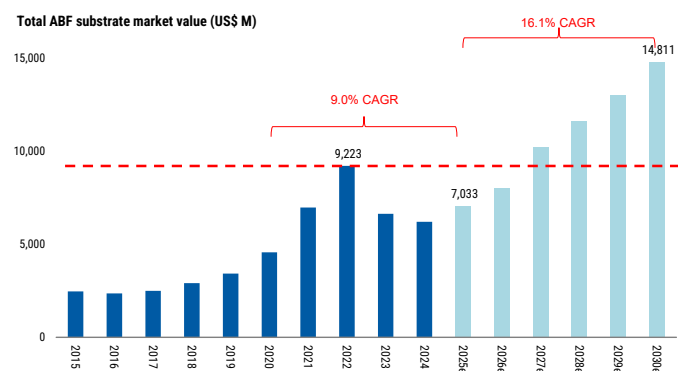
Our Thesis in Six Key Charts

Exhibit 2: Demand for AI is becoming the major growth driver for ABF substrates



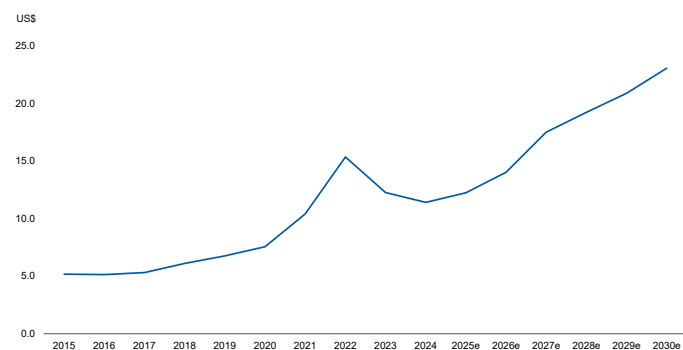
Source: Morgan Stanley Research estimates.

Exhibit 4: This is driving the ABF substrate market to accelerate over the next five years at a 16.1% CAGR...



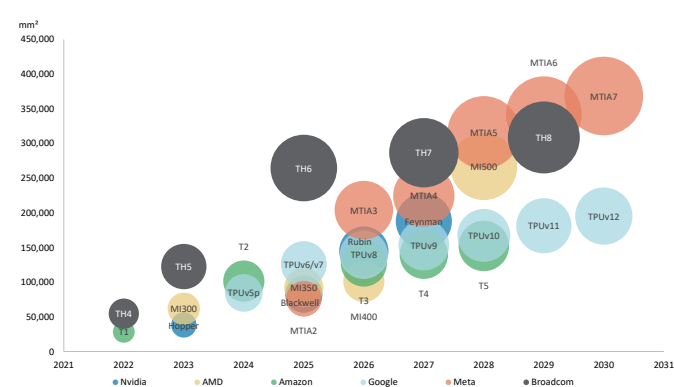
Source: Morgan Stanley Research estimates.

Exhibit 6: ...which should also lead to blended pricing adjustments for ABF substrates in the coming years



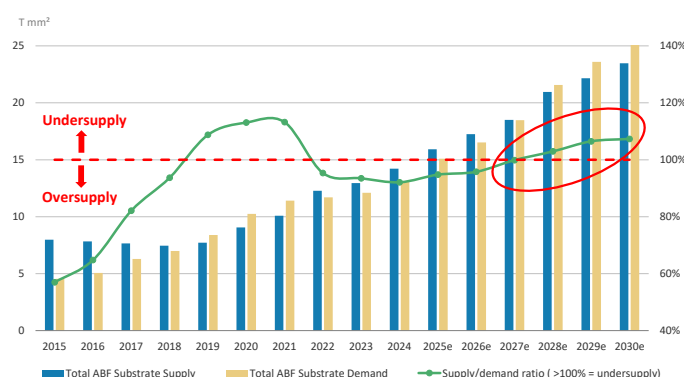
Source: Morgan Stanley Research estimates.

Exhibit 3: Chip size and layer counts are increasing for these advanced AI semiconductors



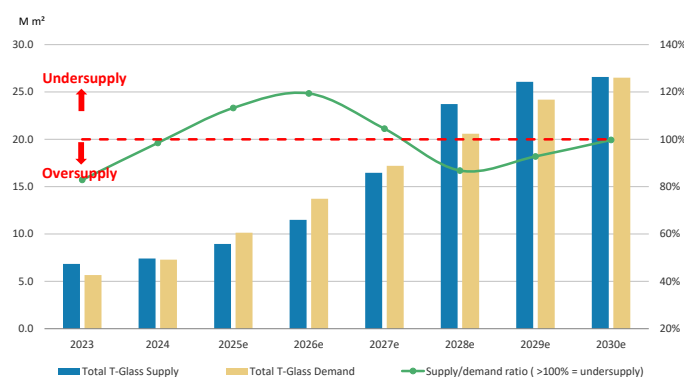
Source: Morgan Stanley Research estimates.

Exhibit 5: ...and will also drive undersupply, starting from 2027 onwards...



Source: Morgan Stanley Research estimates.

Exhibit 7: T-glass supply constraint is real; it will get better after this year, but we think it will mainly affect BT substrates and midrange/low end. There are also some viable alternatives under discussion.



Source: Morgan Stanley Research estimates.

Fundamental Examination of the ABF Substrates Market

Introduction: What are ABF substrates?

Ajinomoto Build-Up Film, or ABF, is the de facto standard interlayer insulating material for high-performance semiconductor substrates, supplied by [Ajinomoto \(2802.T, covered by Tomonobu Tsunoyama\)](#) in Japan.

Today, ABF commands an overwhelming global market share of over 90%, as the key material used for manufacturing advanced semiconductor substrates, such as CPUs, GPUs, XPU, FPGAs, etc.

Substrate makers such as Unimicron, Nan Ya PCB, Kinsus, Ibiden, Shinko, SEMCO, and AT&S procure ABF from Ajinomoto to produce ABF substrates.

ABF was first adopted by Intel in 1999. Initially used mainly for CPUs, it eventually became the industry standard as the go-to material. These substrates enable fine-line routing between silicon and PCB for advanced packages (FC-BGA, CoWoS substrates, etc), and enable good signal integrity and power delivery, supporting high IO counts.

How big is the ABF market? Where is it in the cycle?

The ABF substrate market has gone through a few cycles over the past two decades

There was a prolonged down-cycle from 2012-18 as the PC TAM shrank amid the rise of smartphones and tablets. During this tough period, companies were not able to generate profit and had to compete on price to fill their idle capacity. As a result, during those years, there was no or very limited capacity expansion, and capital expenditure by the major ABF substrate suppliers was kept at a minimum ([Exhibit 3](#)).

Even as general server demand picked up, lifting utilization rates across the group going into 2020, ABF substrate suppliers remained cautious on the longer-term demand outlook and stayed disciplined, hesitant to expand capacity. Hence, capex turned up just slightly in 2019-2020, increasing ~US\$600mn vs. ~US\$2.2bn on average per year in 2012-18.

This underinvestment led to a sudden undersupply, as the shock from Covid-19 drove strong work-from-home demand for PCs, as well as increased data usage and storage for general computing driven by online entertainment. This ultimately led to a significant increase in ABF substrate market value between 2020-2022, driven by the sudden surge in demand, along with meaningful pricing adjustments owing to the lack of supply.

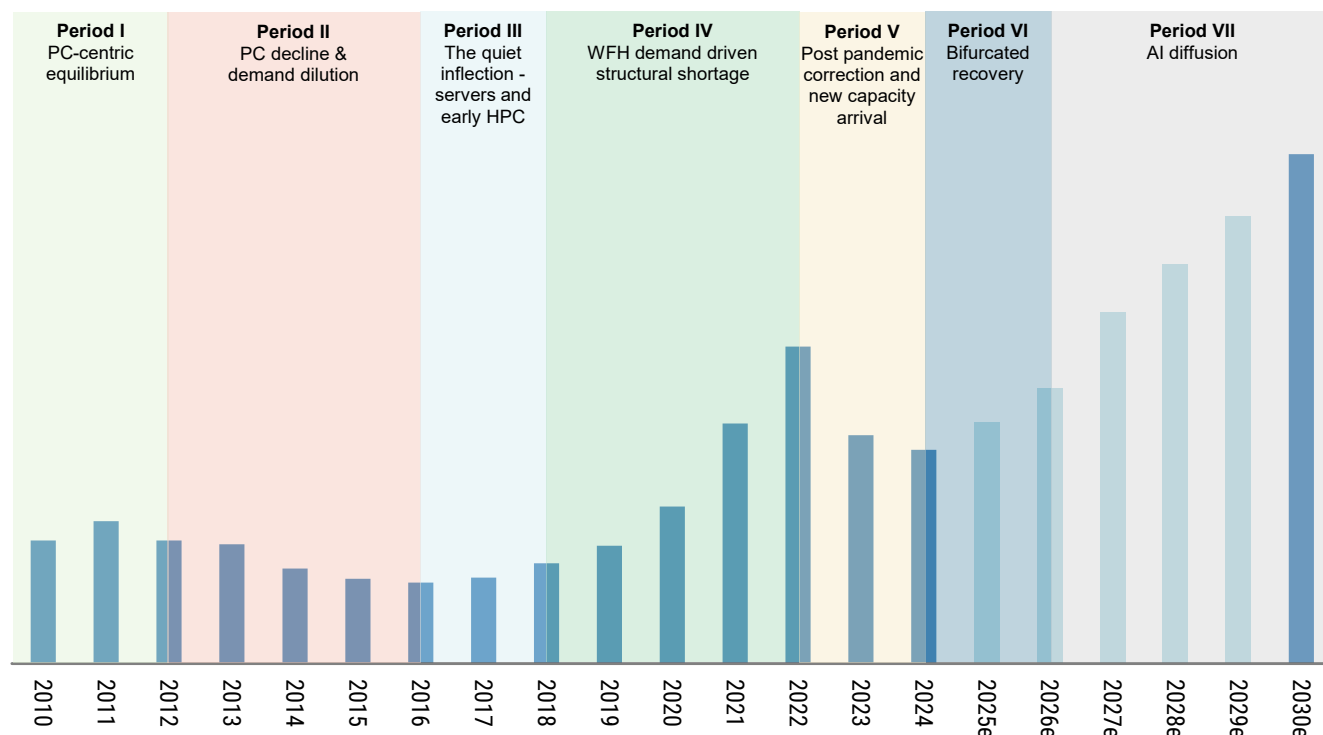
During this 2020-2022 period, a lot of capacity expansion was set in motion, as ABF substrate suppliers and customers came to agreement that necessary investments were needed for future growth. Some subsidies were even given to these suppliers to facilitate this capacity expansion. This is why capex jumped from ~US\$2.9bn in 2020 to US\$4.1bn in 2021 and US\$5.5bn in 2022; even in 2023-24 capex remained elevated, as capacity expansion plans continued.

Unfortunately, demand did not play out as expected. As the world came out of the pandemic, demand for PCs and online entertainment fell sharply. This normalization of demand, coupled with capacity expansion from practically every major ABF substrate supplier, resulted in the ABF substrate market once again becoming oversupplied. This led to a sharp decline in ABF substrate market value in 2023-24, as volumes normalized, driving prices down to normalize against the new supply/demand dynamic.

Fortunately, cloud demand remained strong for general computing, and with AI demand starting to pick up, the down-cycle this time around was much shorter (two years) vs. the prior down-cycle (six years).

In 1H24, the ABF substrate market troughed and we started to see an upward inflection since then, but the pace of the recovery has been moderate and bifurcated. Midrange/low-end markets such as PCs have shown moderate demand recovery, with high-end server and AI-related end demand recovering and growing at a faster pace. This is why [we turned the page on our bearish stance in 1H25](#), but did not turn more bullish.

Periods	Description
Period I (2010-2012): PC-centric equilibrium	ABF substrate demand is primarily driven and dominated by CPU and GPU demand for desktops, notebooks, discrete GPUs, with a meaningfully smaller exposure towards server CPUs.
Period II (2012-16): PC decline and demand dilution	Global PC demand entered a structural down-cycle as consumer behavior changed, with smartphones and tablets absorbing more consumer time spent. This resulted in lower volumes and increased pricing competition as suppliers fought to fill idle capacity.
Period III (2016-18): The quiet inflection - servers and early HPC	During these years, the PC market matured, while cloud computing started to take off. At the same time, server CPU substrates quietly grew larger with more layers. We also started to see early accelerators being adopted in high performance computing.
Period IV (2018-2022): WFH demand driven structural shortage	The Covid-19 pandemic drove massive work from home demand for consumer electronics and cloud computing. This sudden surge in PC demand and increased time spent on gaming and online entertainment, bundled with a continuous increase in substrate package size and layer count, resulted in a structural shortage, which also resulted in longer lead-times and raised prices. It was during this period (2021-22) that every ABF supplier globally started to invest in new capacity.
Period V (2022-2024): Post-pandemic correction and new capacity arrival	As the world came out of the Covid-19 pandemic, WFH demand faded and PC and cloud computing demand were negatively affected. But at the same time, the new investments and capacity additions were just starting to slowly come online, resulting in a drop in utilization rates and pricing.
Period VI (2024-2026): Bifurcated recovery	PC demand normalized at the pre-Covid levels, with some replacement demand driven by Windows upgrade, while the recovery and growth of AI and general server demand for cloud picks up at a meaningfully higher growth trajectory.
Period VII (2026-30): AI diffusion	Demand for AI proliferates from training to inference, driving ABF substrate market growth to re-accelerate over the next five years after a two-year down-cycle following the Covid-19 pandemic. This drives increased demand for AI computing, networking, and cloud computing, while demand for PCs remains soft.

Exhibit 8: We are entering the AI diffusion era

Source: Morgan Stanley Research.

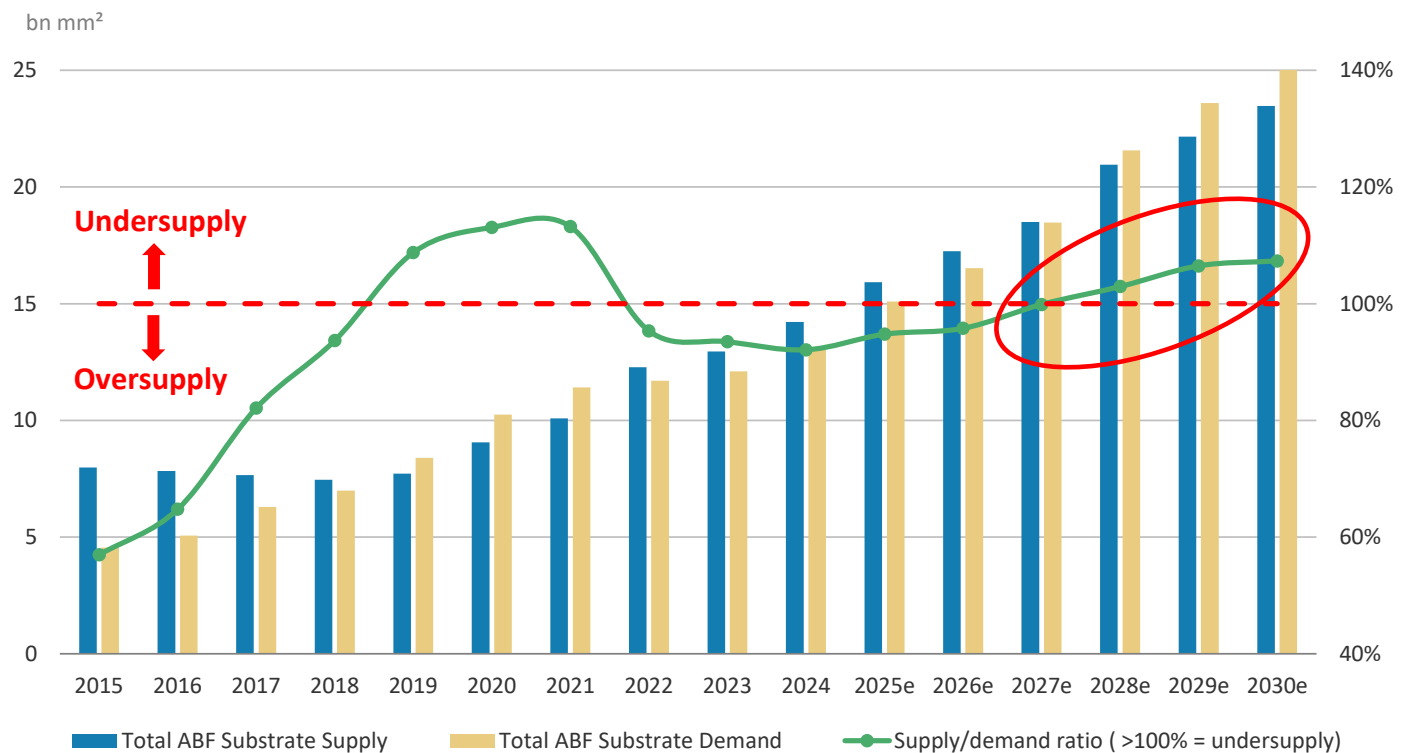
The market will be undersupplied again from 2027 onwards

Based on our bottom-up analysis, we believe the ABF substrate market will enter undersupply again, from 2027 onwards. As a result, we believe the industry will need to expand capacity further to cater to this future demand growth, more than the investments and announcements that we already see today. However, new capacity additions always take time. For a brand new production site to go from start of construction to mass production will typically take 24 months or more.

In 2029-30, we believe the ABF substrate market will be 5-10% undersupplied

For comparison, in the 2020-2022 up-cycle, the market was 10-15% undersupplied, by our estimates. This cycle does not appear to be as severe, but we want to point out that our figure represents the overall ABF substrate supply/demand dynamic, which includes the midrange to low-end segments as well as the high end.

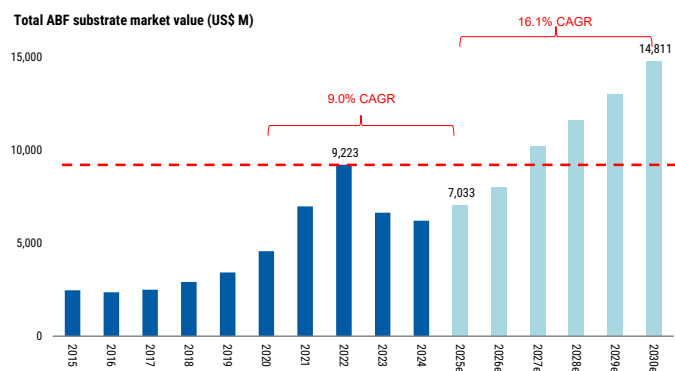
If we solely focus on the high end, the magnitude of undersupply will be greater than 5-10%. We believe our demand forecast is also skewed towards the conservative side. And on the supply side, we have baked in capacity expansion from the top ABF substrate suppliers up until February 3, 2026, even including Ibiden's latest [capital investment plan announced after market on February 3, amounting to approximately 500 billion yen over the three-year period from fiscal year 2026 to fiscal year 2028](#). Even after baking in such a big capital investment from Ibiden, our analysis still points towards ABF substrates being undersupplied until the end of this decade.

Exhibit 9: We estimate that ABF substrates will be undersupplied again from 2027 onwards

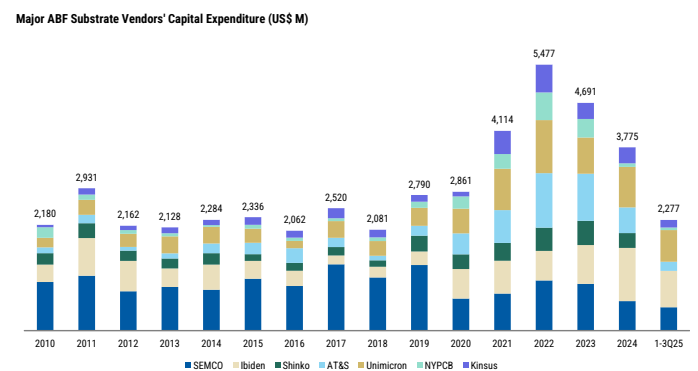
Source: Morgan Stanley Research estimates.

As a result, we are turning positive on ABF substrates – we expect growth to accelerate over the next five years

We are now turning bullish on this segment, especially because we expect growth to accelerate in the coming five years vs. the past five years. On an annual basis, 2025 marks the first year that the ABF substrate market grew, following two years of decline after Covid-19. We estimate 2025 ABF substrate market size grew ~13% y/y, reaching ~US \$7.0bn. Over the next five years, we believe the market's growth will accelerate again, driven primarily by both general computing and AI computing. We project a 16.1% CAGR over 2025-2030, vs. the past five years (2020-2025e) at a 9.0% CAGR.

Exhibit 10: We forecast ABF substrate TAM to show growth acceleration in the coming five years

Source: Prismark, Morgan Stanley Research estimates.

Exhibit 11: Major ABF vendors only engaged in maintenance capex from 2012-2018

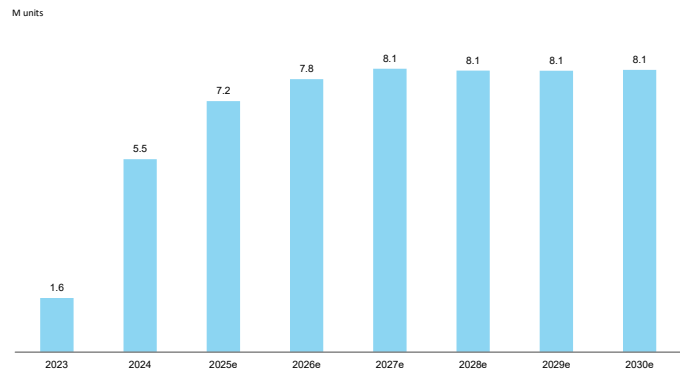
Source: Company data, Morgan Stanley Research.

This growth will be mainly driven by AI demand

We believe the demand drivers will be mainly supported by GPUs, ASICs, and networking chips used in data centers, and to a lesser extent, CPUs as well. Because of the increased demand from computing, we believe the demand for all of these semiconductors will grow and/or remain elevated in the coming years.

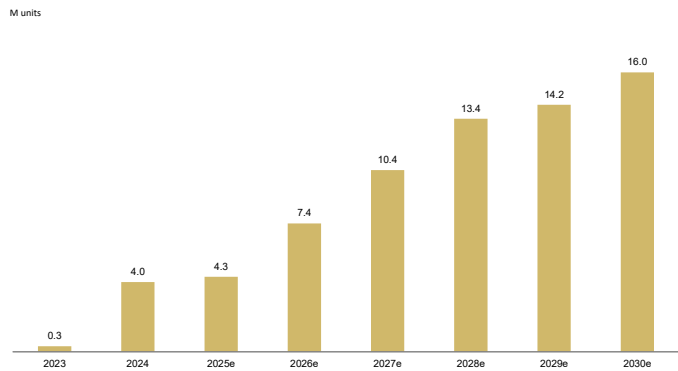
- For GPU units, which is primarily Nvidia GPUs – our Nvidia analyst, Joe Moore, is forecasting AI GPU units to grow 48% y/y in F26 (February 2026 to January 2027) and to grow an additional 2% in F27. Joe does not have a forecast beyond F27, but we model conservatively in the years following, and estimate the market demand to remain elevated, but flattish.
- For AI ASICs, demand for in-house designed chips appears strong in the near term, and so we project units to continue to grow throughout the rest of the decade – we also continue to see rising demand from Meta, Microsoft, Open AI, China, etc., where demand is small today, but where we bake in some growth over the next few years.
- For networking chips, we believe this is highly correlated to computing demand, and with GPU demand remaining elevated and growing ASIC volumes, we believe demand for networking chips will also continue to grow over the next several years.
- And last, for server CPUs, demand for general and storage computing is also highly correlated with overall AI demand. As such, we also forecast demand for server CPUs to rise in the coming years.

Exhibit 12: AI GPU unit forecast, 2023-2030e



Source: Morgan Stanley Research estimates.

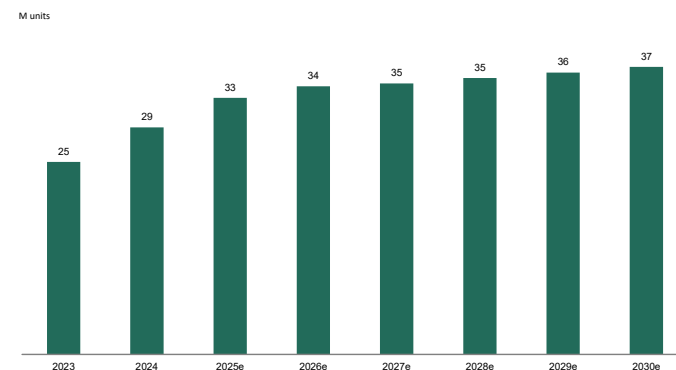
Exhibit 13: AI ASIC unit forecast, 2023-2030e



Source: Morgan Stanley Research estimates.

Exhibit 14: Networking chip forecast, 2023-2030e

Source: Morgan Stanley Research estimates.

Exhibit 15: Server CPU unit forecast, 2023-2030e

Source: Morgan Stanley Research estimates.

Aside from unit growth, larger package size and higher layer count is also a demand tailwind

Increasing demand for semiconductor circuit scale and functionality have been met with increases in the degree of integration of semiconductor dies and in package substrate mounting technology. With deterioration in cost performance owing to increases in the degree of integration of semiconductor dies and advancing process technology, moving to chiplets that contain multiple dies on one package substrate has become indispensable for raising total semiconductor performance.

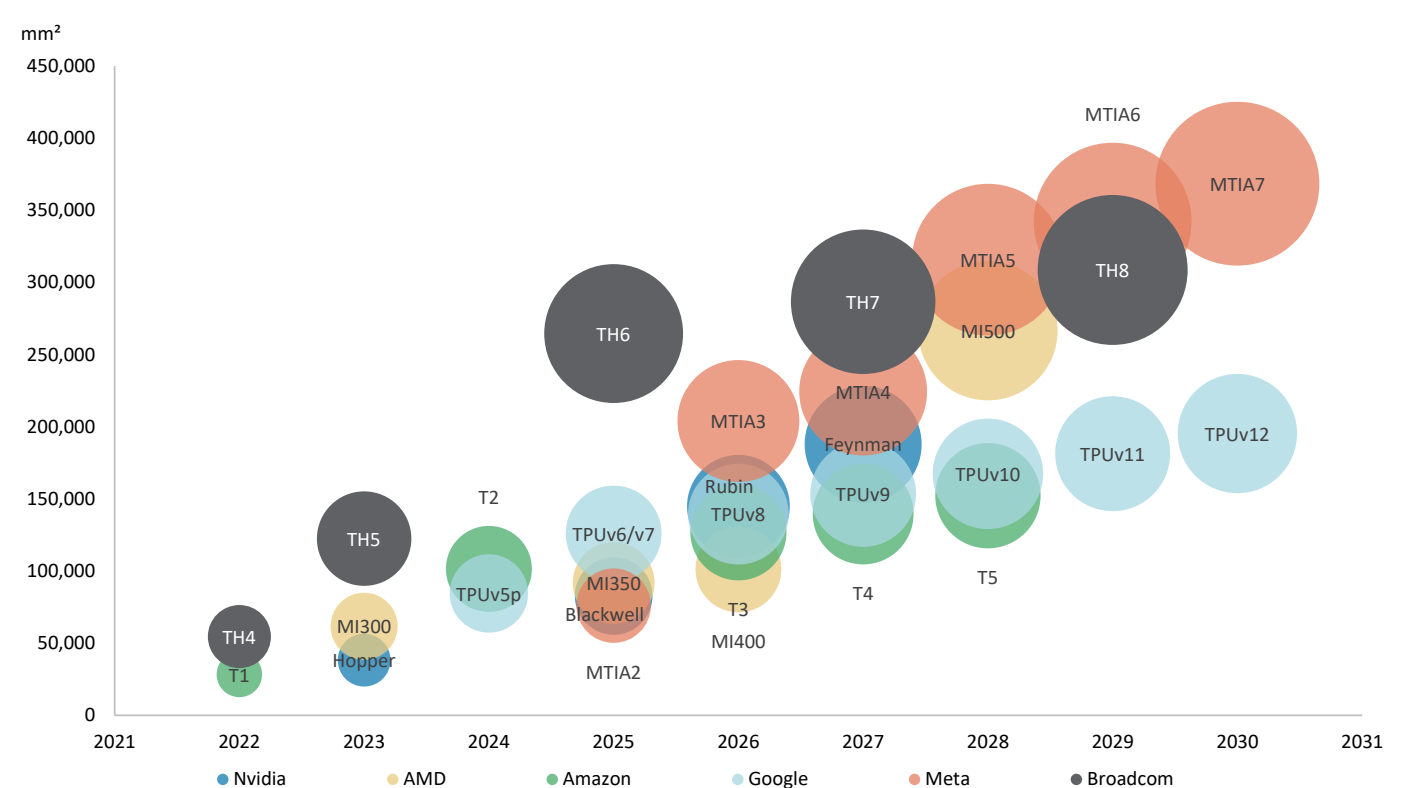
Behind this is that once multiple functions are incorporated into a single semiconductor die, die size expands, and manufacturing complexity and cost rises. Closely-watched methods for mounting multiple semiconductor elements on one package substrate and increasing total semiconductor performance are:

1. 3D, where semiconductor dies are stacked vertically, and
2. 2.5D, where semiconductor elements are connected via silicon or organic interposers on the package substrate.

Larger substrate size and higher layer count are particularly evident, in view of the increase in electrical connections to move data, control signals, and power between the AI accelerator dies and HBM memory stacks.

This trend towards larger substrate size and higher layer count has been clear and it does not appear to be reversing any time soon. So if we take into account the growth from substrate size and layer count, ABF substrate demand will grow even faster than unit growth.

Exhibit 16: Near-term trend points toward larger substrate sizes and higher layer count



Source: Morgan Stanley Research. Note: Bubble size indicates ABF substrate demand, including area size and layer count.

Ajinomoto (2802.T)

Tomonobu Tsunoyama

High-speed Development System leverages strength of de facto standards

Ajinomoto developed ABF in the 1990s by combining epoxy resin with amino acid derivatives, creating a film with superior insulation and processability. In 1999, ABF was adopted by major semiconductor manufacturers such as Intel, which established its position as the industry standard material. Initially used mainly for laptop PCs, demand expanded with the growth of cloud services and general-purpose servers, and now includes next-generation applications such as AI, 5G, and autonomous driving.

ABF enables advances in semiconductor packaging, including finer wiring and high-speed communication. Ajinomoto's strength lies in leveraging ABF's de facto standard status to build an ecosystem with raw material suppliers, equipment makers, substrate manufacturers, and CPU producers, ensuring continuous adaptation to rapid semiconductor technology evolution and maintaining its leadership position.

Multilayered, larger devices mean use of ABF has increased

Current ABF demand growth is driven by HPC applications, including semiconductors for servers, generative AI, and deep learning. HPC usage requires larger and more multilayered package substrates, which has led to expansion in both ABF volumes and pricing through higher value-added products. Compared with traditional PC-related semiconductor packages, AI chips consume roughly 10 times more ABF, accelerating growth momentum.

Advancing material development for optical-electrical hybrid packaging

A key emerging challenge is managing increased power consumption driven by surging data traffic, particularly with rapid AI adoption. Optical-electrical hybrid packaging is gaining attention as a solution, combining efficient optical signal transmission with conventional electrical signals in advanced semiconductor packages. Industry forecasts suggest that CPUs and GPUs may adopt this technology as a standard feature after 2030. Ajinomoto is leveraging its existing network to advance material development in this field, in order to establish its position as an innovation provider.

FM business growth boosts overall profitability for Ajinomoto

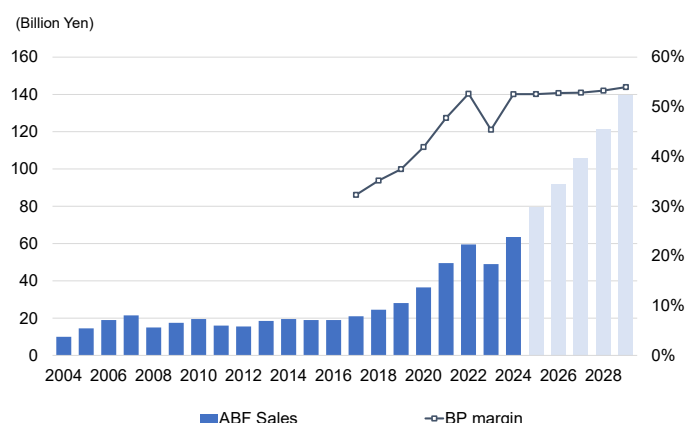
In F3/25, the Functional Materials (FM) segment posted sales of ¥76.5bn (+26% YoY) and BP of ¥40.2bn (+46% YoY), with a BPM of 53%, reflecting exceptional profitability in a growth market where it holds a de facto standard position. Ajinomoto's overall BPM for F3/25 was 10%.

Over the past five years, the FM segment generated BP CAGR of 24%, compared to 10% for the company overall. Looking ahead, we expect the FM segment to achieve CAGR of 16% over the next five years, outpacing Ajinomoto's projected 10% and driving overall profit growth.

The new plant has accelerated ABF growth

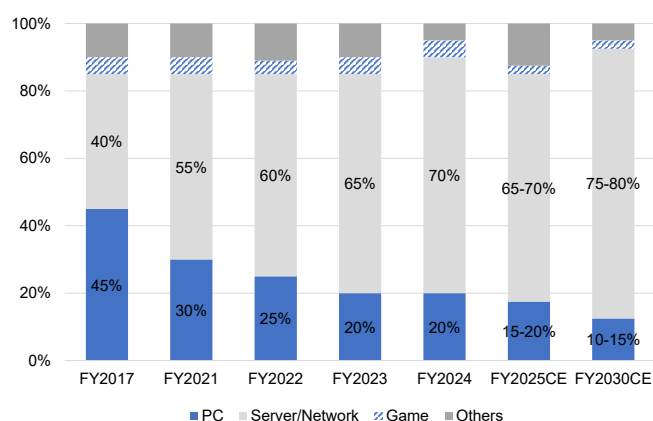
Ajinomoto began operation of its new ABF varnish plant in October 2025. This facility is designed to produce the highest spec products within ABF, including those for AI applications. As a result, sales in the Functional Materials (FM) business for FY2025 Q3 (October–December) rose 41% year-on-year, accelerating from the 21% increase recorded in the first half (April–September). The business profit margin also improved to 57% (from 53% in the first half). Following the Q3 earnings announcement, the market confirmed that the growth of Ajinomoto's ABF business had accelerated due to the new plant's operation, leading to a sharp rise in the stock price.

Exhibit 17: Trends in ABF sales and FM business profit margin

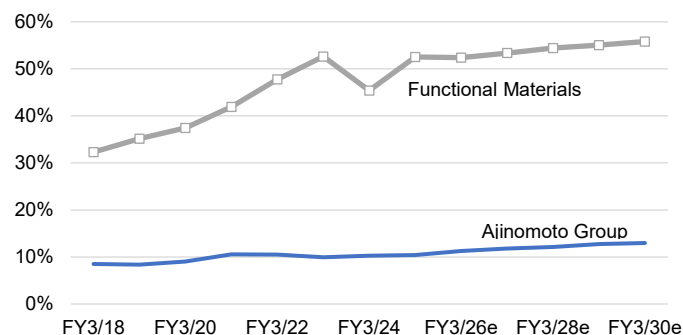


Source: Company materials; some estimates by Morgan Stanley Research

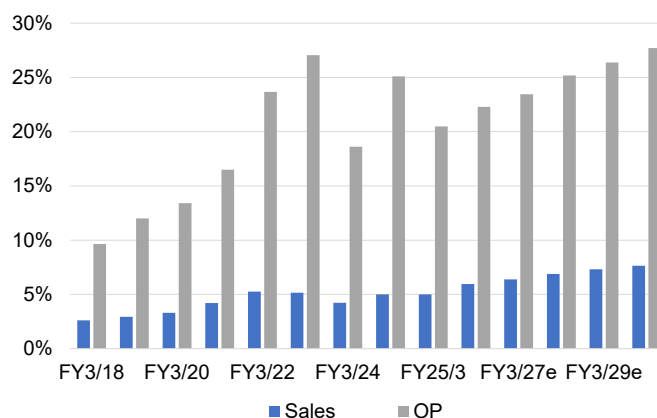
Exhibit 18: ABF sales volume breakdown by application



Source: Morgan Stanley Research. CE = company estimates

Exhibit 19: Functional Materials profit margin

Source: Company materials, Morgan Stanley Research; e = Morgan Stanley Research estimates.

Exhibit 20: Revenue breakdown for Functional Materials

Source: Company materials, Morgan Stanley Research; e = Morgan Stanley Research estimates.

Implications of T-glass supply constraint

What is T-glass and why is it important?

T-glass is a low-CTE (Coefficient of Thermal Expansion) woven glass cloth manufactured by Nitto Boseki (3110.T, covered by Yoshihito Hasegawa). It is mainly used in semiconductor package substrates. Because of its low-CTE characteristic, it can reduce cracking, mechanical stress, warpage, etc. After Nitto Boseki produces T-glass, it is sent to makers of substrate Copper Clad Laminate (CCL), including Mitsubishi Gas Chemical (4182.T, covered by Ryoichi Watanabe) and Resonac (4004.T, covered by Ryoichi Watanabe), for BT substrate CCL and ABF substrate CCL production, respectively.

T-glass supply has been constrained since 2025 because of the strong demand for AI chips, and supply has not increased meaningfully in the past few years. Nitto Boseki has a near monopoly position on T-glass currently, but recently there are several new entrants trying to penetrate this market, such as Grace Fabric (603256.SS, not covered), Taiwan Glass (1802.TW, not covered), Taishan Fiberglass (private, a wholly-owned subsidiary of Sinoma Science & Technology (002080.SZ), etc. It takes time for this new capacity to come online, to pass qualification, and to bring up production yield, so in the near term, T-glass supply will likely continue to remain constrained.

Copper Clad Laminate (CCL) implications

Ryoichi Watanabe

CCL to benefit from ASIC market expansion, price hikes to gain traction from April onward

Two CCL applications are PCBs and substrates, and key Japanese CCL makers that focus on substrate applications are Resonac (4004.T) and Mitsubishi Gas Chemical (4182.T).

Before the mid-1990s, expensive ceramics were the mainstream substrate material. Bismaleimide triazine (BT) resin, developed by Mitsubishi Gas Chemical, achieved thinness, heat resistance, and electrical insulation comparable to ceramics while also being cheap and easy to process. As a result, it made rapid market inroads in the late 1990s as an MPU substrate material.

From around 2010, Mitsubishi Gas Chemical focused its efforts on CCL for small packages for smartphones (CSP, AiP, etc.; so-called BT substrates), an area where growth was expected, and it currently has a market share of around 60% in this area.

Resonac (then Hitachi Chemical), by contrast, had positioned CCL for PCBs as its main product through the 1990s. From 2000 onward it began focusing on substrate applications, and it has achieved a market share of around 70% in the large FC-BGA (ABF substrate) space in particular. For GPU substrates alone, it currently appears to hold a 100% share. We think its success in GPUs can be largely attributed to the 2022 launch of a grade with a coefficient of thermal expansion of 3ppm/K or less (MCL-E-795G). The glass cloth used in such low-thermal-expansion grades is T-glass.

Meanwhile, Mitsubishi Gas Chemical has also developed a next-generation low-thermal-expansion material (HL832RS), which it began promotion in 2024. At its F3/26 3Q earnings briefing, it expressed the view that this new product will enjoy substantial growth in AI ASIC application from F3/27. We think this product accounts for less than 1% of total BT materials as of F3/26 but could expand to just under 10% of the mix in F3/27. HL832RS includes a grade that uses T-glass as well as a grade that uses general-purpose E-glass, and a key feature is that both offer low thermal expansion.

To summarize our views on CCL for FC-BGA, we expect Resonac to retain a market share of around 70%, mainly for GPUs and CPUs, while Mitsubishi Gas Chemical expands its share from around 10%, with companies other than these two losing share.

Meanwhile, for small packages as well, there are areas in which the market is expanding rapidly, such as retimers for AI servers, so we still think the market will expand even if the smartphone market remains sluggish.

The current market is holding up well

Mitsubishi Gas Chemical's Information & Advanced Materials business (part of its Specialty Chemicals segment) recorded sales of ¥26bn (+41% YoY) in F3/26 3Q. Sales of BT materials, which we estimate account for around two-thirds of the business's sales, appear to have risen 50% YoY.

Looking toward F3/27, we think growth will continue, driven by the expansion of new products for ASICs and the uptake of price hikes reflecting tight supply/demand.

Meanwhile, Resonac's semiconductor back-end process materials sales in F12/25 4Q were ¥68.9bn (+25% YoY), of which CCL accounts for around 30% and is the largest product by sales. In F12/26, we expect the strong performance to continue as the GPU and ASIC markets expand and price hikes filter through. The company also handles NCF and TIM as AI back-end materials, but it said at its February 14 earnings briefing that it expects AI CCL (excluding PC-related applications, etc.) to be the biggest area of growth in F12/26.

As the performance of the chips increases, and substrate sizes and layer count

increase, it will put more pressure focus on thermal and warpage issues: To solve these issues, substrate makers are using more low-CTE (Coefficient of Thermal Expansion) glass fabrics, so that the substrate's effective CTE is closer to that of silicon, since organic substrates and the silicon atop expand at different rates when temperature rises. The resulting warpage issue can cause solder joint failures or simply uneven surface during assembly.

Smaller CTE mismatch can reduce cracking, mechanical stress, warpage, etc, during thermal cycling throughout the manufacturing process and when in operations. Increasing the core thickness can also reduce probability of warpage.

This is why we typically see a larger or thicker CCL core for ABF substrates used in AI accelerators, higher-end server CPUs and networking chips.

And this is why this low-CTE glass fabric, also known as T-glass, is currently undersupplied, because a thicker CCL core means more layers of T-glass is used per substrate.

Microchannel lids

Sharon Shih

Microchannel lids could help solve the thermal problem at chip level

A microchannel lid is an advanced thermal lid in which a network of micro-scale channels is fabricated within a metal cap and coolant is actively circulated through these channels. We believe microchannel lids could be a potential thermal design for upcoming chip platforms because TDP for chips continues to increase and exceed the thermal capabilities of existing solutions. However, we expect the adoption of MCL to take time – manufacturing yield, leakage detection and long-term reliability requirements remain challenges.

A microchannel lid reduces warpage by addressing both the thermal and mechanical drivers of deformation.

1. By significantly lowering the peak die temperature and minimizing lateral temperature gradients, it reduces thermally induced strain arising from CTE mismatch between the die, package, and substrate.
2. The lid itself acts as a stiff structural layer, increasing the bending rigidity of the die–lid stack and shifting the neutral axis closer to the die, which lowers curvature under thermal loading.

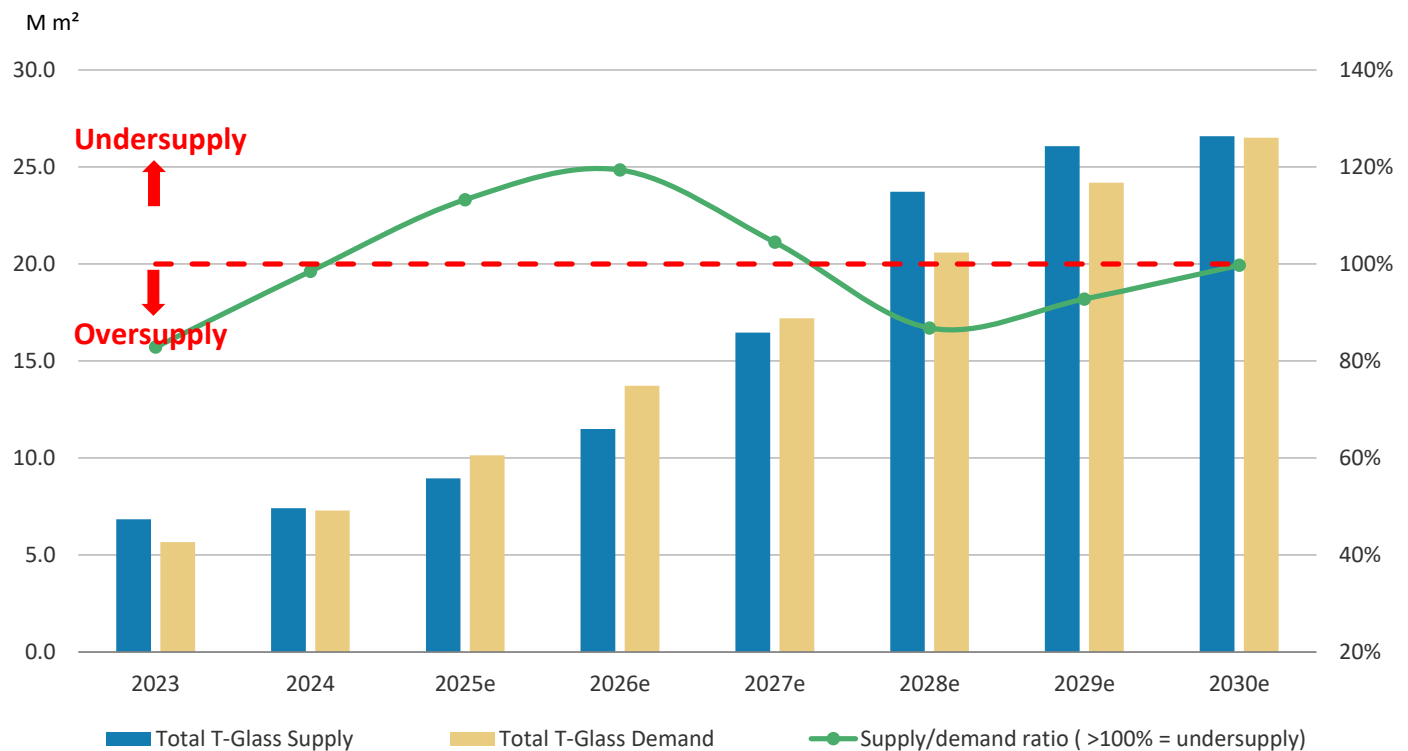
Together, the combination of uniform cooling, reduced thermal excursions, and added stiffness makes microchannel lids particularly effective at mitigating both static and transient warpage in large, high-power packages.

Based on our analysis, we think there is a ~20% supply/demand gap this year: We estimate that ~20% of T-glass supply from Nitto Boseki goes to Mitsubishi Gas Chemical (MGC) for BT substrate CCL, with the majority going to Resonac for ABF substrate CCL production. We believe T-glass undersupply will have a bigger impact on BT substrates, as the priority will be to fulfill demand for higher-end ABF substrates, for GPU, ASIC, CPU, etc.

This is the reason why prices for BT substrates started to move in 2H25, as this constraint started to materialize last year. Even for the ABF substrate market, we are seeing pricing adjustments for ABF substrate CCL recently – [on January 16, 2026, Resonac announced that it would raise prices 30% or more](#) on shipments from March 1, 2026.

We believe both BT substrate and ABF substrate makers will continue to reflect higher raw material prices to their end customers this year: We see some possibility of additional price increases that will directly benefit profits. We will also continue to hear about T-glass constraints throughout the year and even likely into next year.

However, we tend to think that the high-end AI GPUs and ASICs should have T-glass priority, and will not have their demand capped by T-glass constraint. *It will primarily be the other lower-end, non-AI segments that will face bigger impact.*

Exhibit 21: We estimate there is a ~20% supply/demand gap for T-glass in 2026e

Source: Morgan Stanley Research estimates.

IC vendors' abilities to secure T-glass supply

Charlie Chan

This will be a decisive factor in determining market shares

Substrate supply—particularly BT and ABF—has become one of the primary bottlenecks across several high-end IC categories, including AI ASICs, gaming GPUs, memory controllers, and baseboard management controllers (BMCs). For now this stems mainly from the availability of upstream T-glass supply, but customers are mindful that as demand for advanced-node products continues to grow, requiring higher layer-count and more sophisticated substrate solutions could lead to ABF substrate capacity constraints.

In this environment, vendors that are able to secure incremental substrate supply are likely to capture disproportionate market share gains. This is particularly relevant amid intensifying competition in this AI arms race era, where time to market and volume scalability will be key differentiators. Access to substrates is increasingly becoming more strategic, not merely operational.

Against this backdrop, we observe MediaTek taking proactive steps to lock in additional substrate capacity. The company's ability to secure sufficient supply will be a decisive factor in determining whether it can achieve shipments exceeding 2.5 million units of its 2nm "Zebrafish" platform in 2027. In our view, substrate allocation—not end demand—will likely be the binding constraint. Execution on supply chain partnerships will be critical to realizing its targeted scale.

Exhibit 22: Google TPU volume forecast by vendor

k Units	2023	2024	2025e	2026e	2027e	2028e
v5	500	2,400	250			
v6 (Trillium)			1,000			
v7 (Ironwood/Sunfish, by Broadcom)			500	3,200	3,500	3,500
v8 (Zebrafish; 3nm, by MediaTek)				400	2,500	2,500
v9 (Pumafish; 2nm, by Broadcom)						500
v10 (Humafish; 2nm, by MediaTek)						500
Total	500	2,400	1,750	3,600	6,000	7,000

Source: Morgan Stanley Research estimates.

We are seeing new T-glass suppliers trying to enter the market – such as Grace Fabric, Taiwan Glass, Taishan Fiberglass, etc. Nitto Boseki should also have new T-glass supply slowly ramping up from 2H26 onwards, which should help alleviate these constraints over the next few years. But based on our analysis, we do think 2027 will likely remain constrained ([Exhibit 21](#)).

However, this does not take into account alternative solutions that substrate makers are working on...

...such as replacing some T-glass layers with more standard "electrical grade" E-glass fiber cloth, which is cheaper and more widely available. If proven viable, this could also help alleviate some supply constraints for T-glass.

Nitto Boseki (3110.T)

Yoshihito Hasegawa

Nittobo's subsidiary in Taiwan, NITTOBO ASIA Glass Fiber, held a ceremony on February 11 to commemorate the start of full-scale mass production of T-glass at its second plant. The melting furnace installed at the second plant is scheduled to ramp up production gradually from this spring through year-end. By early 2027, when refurbishment work on the existing melting furnaces in Japan is completed, Nittobo's T-glass production capacity is expected to double compared with three years ago.

We believe that, depending on the yield achieved at NITTOBO ASIA Glass Fiber, supply capacity could increase more than twofold. Taking into account the medium-term demand outlook for T-glass, the company's strong position within the industry, and the long lead time required to bring new capacity into mass production, we infer that Nittobo is accelerating its consideration of the next phase of capacity expansion. Potential options include the construction of a new plant, the addition of new melting furnaces, and the repurposing of production facilities used for other types of specialty glass, such as NE glass. In any case, the next capital investment plan is likely to be unveiled in the not-too-distant future.

At this point, we would like to once again highlight the key factors behind the competitiveness of T-glass. Nittobo discloses the composition of its T-glass on its website (SiO₂: 64–66, Al₂O₃: 24–26, MgO: 9–11). As can be seen from the E-glass market, the glass yarn manufacturing process itself is not particularly complex, and if yield were not a concern, T-glass could theoretically be produced in large volumes without difficulty.

The real challenge lies in mass-producing yarn completely free of air bubbles. The presence of bubbles can allow water to penetrate during the substrate drilling process, leading to fatal defects. Therefore, while various new entrants have been able to obtain product qualifications from customers, we believe they have not yet been able to supply sufficient volumes, because the hurdle to improving yields remains high.

ABF substrate pricing inflected in 2025 and we believe it will continue to rise in the coming years: We believe blended ABF substrate pricing will continue to increase, driven by increasing unit demand and mix shift ([Exhibit 23](#)). ASP of the substrates for higher-end AI-related chips (GPU, ASIC, networking, etc.) is higher than that of traditional PC CPU or PC GPU. As we continue to see the demand for these higher-end chips grow, blended ASP will continue to rise.

Moreover, we are also seeing cost tailwinds, as we see prices of CCL for ABF substrates and copper prices increasing. As we mentioned above, Resonac has announced raising ABF CCL prices 30% or more starting from March 1, 2026. And at the same time, copper prices have also been increasing, with prices up ~40% over the past year ([Exhibit 24](#)), as of the end of January 2026.

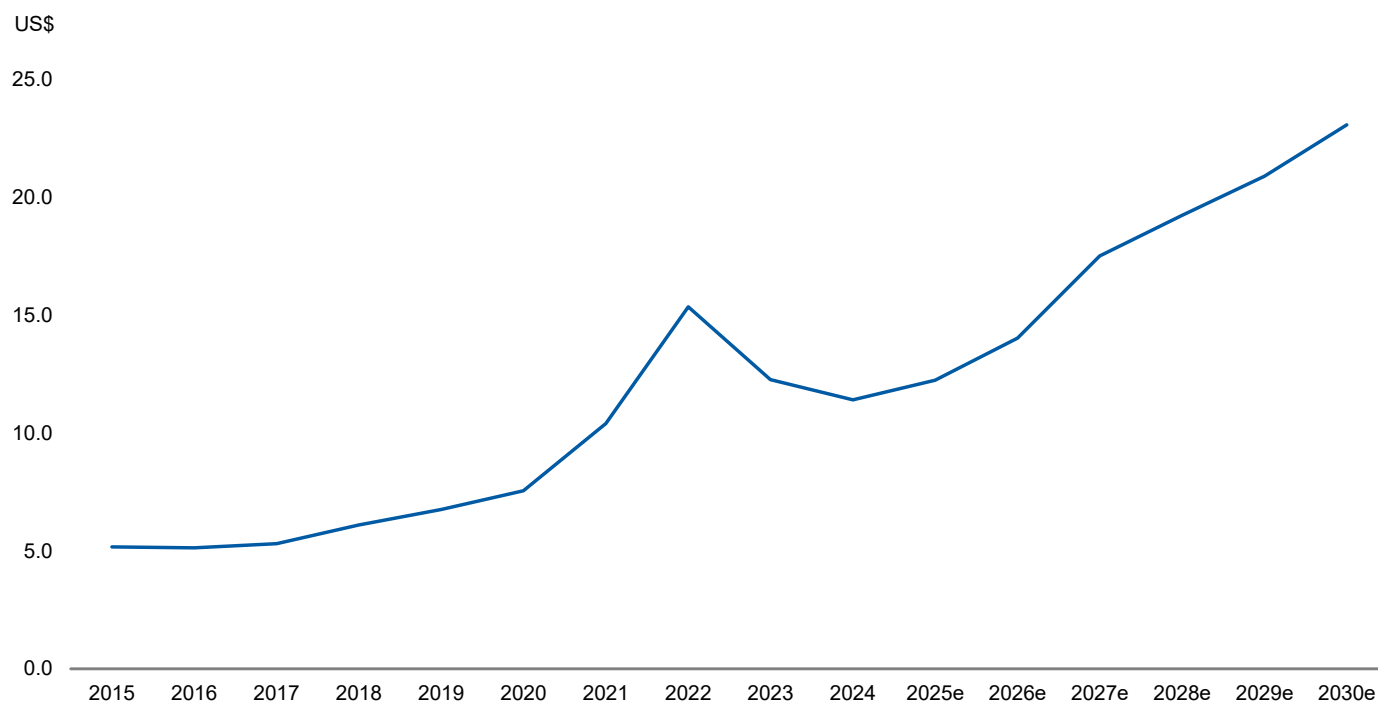
As a result, we expect that ABF substrate makers will look to reflect these higher costs to their end customers, with a possibility of additional price increases that will directly benefit profits.

Based on our analysis, the increase in CCL prices and copper prices will drive ABF substrate pricing to increase 5-10% ([Exhibit 25](#)): We believe ABF substrate makers will reflect the higher costs to end customers over time – our checks indicate that ABF substrate makers will likely hike prices 3-5% each quarter in the first half of 2026, with further price hikes likely in the second half. And heading into 2027, we think there could be a chance of further price increases driven by actual ABF substrate supply/demand tightness.

However, the magnitude of the potential price hikes from CY27 onwards will likely not be able to match what we saw back in the Covid-19 up-cycle, because that was a period where there was a very unexpected WFH demand spike.

In addition, our supply chain checks are suggesting more discussions around subsidies from end customers to ABF substrate suppliers, to secure capacity: This also gives us more confidence that the ABF substrate market is headed towards undersupply; some customers are aware of this and are making plans today to ensure that this will not become a potential bottleneck in the coming years.

Exhibit 23: We forecast ABF substrate blended ASP to continue to increase in coming years, driven by mixture of higher-end substrates and higher material costs



Source: Morgan Stanley Research estimates.

Exhibit 24: Copper price has risen ~40% over the past year (as of January 28, 2026)



Source: FactSet, Morgan Stanley Research.

Exhibit 25: Raw material price increase should drive ABF substrate makers to raise prices by ~10%, by our estimate

If including CCL/PP & Copper price increase (from Mar 1)	PC CPU	AI GPU / ASIC
Raw materials	2.4	35.8
Ajinomoto Build-Up Film (ABF)	0.6	6.0
CCL+Prepreg	1.1	23.4
Copper & plating metals	0.5	3.4
Others (photoresist, solder masks, etc)	0.2	3.0
Direct Labor	0.5	5.0
Depreciation & Amortization	0.3	10.0
Other manufacturing overhead	0.3	5.0
Total	3.5	55.8
<i>Increase in total ABF substrate price would be</i>	<i>9.4%</i>	<i>11.5%</i>

Bill of materials (%)	PC CPU	AI GPU / ASIC
Raw materials	68%	64%
Ajinomoto Build-Up Film (ABF)	18%	11%
CCL+Prepreg	31%	42%
Copper & plating metals	13%	6%
Others (photoresist, solder masks, etc)	6%	5%
Direct Labor	14%	9%
Depreciation & Amortization	9%	18%
Other manufacturing overhead	9%	9%
Total	100%	100%

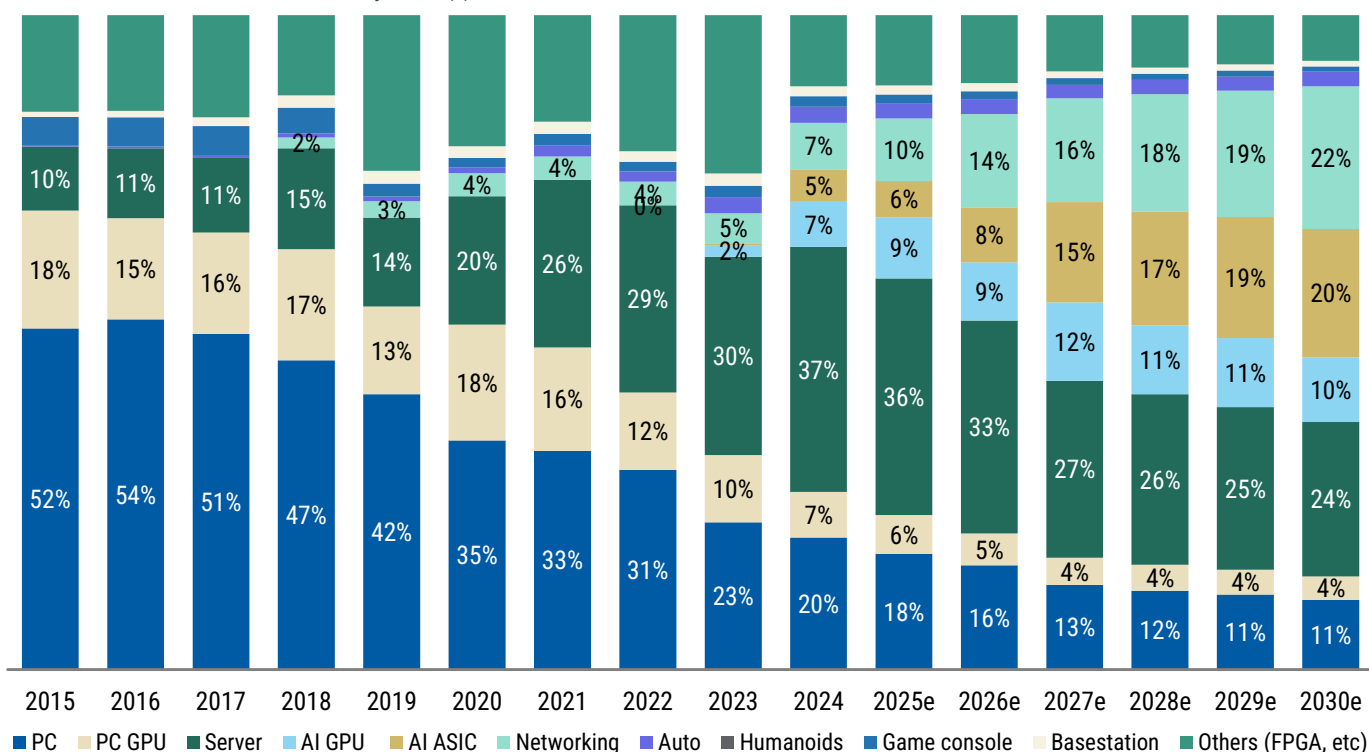
Source: Morgan Stanley Research estimates.

Breaking down ABF substrate end demand

PCs were once the biggest end market for ABF substrates, in value terms. Back in 2015 ABF substrates used in relation to the PC market accounted for ~70% of the total ABF substrate value globally, mainly supported by demand for PC CPUs and PC GPUs. But end demand has shifted meaningfully since. We estimate that in 2025, ABF substrates used in the PC industry had declined to account for only ~24% of the market, and by 2030, we estimate it will drop further, to less than ~15%.

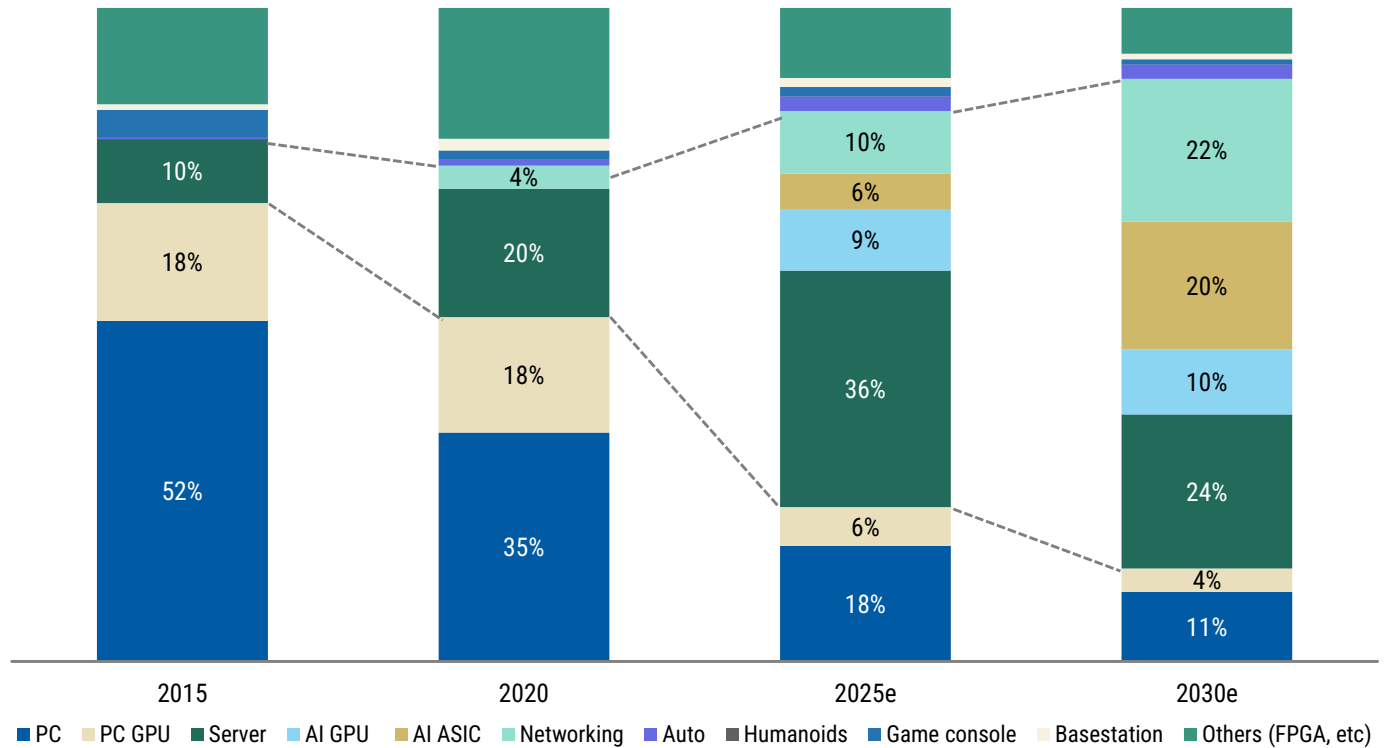
On the other hand, server, AI GPU, AI ASIC, and networking-related ABF substrate demand is on the rise. In 2015, we estimate that these segments combined were only ~10% of the market, but that rose to ~60% by 2025 and we expect it will account for ~75% of the entire ABF substrate market value by 2030. In other words, server, AI GPU, AI ASIC, and networking-related applications have been the main growth driver for ABF substrate demand over the past decade and will remain a tailwind to drive even further growth in the coming years. This is the main reason why we think the ABF substrate market growth will accelerate over the next five years.

Exhibit 26: ABF substrate demand by end applications, 2015-2030E



Source: Morgan Stanley Research estimates.

Exhibit 27: PC-related ABF demand is declining from ~70% in 2015 to <15% by 2030, while server, AI GPU, AI ASIC, and networking-related ABF demand is rising from ~10% in 2015 to ~75% by 2030



Source: Morgan Stanley Research estimates.

Which companies are manufacturing ABF substrates?

There are seven major companies globally that supply ABF substrates:

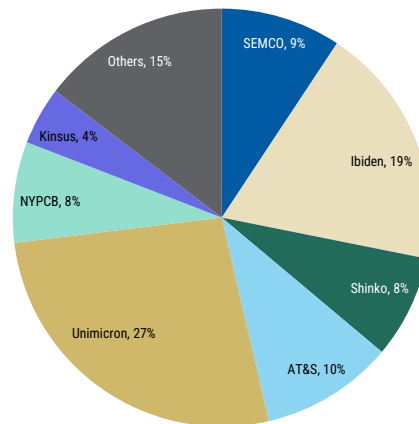
- Ibiden and Shinko in Japan;
- Unimicron, NYPCB, and Kinsus in Taiwan;
- SEMCO in Korea; and
- AT&S in Austria.

There are also other suppliers like Kyocera and Toppan in Japan, and LG Innotek in Korea.

In addition, we've recently seen new market entrants like Zhen Ding In Taiwan, as well as Shennan Circuits, Shenzhen Fastprint, and Access Substrates in China.

Exhibit 28: We estimate the top seven ABF substrate suppliers account for ~85% of the market

ABF Substrate Revenue Mix by Suppliers (2024)



Source: Company data, Morgan Stanley Research estimates.

Differences and similarities with this cycle vs. 2020-2022

The major differences

This ABF up-cycle has clearer demand visibility, driven by enterprise demand: AI is the main driver of this demand cycle, whereas the last up-cycle was mainly driven by WFH demand, with a big portion of the demand increase driven by PCs ([Exhibit 27](#)). As a result, in this cycle, ABF substrate suppliers should have better end demand clarity since they will just be able to directly engage with the end customers (i.e., Nvidia, AMD, Intel, Google, Amazon, Broadcom, etc). This should also make it easier for them to get comfort investing in capacity to support their future demand roadmap.

This is also why we think that while the industry will head into undersupply from CY27 onwards, we don't think pricing and profit expansion for the ABF substrate suppliers will be milder – it will not replicate that of the 2020-2022 cycle.

There are more players now: Since the 2020-2022 up-cycle, we see more suppliers investing in this market to try to become a more meaningful supplier. AT&S in Austria is one such supplier; it has made a point of investing in more capacity, aiming to be a top three supplier globally, but we also see new entrants in the Greater China region, like Zhen Ding, Shennan Circuits, and Shenzhen Fastprint.

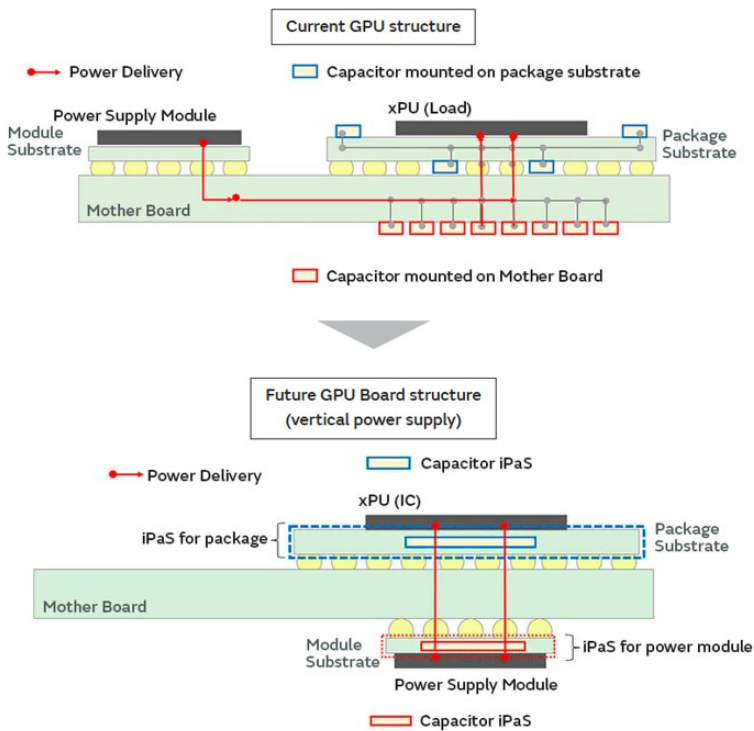
However, the capability of these new entrants remains to be seen, and we will have to closely monitor the progress of these new entrants. For now, they do not pose any real threat to drive the industry into oversupply ([Exhibit 28](#)).

Increasing demand for substrate with embedded passives: Our supply chain checks indicate that this could become more prevalent in future generations of advanced semiconductor substrates. By embedding passives into the substrate, it allows a higher degree of freedom in the PCB design, reduces impedance, and enhances power integrity, which is extremely critical for AI applications.

This could prove more difficult to produce, as it introduces new failure points, potentially resulting in lower yields, which would put increased pressure on ABF substrate capacity.

This also raises the repair and maintenance costs of the vendors and customers, because if a single embedded passive component is defective, the entire package might need to be scrapped.

Exhibit 29: Murata's Capacitors embedded substrates (Capacitor Type iPaS™)



Source: Murata.

T-glass supply constraint and subsequent CCL price increase: This was not an issue back in the 2020-2022 up-cycle, as demand was primarily driven by PCs and to a lesser extent servers. The substrate cores for PC CPUs are much thinner than the ones we see today in advanced AI chips, which is why T-glass is now in constraint.

A more elevated copper price environment: Copper prices also increased sharply from March 2020 to May 2021, up ~114% during that period from trough to peak, but in today's environment (as of January 28, 2026) copper prices are even higher, ~27% above the peak in May 2021.

Within the substrate segment, we see companies placing a bigger focus on ABF substrates over BT substrates: Based on our supply chain checks, companies like Unimicron are placing even more focus on ABF substrates in this cycle vs. the last cycle, de-emphasizing some lower-end, lower-margin BT substrate products, which will likely cause some share shifts within the BT substrate market.

The similarities

ABF substrate size and layer count are increasing: This is the same tailwind that we saw back in the 2020-2022 up-cycle as well, but we are seeing a more accelerated impact with the rise of advanced semiconductors driven by AI. This rise in demand for the next few years is beyond what the industry was able to grasp in the last cycle, and so we do believe the capacity expansion put in place then is not enough to support the growing AI opportunities.

We believe ABF substrate suppliers will start to raise prices to end customers: This will kick in from 1H26; initially it will mainly be to reflect higher input costs (CCL, copper, etc.), and prices will adjust over time. But moving into 2027, we believe the ABF substrate market will head into undersupply again, and the industry supply/demand dynamics will start to manifest in substrate prices.

We are also hearing discussions around subsidies from end customers to ABF substrate suppliers, to secure capacity. Similar discussions were heard in the previous up-cycle as well, and should drive increased confidence that this means end customers are aware of the growing demand and are taking necessary steps to ensure that ABF substrates do not become a bottleneck for their future chip growth.

Why it is not too late

This undersupply will continue into the end of the decade

Based on our bottom-up supply/demand analysis, we believe ABF substrate capacity will head into undersupply from 2027 onwards and last into 2030 ([Exhibit 9](#)). While stocks have already started to move, fundamentally, we are at the beginning of the up-cycle and earnings are just starting to inflect upwards ([Exhibit 32](#)).

Over the next few quarters, we expect ABF substrate suppliers to raise pricing to end customers to reflect higher costs, and moving into 2027, we think there could be further price hikes, driven by a shortage of ABF substrate capacity.

With multiple years of undersupply on the horizon, we believe it is not too late to position in these stocks. We believe earnings will continue to improve over the next several years, and consensus numbers will continue to rise, which will all help drive share prices to continue to move higher.

While the market is currently aware that an up-cycle is on the horizon, we don't think many investors fully understand how strong this AI-driven up-cycle could be. We think they are mainly just pricing in the price hikes from increased material costs at this stage – not that we are just entering a multi-year AI-driven up-cycle, or the potential for further price hikes driven by capacity constraints.

Valuation is still not too expensive

While the stocks have already started to move up, we see further upside for them – valuation is still far from the previous up-cycle's peak.

In the 2020-2022 up-cycle, the top six ABF suppliers' P/E multiple ultimately peaked at 47.5x as earnings started to come down, but today it's still only at ~32x, on relatively low

future earnings potential. This gives us confidence that there is more upside left for these stocks, given our belief that earnings estimates will rise.

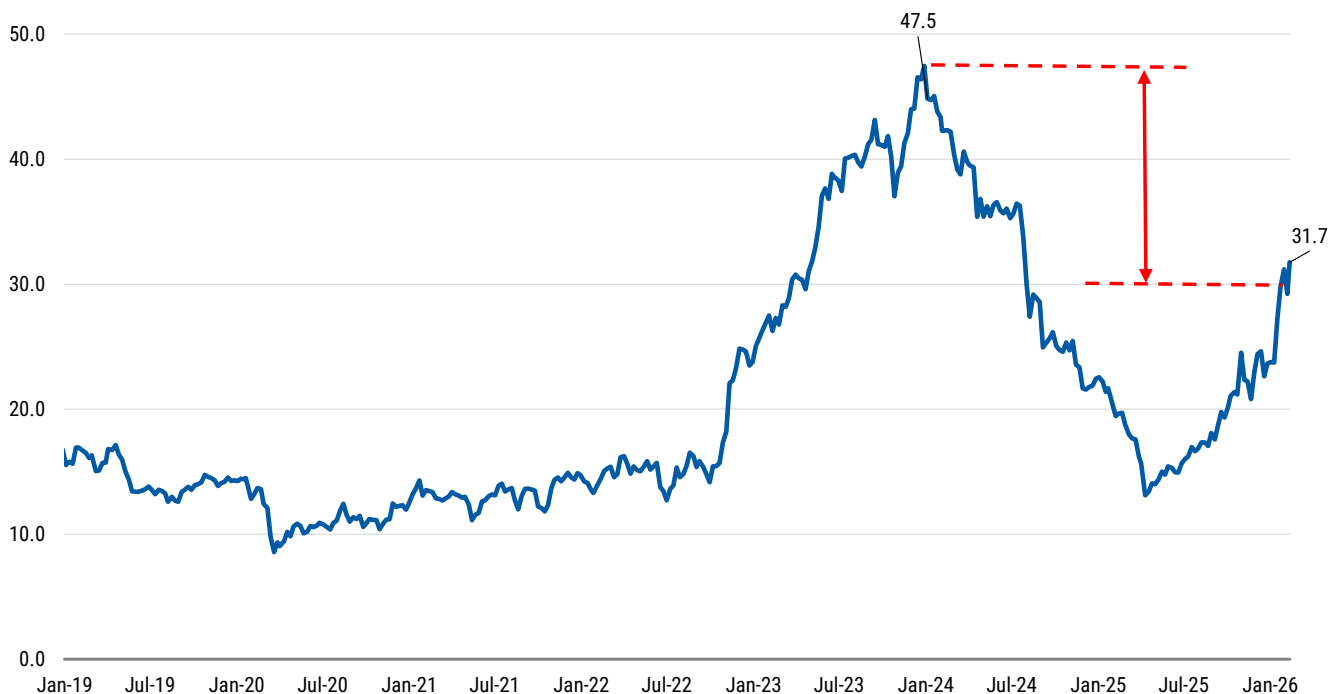
The last cycle's peak multiple can provide a good benchmark for how these stocks could trade in this cycle, but we do see a lot of structural differences this cycle as well. In the previous 2020-2022 cycle, consumer PC demand was the major driver of the ABF substrate shortage, but in this cycle, it is AI-driven. And customers are putting more focus on substrate supply as it is becoming more of a strategic component.

Combined, we think this could help drive a stronger multiple expansion, but also a more sustainable re-rating, as we have seen across so many different segments since 2023.

In short, we believe the real multiple is lower because we expect earnings estimates to rise. We also believe this AI-driven up-cycle can support an even higher multiple compared to history, as we have witnessed across multiple tech segments since 2023.

Exhibit 30: The top six ABF suppliers' P/E today is still below the previous peak, suggesting more room for share price appreciation

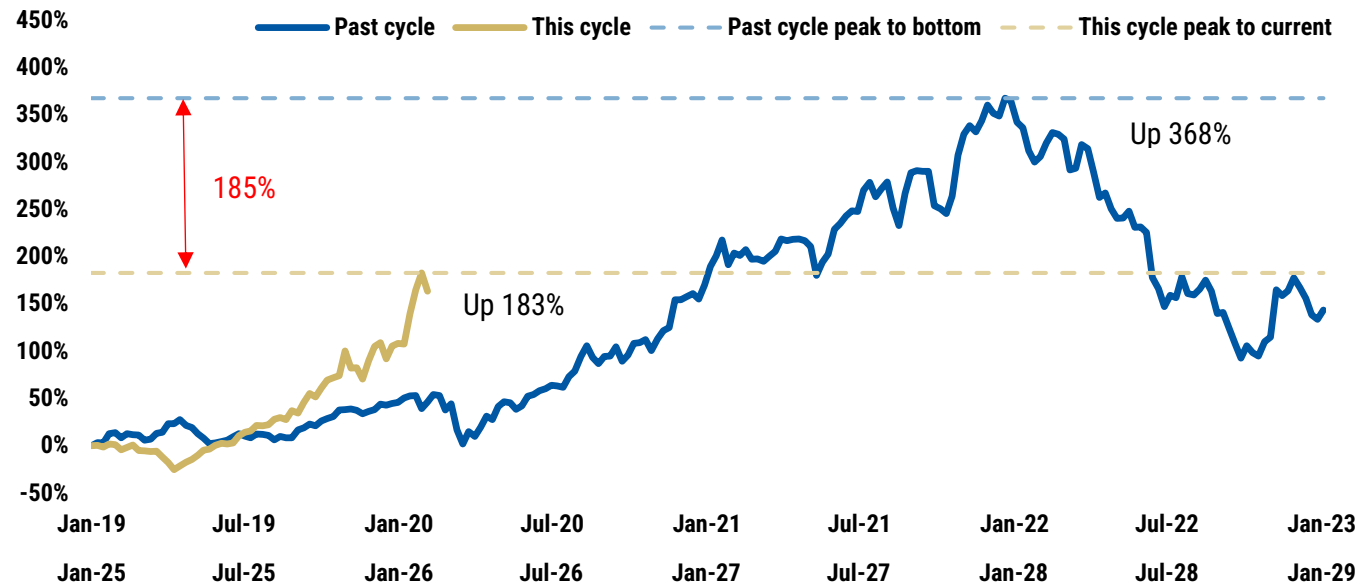
Top 6 ABF Suppliers P/E



Source: FactSet, Morgan Stanley Research. Note: Top 6 ABF suppliers include Unimicron, Ibiden, SEMCO, NYPCB, AT&S, Kinsus.

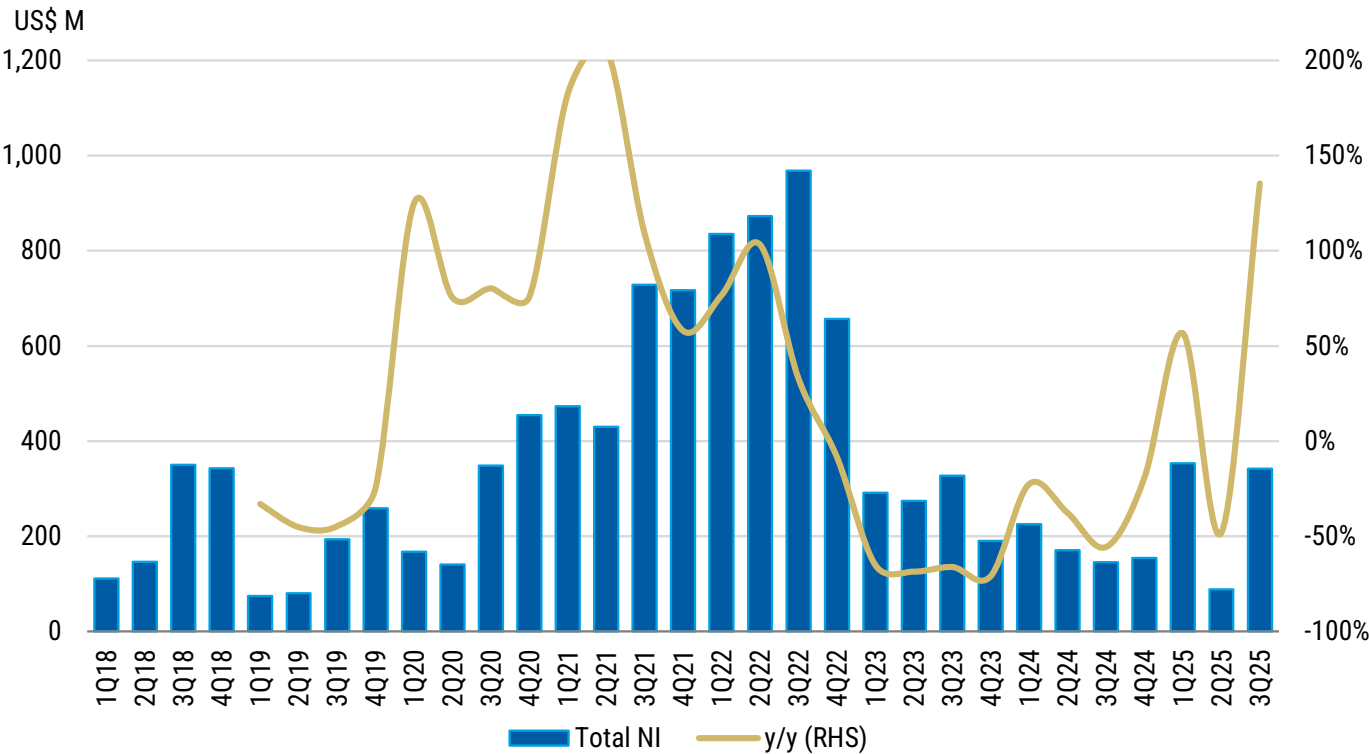
Exhibit 31: This cycle vs. last cycle – share price chart

Top 6 ABF Suppliers Price Performance



Source: FactSet, Morgan Stanley Research.

Exhibit 32: Top six ABF suppliers' quarterly net income (1Q18-3Q25)



Source: FactSet, Morgan Stanley Research.

Where we think there could be risks

We see several potential risks.

1. If demand for AI is not as sustainable as we have modeled, then the supply/demand dynamics will be different from our analysis, causing pricing and profitability to play out differently vs. our base case scenario.
2. T-glass supply is more constrained than expected, resulting in demand downside for a lot of the higher-end AI-related chips.
3. We see incrementally more capacity expansion announcements from the top ABF substrate incumbents, which could change the supply/demand dynamics from 2028 onwards.
4. Better yield ramp and supply from "new" ABF substrate suppliers such as Zhen Ding or Shennan Circuits, which would also lead to a different supply/demand outcome vs. our base case.
5. Weaker demand from non-AI segments owing to accelerated rise of input costs (i.e., memory).
6. Adoption of CoWoP, which would displace ABF substrates, would lead to lower demand for usage of ABF substrates.

Turning Bullish on Unimicron and NYPCB after Nearly Four Years

Unimicron (3037.TW)

Upgrade to OW (from EW)

Our new price target is NT\$500 (up from NT\$120.75): This implies 32x our 2027 base case EPS estimate and 5.9x 2027e base case P/B. This is higher than the 8-20x range in which it traded during the 2020-2022 up-cycle, but we think it's fair because we have also witnessed a re-rating of AI component stocks that enjoy value upgrade in their products for AI servers.

Moreover, this cycle is different and much stronger, driving Unimicron to generate significant growth in 2025-28e, which also supports a higher multiple.

We forecast earnings in 2025-28 to expand at a 105% CAGR (+136% y/y in 2026 and +108% y/y in 2027e, per our estimates).

Given the strong growth, Unimicron's 2027e PEG, based on our price target, looks attractive at only 0.3x.

Nan Ya PCB (8046.TW)

Double upgrade to OW (from UW)

Our 12-month price target is NT\$515 (up from NT\$165): This implies 40x our 2027 base case EPS estimate and 5.9x 2027e base case P/B. This is higher than the 8-25x range in which it traded during the 2020-2022 up-cycle, but here too, we think it's fair because we have also witnessed a re-rating of AI component stocks that enjoy value upgrade in their products for AI servers.

And again, this cycle is different and much stronger, driving NYPCB to generate significant growth in 2025-28e, which also supports a higher multiple.

We forecast earnings in 2025-28 to expand at a 113% CAGR (+200% in 2026 and +73% y/y in 2027, per our estimates).

Given the strong growth, NYPCB's 2027e PEG, based on our price target looks attractive at only 0.5x.

ABF Earnings Calendar

Exhibit 33: ABF Earnings Calendar

Company	C4Q25 Result Date	Earnings Call
Unimicron	Feb. 24, 2026	Feb. 25, 2026
Nan Ya PCB	Feb. 24, 2026	No

Source: Company data, Morgan Stanley Research.

Ibiden (4062.T), UW, Price Target ¥6,500

Shoji Sato

We expect Ibiden to maintain a high market share in ABF package substrates for NVIDIA's Blackwell and Rubin platforms and to sustain its technological advantage over Taiwanese and Korean competitors over the medium to long term. However, we expect equity market expectations for Ibiden's earnings to be lowered after the F3/26 full-year results announcement in early May. The February 13 closing price corresponds to 4.3x our forecast F3/27-end BPS of ¥1,989, well above the post-2020 average one-year forward P/B of 1.9x. We believe there is a high risk of a share price decline of around 25% over the next 12–18 months.

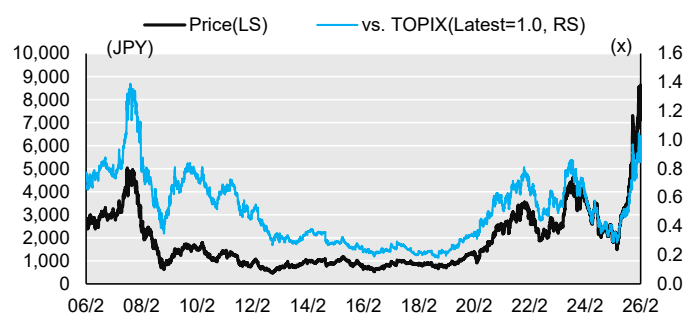
Exhibit 34: Ibiden's stock price and rating history



Note: OW=Overweight, EW=Equal-weight, UW=Underweight
Source: FactSet, Morgan Stanley Research

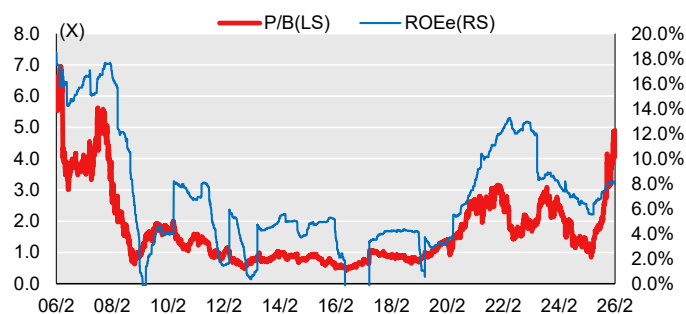
Following Ibiden's F3/26 3Q earnings announcement on February 2, we revised our earnings forecasts and downgraded our investment rating to Underweight, while maintaining our price target at ¥6,500. Although Ibiden's earnings bottomed out in F3/24 and we forecast that earnings improvement and expansion will continue, we expect that after the announcement of F3/26 full-year results in early May and the release of company guidance for F3/27 and F3/28, market consensus for F3/27 and F3/28 will be lowered. Our price target of ¥6,500 implies 25% downside from the February 13 closing price of ¥8,640.

Exhibit 35: Share Price and Relative to TOPIX

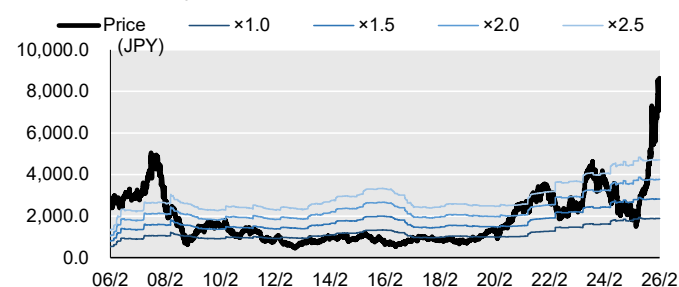


Source: FactSet

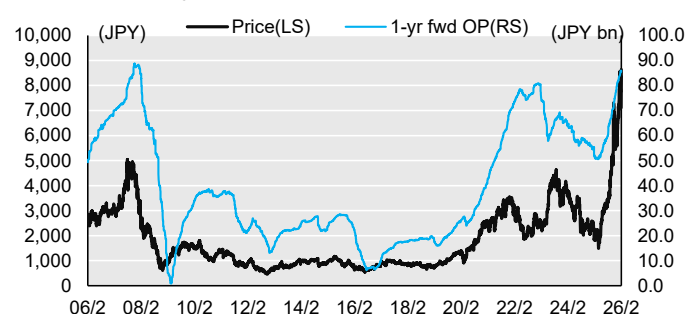
Exhibit 36: Actual P/B and 1FY forward ROE



Source: FactSet

Exhibit 37: One-year Forward P/B Bands

Source: FactSet

Exhibit 38: One-year Forward Consensus OP and Share Price

Note: Consensus is based on FactSet
Source: FactSet, Morgan Stanley Research

The F3/26 3Q OP of ¥12.0bn announced after market close on February 2 (our estimate: ¥14.2bn; FactSet consensus: ¥13.6bn) declined ¥2.9bn QoQ. Ibiden left its F3/26 company-wide guidance unchanged at sales of ¥420bn and OP of ¥61.0bn, but lowered its F3/26 outlook for the Electronics segment, whose main product is ABF package substrates: sales were revised from ¥255bn to ¥250bn, OP from ¥48.0bn to ¥46.5bn, and pre-depreciation OP from ¥107.0bn to ¥102.0bn. Compared with the guidance released on October 30, we believe the main factors behind the revision were: (1) weaker-than-expected sales to Intel and new customers, and (2) delays in the production ramp-up at the Ono plant.

On October 31 in 2025, assuming an exchange rate of ¥145/US\$, Ibiden presented OP outlooks of ¥61.0bn for F3/26 (our estimate: ¥61.8bn; FactSet consensus: ¥64.1bn), ¥72.0bn for F3/27 (¥77.0bn; ¥88.0bn), and ¥90.0bn for F3/28 (¥106.4bn; ¥110.5bn). Our forecasts assume ¥150/US\$, and we estimate that OP changes ¥0.8bn per year for every ¥1/US\$ move. Ibiden also explained that even if procurement constraints for T-Glass/CCL persist, it can still achieve the F3/27 sales target of ¥480bn presented on October 31. We assume that T-Glass/CCL will remain a bottleneck for expanding ABF package substrate production until around the end of C2026. Accordingly, we expect that the F3/27 company OP guidance to be announced in early May will remain unchanged at ¥72.0bn, assuming an exchange rate of ¥145/US\$.

On February 2, Ibiden announced a ¥500bn capex plan over three years from F3/27 to F3/29, including ¥220bn of investment in EMIB T at the Gama plant. If mass production of EMIB T were to start in F3/27 instead of Ibiden's assumed F3/28, earnings from F3/27 onward would significantly exceed our forecasts. However, given equipment lead times and the high technological difficulty of EMIB T production, we assume that mass production will begin in 2H F3/28.

Exhibit 39: Consolidated Income Statement

(Y bn)	F3/25				F3/26			
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Qe
Income Statement								
Net Sales	88.2	93.4	88.8	99.1	97.5	98.0	103.1	119.5
COGS	62.6	60.6	62.6	67.7	62.4	65.2	75.1	85.5
Gross profit	25.6	32.8	26.2	31.4	35.1	32.8	28.1	34.0
SG&A	14.3	15.5	17.3	18.6	17.4	17.9	16.1	16.7
Operating Profit	11.3	17.2	6.3	12.8	17.6	14.9	12.0	17.3
Non-operating Income/Loss	1.6	-0.6	0.0	-0.8	-0.2	0.0	-0.6	-1.2
Recurring Profit	12.9	16.7	6.4	12.0	17.4	14.9	11.3	16.1
Extraordinary Income/Loss	-0.9	-1.4	-0.3	6.2	-0.4	0.0	-0.1	-0.5
Pretax Profit	11.9	15.2	6.1	18.2	17.0	14.9	11.2	15.6
Income Tax	3.0	3.4	1.7	9.3	4.2	5.5	2.2	4.2
(Effective Tax Rate)	25%	23%	28%	51%	24%	37%	20%	27%
Minority Interest	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Net Profit	8.8	11.7	4.3	8.9	12.7	9.3	8.9	11.3
Capex	40.3	43.2	27.0	46.8	20.4	14.4	12.5	19.7
Depreciation and Amortization	10.2	12.7	15.3	16.0	11.7	13.8	19.1	21.0
R&D Expenses	5.4	6.5	8.3	7.3	7.8	8.1	6.7	7.0
EBITDA	21.5	29.9	21.6	28.8	29.3	28.7	31.0	0.0
ROE	-	-	-	-	-	-	-	-
Average Share Outstanding (Diluted)	-	-	-	-	-	-	-	-
EPS (Diluted, Yen)	-	-	-	-	-	-	-	-
BPS (Yen)	-	-	-	-	-	-	-	-
DPS (Yen)	-	-	-	-	-	-	-	-
USD/JPY: Average Exchange Rate	152.0	154.0	147.0	155.0	145.0	145.0	151.0	150.0
EUR/JPY: Average Exchange Rate	164.0	166.0	162.0	156.0	161.0	169.0	174.0	170.0
YoY % Change								
Net Sales	-7%	0%	-4%	10%	10%	5%	16%	21%
Operating Profit	38%	8%	-51%	20%	56%	-13%	88%	35%
Net Profit	21%	10%	-55%	117%	44%	-20%	109%	27%
Sequential % Change								
Net Sales	-2%	6%	-5%	12%	-2%	1%	5%	16%
Operating Profit	6%	52%	-63%	101%	38%	-15%	-20%	45%
Net Profit	115%	33%	-64%	108%	43%	-27%	-4%	27%
As % of Sales								
Gross profit	29.0%	35.1%	26.6%	31.6%	36.0%	33.5%	27.2%	28.5%
Operating Profit	12.8%	18.4%	7.1%	12.9%	18.1%	15.2%	11.6%	14.5%
Net Profit	10.0%	12.5%	4.8%	9.0%	13.1%	9.5%	8.7%	9.5%
Capex	45.7%	46.3%	30.4%	47.2%	21.0%	14.7%	12.2%	16.4%
Depreciation and Amortization	11.6%	13.6%	17.2%	16.2%	12.0%	14.0%	18.5%	17.6%

Note: Shares outstanding measured in million shares. DPS estimates are based on the dividend ratio in the past. e=Morgan Stanley Research estimates, Ce=Company estimates.

Source: Company data, Morgan Stanley Research

Exhibit 40: Consolidated Balance Sheet and Cash Flow Statement

(Y bn)	F3/25				F3/26			
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Qe
Balance Sheet								
Total Assets	1,147.7	1,100.0	1,105.7	1,081.7	1,079.7	1,037.9	1,054.1	
Total Current Assets	590.2	539.3	523.2	549.6	532.4	487.5	504.7	
Cash and Cash Equivalents	425.3	374.2	358.4	390.7	374.2	325.5	340.7	
Marketable Securities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Account Receivables	65.9	60.4	68.8	68.2	65.1	68.6	72.4	
Inventory	68.1	69.2	73.1	67.2	67.6	68.4	73.4	
Total Long Term Assets	557.5	560.7	582.4	532.1	547.3	550.5	549.4	
Net PP&E	444.5	460.1	476.0	460.1	465.5	465.2	459.9	
Construction in Progress	268.7	233.8	192.6	202.0	205.0	150.2	129.8	
Investments in Securities	102.3	90.7	95.8	58.8	70.5	73.7	77.8	
Total Current Liabilities	366.0	319.1	294.4	327.7	305.9	261.8	259.2	
ST Debt & Current Portion of LT Debt	90.0	70.0	60.0	90.0	90.0	65.0	65.0	
Accounts Payable	44.2	42.4	48.9	38.8	36.2	38.4	45.7	
Advance received	77.0	88.6	87.1	92.1	89.8	77.6	76.9	
Total Long Term Liabilities	267.2	283.6	295.0	256.7	258.2	244.0	245.3	
Long Term Debt	253.4	273.2	283.1	253.0	252.9	237.7	237.6	
Net Assets	514.6	497.3	516.3	497.3	515.6	532.2	549.5	
Minority Interests	6.9	6.9	6.8	6.8	7.0	7.1	7.1	
Stock Acquisition Rights	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Net Debt	-184.2	-121.7	-111.1	-106.5	-101.8	-96.4	-116.0	
Days Receivable Outstanding (Average)	68	62	66	63	62	62	62	
Days in Inventory (Average)	97	103	100	95	99	95	86	
Days in Accounts Payable (Average)	64	65	64	59	55	52	51	
Cash Conversion Cycle (Average)	101	100	102	99	106	105	97	
Cash Flow Statement								
Net Cash from Operations	8.4	36.4	23.3	50.7	9.6	27.1	36.0	
Net Income	8.8	11.7	4.3	8.9	12.7	9.3	8.9	
Depreciation and Amortization	10.2	12.7	15.3	16.0	11.7	13.8	19.1	
Changes in Working Capital	-0.1	-1.4	-3.1	-5.1	0.8	-1.7	1.8	
Others	-	-	-	-	-	-	-	
Net Cash from Investing Activities	-26.7	-82.4	-39.1	-16.1	-23.3	-36.9	-20.2	
Purchase of PP&E	-26.1	-82.2	-38.0	-51.2	-23.3	-36.3	-20.0	
Sale/Purchase of Investments	0.0	0.0	0.0	34.6	0.0	0.0	0.3	
Other-net	-0.6	-0.1	-1.0	0.5	0.0	-0.6	-0.6	
Free Cash Flow	-18.3	-45.9	-15.7	34.6	-13.7	-8.8	15.8	
Net Cash from Financing Activities	-2.8	-0.3	-3.7	-0.3	-2.9	-40.7	-3.8	
Change in Interest-bearing Debt	0.0	-0.2	0.0	0.2	0.0	-40.0	0.0	
Change in Treasury Stock	0.1	0.0	-0.6	0.0	0.0	0.0	0.0	
Payment of Cash Dividends	-2.8	0.0	-2.8	0.0	-2.8	0.0	-4.2	
Other-net	-0.1	-0.2	-0.3	-0.5	-0.2	-0.7	0.4	
Net Effect of Exchange Rate Changes	2.8	-4.8	3.6	-2.1	0.2	1.7	3.3	
Net Change in Cash	-18.3	-51.1	-15.8	32.2	-16.4	-48.8	15.2	
Cash at End of Period	425.3	374.2	358.4	390.7	374.2	325.5	340.7	

Note: Net debt = Interest bearing debt - Cash & cash equivalents - Short-term marketable securities - Investments in securities; e=Morgan Stanley Research estimates, Ce=Company estimates

Source: Company data, Morgan Stanley Research

F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26e	F3/27e	F3/28e	F3/26Ce		
										1H	2He	
291.1	296.0	323.5	401.1	417.5	370.5	369.4	418.1	499.7	602.5	195.5	224.5	420.0
232.8	227.8	238.0	281.1	290.0	268.0	256.1	288.1	343.9	409.6	-	-	-
58.3	68.2	85.4	120.1	127.5	102.5	113.3	130.0	155.8	192.9	-	-	-
48.2	48.5	46.8	49.3	55.2	54.9	65.7	68.2	78.8	86.5	-	-	-
10.1	19.7	38.6	70.8	72.4	47.6	47.6	61.8	77.0	106.4	32.6	28.4	61.0
2.5	1.7	2.1	3.6	3.8	3.6	0.3	-2.1	-3.3	-3.3	-	-	-
12.6	21.4	40.7	74.4	76.2	51.1	47.9	59.7	73.7	103.1	32.3	24.7	57.0
-3.3	-3.8	-8.6	-15.1	-4.5	-3.7	3.6	-0.9	-2.0	-2.0	-	-	-
9.3	17.5	32.2	59.3	71.7	47.4	51.5	58.8	71.7	101.1	-	-	-
5.8	6.0	6.2	17.7	19.3	15.7	17.4	16.1	19.4	27.3	-	-	-
62%	34%	19%	30%	27%	33%	34%	27%	27%	27%	-	-	-
0.3	0.2	0.2	0.3	0.2	0.3	0.3	0.4	0.2	0.2	-	-	-
3.3	11.3	25.7	41.2	52.2	31.5	33.7	42.3	52.1	73.6	22.1	14.9	37.0
22.9	57.1	78.2	60.6	131.3	146.6	157.3	67.0	160.0	160.0	-	-	67.0
25.1	24.2	35.4	52.7	54.9	46.0	54.2	65.5	79.5	95.0	-	-	65.5
14.8	16.2	16.8	15.7	19.7	20.0	27.5	29.6	34.5	35.0	-	-	-
35.3	43.9	74.0	123.5	127.3	93.6	101.8	127.3	156.5	201.4	-	-	126.5
1.2%	4.2%	8.8%	12.1%	13.3%	6.9%	6.8%	8.3%	9.5%	12.3%	-	-	-
279.4	279.5	279.4	279.2	279.3	279.4	279.4	289.5	289.5	289.7	-	-	-
11.8	40.5	92.0	147.7	186.9	112.7	120.7	146.2	180.04	254.1	-	-	132.5
989	980	1,152	1,328	1,524	1,796.1	1,840.0	1,988.7	2,206.8		-	-	-
17.5	17.5	17.5	20.0	25.0	20.0	20.0	30.0	35.0	40.0	15.0	10.0	25.0
109.0	108.0	105.6	111.0	133.0	142.0	152.0	147.5	150.0	150.0	145.0	151.0	148.0
127.0	120.0	120.3	129.0	139.0	154.0	162.0	167.5	170.0	170.0	165.0	177.0	171.0
-3%	2%	9%	24%	4%	-11%	0%	13%	20%	21%	-	-	14%
-39%	94%	96%	83%	2%	-34%	0%	30%	25%	38%	-	-	28%
-71%	243%	127%	60%	27%	-40%	7%	26%	23%	41%	-	-	10%
-	-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-	-	-
20.0%	23.0%	26.4%	29.9%	30.5%	27.7%	30.7%	31.1%	31.2%	32.0%	-	-	-
3.5%	6.7%	11.9%	17.7%	17.3%	12.8%	12.9%	14.8%	15.4%	17.7%	-	-	14.5%
1.1%	3.8%	7.9%	10.3%	12.5%	8.5%	9.1%	10.1%	10.4%	12.2%	-	-	8.8%
7.9%	19.3%	24.2%	15.1%	31.4%	39.6%	42.6%	16.0%	32.0%	26.6%	-	-	16.0%
8.6%	8.2%	10.9%	13.1%	13.2%	12.4%	14.7%	15.7%	15.9%	15.8%	-	-	15.6%

Exhibit 41: Sales and OP breakdown by Segment

(Y bn)	F3/25				F3/26			
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Qe
Sales by Segment								
Net Sales	88.2	93.4	88.8	99.1	97.5	98.0	103.1	119.5
Electronics	44.9	53.3	47.2	51.8	56.3	57.3	58.4	74.0
Ceramics	23.6	19.6	18.9	22.0	19.6	20.1	20.9	20.5
Others	19.7	20.5	22.7	25.3	21.6	20.7	23.9	25.0
YoY % Change								
Electronics	-7%	0%	-4%	10%	10%	5%	16%	21%
Ceramics	-11%	2%	3%	22%	25%	7%	24%	43%
Ceramics	-3%	-17%	-24%	-7%	-17%	3%	10%	-7%
Others	0%	19%	2%	4%	10%	1%	5%	-1%
Sequential % Change								
Electronics	-2%	6%	-5%	12%	-2%	1%	5%	16%
Electronics	6%	19%	-12%	10%	9%	2%	2%	27%
Ceramics	0%	-17%	-3%	16%	-11%	2%	4%	-2%
Others	-19%	4%	11%	12%	-15%	-4%	15%	5%

(Y bn)	F3/25				F3/26			
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Qe
Operating Profit by Segment								
Total Operating Profit	11.3	17.2	6.3	12.8	17.6	14.9	12.0	17.3
Electronics	5.3	12.7	1.9	6.9	14.0	11.4	7.6	12.8
Ceramics	4.3	2.8	2.3	2.9	2.1	1.7	2.1	2.0
Others	1.7	1.7	2.2	2.9	1.5	1.8	2.3	2.5
Operating Profit Margin								
Electronics	12.8%	18.4%	7.1%	12.9%	18.1%	15.2%	11.6%	14.5%
Electronics	11.8%	23.8%	4.1%	13.4%	24.9%	19.9%	13.0%	17.3%
Ceramics	18.2%	14.3%	11.9%	13.1%	10.7%	8.7%	9.9%	9.8%
Others	8.6%	8.5%	9.6%	11.7%	7.0%	8.7%	9.6%	10.0%

Note: e=Morgan Stanley Research estimates, Ce=Company estimates
Source: Company data, Morgan Stanley Research

F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26e	F3/27e	F3/28e	F3/26Ce	F3/26Ce
										1H	2He
291.1	296.0	323.5	401.1	417.5	370.5	369.4	418.1	499.7	602.5	195.5	224.5
116.0	132.2	166.1	237.0	250.7	190.7	197.2	245.9	322.1	419.1	113.5	136.5
102.5	88.4	87.4	90.7	89.9	96.5	84.1	81.1	83.8	86.3	39.7	40.3
72.7	75.4	70.0	73.5	76.9	83.3	88.1	91.1	93.9	97.1	42.2	47.8
-3%	2%	9%	24%	4%	-11%	0%	13%	20%	21%	8%	20%
0%	14%	26%	43%	6%	-24%	3%	25%	31%	30%	16%	38%
-10%	-14%	-1%	4%	-1%	7%	-13%	-4%	3%	3%	-8%	-2%
2%	4%	-7%	5%	5%	8%	6%	3%	3%	3%	5%	0%
-	-	-	-	-	-	-	-	-	-	4%	15%
-	-	-	-	-	-	-	-	-	-	15%	20%
-	-	-	-	-	-	-	-	-	-	-3%	2%
-	-	-	-	-	-	-	-	-	-	-12%	13%

F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26e	F3/27e	F3/28e	F3/26Ce	F3/26Ce
										1H	2He
10.1	19.7	38.6	70.8	72.4	47.6	47.6	61.8	77.0	106.4	32.6	28.4
2.5	14.9	27.8	55.1	60.6	27.3	26.8	45.8	60.5	88.2	25.4	21.1
3.0	-1.0	4.6	8.7	6.1	13.4	12.2	7.9	7.9	9.0	3.8	2.7
4.6	5.8	6.2	7.0	5.6	6.9	8.6	8.1	8.5	9.2	3.3	4.7
3.5%	6.7%	11.9%	17.7%	17.3%	12.8%	12.9%	14.8%	15.4%	17.7%	16.7%	12.7%
2.2%	11.3%	16.7%	23.3%	24.2%	14.3%	13.6%	18.6%	18.8%	21.1%	22.4%	15.4%
2.9%	-1.1%	5.3%	9.6%	6.8%	13.8%	14.5%	9.7%	9.5%	10.4%	9.7%	6.6%
6.4%	7.7%	8.8%	9.5%	7.3%	8.3%	9.7%	8.9%	9.1%	9.5%	7.8%	9.8%

Reiterate OW on Samsung Electro-Mechanics (009150.KS)

Shawn Kim

SEMCO's fundamentals

MLCC and ABF – dual AI beneficiaries

SEMCO's component division operates with a relatively high fixed-cost structure, with material costs accounting for only the low-to-mid 20% range as a proportion of sales. As a result, operating margin exhibits significant sensitivity to utilization rates. Over the past decade, divisional operating margins have shown a strong correlation with MLCC utilization, demonstrating clear cyclical co-movement.

However, compared to prior peak cycles in 2017-2018 and 2020-2021, recent margin upside has been more limited despite improving utilization. While some pressure stems from a higher raw material cost ratio, the more structural factor appears to be constrained blended ASP expansion. Although product mix has improved, price declines in IT-focused MLCCs have capped overall ASP gains.

With MLCC utilization now nearing 90-95%, incremental earnings upside from further utilization improvement alone appears limited. Thus, the key variable for earnings to rise is pricing for high-spec MLCCs used in AI servers. In this context, Murata's reference to potential price hikes may signal that the anticipated pricing cycle is beginning to materialize, which could be a catalyst for earnings upgrades.

Structurally, AI servers require significantly higher MLCC capacitance and voltage vs. conventional servers, resulting in materially higher content per unit. Demand is therefore shifting toward ultra-high-capacitance, high-temperature MLCCs, as well as high-layer products that consume disproportionate production capacity. In parallel, AI servers require large-area, complex, high-layer ABF (FC-BGA) substrates, further tightening supply dynamics in advanced packaging materials.

Beyond 2026, we expect AI infrastructure spending to remain the dominant growth engine, driven by continued architectural upgrades (e.g. next-gen GPUs and ASICs). This should further increase MLCC and advanced substrate content per server, reinforcing structurally higher demand intensity.

In the substrate division, utilization is currently ~70% and is expected to approach 80% by end of 2026 and full capacity by end of 2027, implying meaningful margin expansion. SEMCO is also evaluating additional capacity expansion, with new capacity reportedly largely pre-booked. Given supply commitments via customer agreements, we expect incremental ROIC on new facilities to be solid.

Separately, new growth initiatives including glass substrates, all-solid-state batteries, and differentiated camera modules (i.e., physical AI) should begin contributing more meaningfully over time, reinforcing long-term growth optionality. While consumer electronics demand remains muted, the company's revenue mix shift toward higher-content industrial, server and automotive applications is likely to support structurally higher utilization and improved margin resilience across cycles.

AI substrate and socket demand is accelerating with higher complexity

This is a result of the shift toward large-area, high-layer-count, fine-line, high-frequency, and high-thermal-resistance substrates. AI server demand accelerates demand for memory substrates and ABF, where we see sustained strength in multi-layer board supporting a likely ABF up-cycle. Across the PCB supply chain where IT-driven inflation is affecting substrates costs, differentiation will increasingly depend on product mix (high-end/AI is better) and the ability to pass through not only raw material costs but also higher processing costs.

On the materials side, tight supply and successive price increases led by Japanese suppliers are likely to enhance SEMCO's relative pricing power and competitiveness.

Geopolitical beneficiary

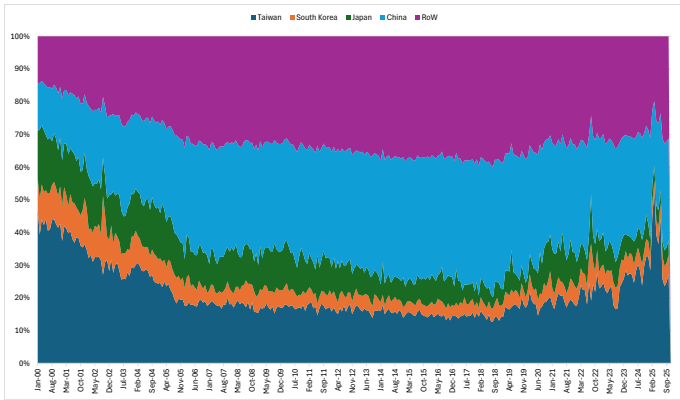
During periods of heightened geopolitical tension, buyers tend to shift orders toward suppliers with stronger supply stability. The tightening controls on rare-earth exports are increasingly affecting Japan's electronic components industry, accelerating a broader restructuring of the global supply chain. With an increase in stricter export controls of key minerals (such as gallium, germanium, and graphite), SEMCO's electronic component competitors may face production risks owing to their heavy dependence on China for critical materials such as ceramic powders and rare-earth additives for internal electrodes. While diversifying raw material sourcing is underway, it is likely time-consuming and favors SEMCO as an emerging as a key alternative supplier.

High frequency data updates

According to the latest export data, Japan's substrate export growth (November 2025: 7.6% YoY vs. October 2025: 7% YoY vs. November 20-24: 3.0% YoY) has been relatively weaker compared with Korea (December 2025: 36.3% YoY vs. November 2025: 18% YoY vs. December 2024: -6.7% YoY) and Taiwan (December 2025: 14% YoY vs. November 20-25: 17% YoY vs. December 2024: -6.0% YoY).

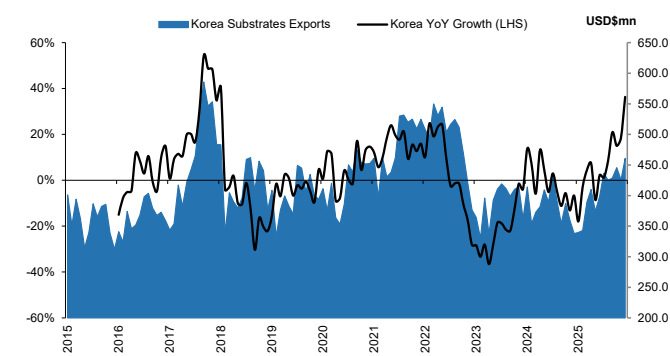
Korea's substrate exports are heavily concentrated in China, which accounts for up to 55% of value, reflecting strong linkage to Chinese assembly and packaging ecosystems. In contrast, the US sources a significant portion of its substrate imports from Taiwan (around 30%), highlighting Taiwan's central role in advanced packaging and AI-related supply chains, while Japan's substrate exposure appears more limited and less leveraged to current growth markets.

Exhibit 42: US – Substrate Import Breakdown



Source: US Customs, Morgan Stanley Research

Exhibit 43: Korea – Substrates Exports



Source: Korea Trade, Morgan Stanley Research

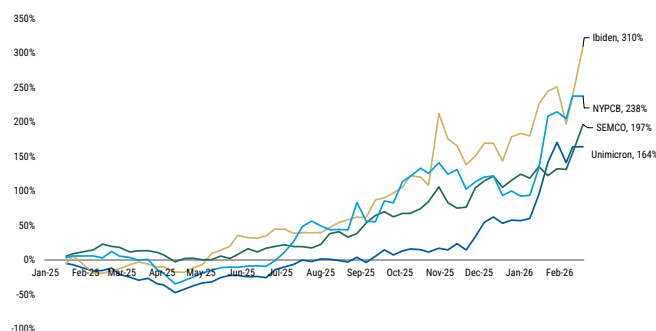
Reconciling Our Positive View with Shoji Sato's Underweight on Ibiden

We see room for Unimicron, NYPCB, and SEMCO to outperform Ibiden from here:

Ibiden's stock has outperformed both Unimicron and SEMCO since January 2025, up ~310% even after the recent correction, while Unimicron's stock is up ~164%, NYPCB is up ~238%, and SEMCO is up ~197% ([Exhibit 44](#)). From a valuation perspective, Ibiden is also the most expensive at 45.3x NTM P/E, vs. Unimicron at 35.4x, NYPCB at 36.9x, and SEMCO at 24.5x ([Exhibit 45](#)).

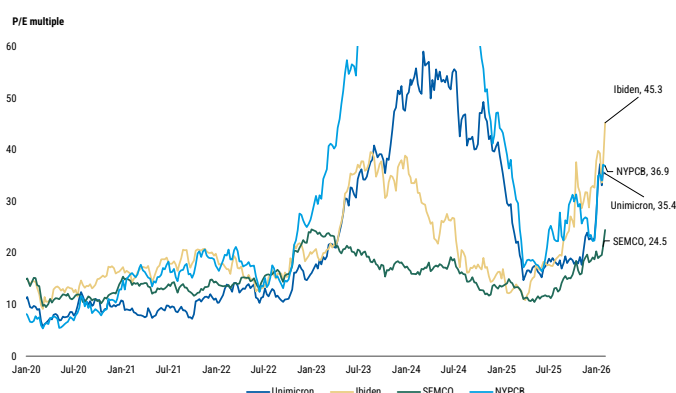
Exhibit 44: Since January 2025, Ibiden has outperformed Unimicron, NYPCB, and SEMCO, but we expect the latter trio to outperform from here

Relative share price performance since Jan 2025



Source: FactSet, Morgan Stanley Research.

Exhibit 45: NTM P/E multiple for the four ABF substrate stocks under Morgan Stanley coverage (Unimicron, NYPCB, Ibiden, and SEMCO)



Source: FactSet, Morgan Stanley Research. Note: earnings based on next 12-months consensus earnings from FactSet.

Exhibit 46:

Global ABF substrate suppliers comparison table

Ticker	Company	Closing price 2/11/2026	Rating	Price target	Market Cap (US\$ mn)	EPS (LC)		P/E (X)		P/B (X)		P/S (X)		EV/EBITDA		ROA (%)		ROE (%)	
						26E	27E	26E	27E	26E	27E	26E	27E	26E	27E	26E	27E	26E	27E
Taiwan Peers																			
3037.TW	Unimicron	369.0	OW	500.0	18,372	10.33	16.95	35.7	21.8	5.2	4.4	3.6	3.0	14.8	11.3	6%	9%	15%	20%
8046.TW	NYPCB	416.0	OW	515.0	8,557	10.09	17.56	41.2	23.7	5.4	4.9	5.0	4.0	18.1	12.9	9%	13%	13%	21%
3189.TW	Kinsus	265.5	NC	--	4,304	7.55	12.68	35.2	20.9	3.3	2.9	2.8	2.3	11.0	9.2	5%	7%	9%	14%
Average								37.4	22.1	4.6	4.1	3.8	3.1	14.7	11.1	7%	9%	12%	18%
Median								35.7	21.8	5.2	4.4	3.6	3.0	14.8	11.3	6%	9%	13%	20%
Global Peers																			
4062.T	Ibiden	9,638	UW	6,500	17,540	201.96	271.70	47.7	35.5	4.7	4.3	5.7	4.9	16.3	13.0	5%	6%	10%	12%
009150.KS	SEMCO	358,000	OW	430,000	18,795	13,964	17,683	25.6	20.2	2.7	2.4	2.1	1.9	10.4	9.0	7%	8%	10%	12%
ATS.VI	AT&S	50.9	NC	--	2,333	0.33	1.98	154.7	25.7	2.4	2.2	0.9	0.8	6.9	5.7	0%	2%	2%	9%
Average								76.0	27.1	3.3	3.0	2.9	2.5	11.2	9.2	4%	5%	7%	11%
Median								47.7	25.7	2.7	2.4	2.1	1.9	10.4	9.0	5%	6%	10%	12%

Source: FactSet, Morgan Stanley Research. Note: earnings based on next 12 months consensus earnings from FactSet. Note: Ibiden and SEMCO are priced as of the close on February 19, 2026.

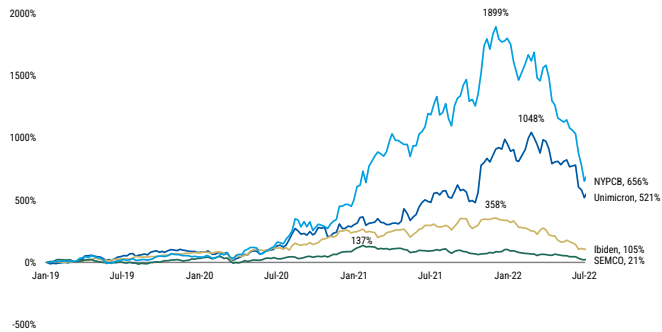
In the last up-cycle, Taiwanese companies also showed more price expansion, which led to better profitability and better share price performance: We think this could happen again because Taiwanese companies are typically more flexible on pricing adjustments vs. their Japanese peers. As a result, in the last up-cycle, Unimicron and NYPCB's stocks strongly outperformed – [Exhibit 47](#):

- NYPCB's stock peaked at ~1900% on December 3, 2021 (vs. January 1, 2019).
- Unimicron's stock peaked at ~1048% on February 25, 2022 (vs. January 1, 2019).
- Ibiden's stock appreciated ~358% at the peak on December 3, 2021 (vs. January 1, 2019).

We believe this is mainly due to flexibility on pricing adjustments and differences in profitability as a result of the pricing adjustments ([Exhibit 48](#)).

Exhibit 47: Taiwanese ABF suppliers' shares outperformed vs. regional ABF suppliers in the last up-cycle

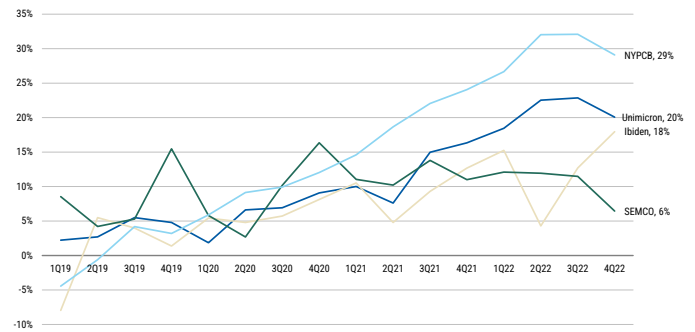
Relative share price performance since Jan 2019 to Jul 2022



Source: FactSet, Morgan Stanley Research.

Exhibit 48: Taiwanese ABF substrate makers showed greater profitability during the last 2020-2022 up-cycle

Net Profit Margin Trend (1Q19-4Q22)



Source: FactSet, Morgan Stanley Research.

Our view on medium-/long-term growth prospects is aligned with Shoji Sato's: We don't see a difference in the growth prospects between our view and Shoji Sato's. But Shoji Sato's Underweight rating on Ibiden mainly reflects overly bullish expectations after strong stock outperformance since early 2025, while we are now turning bullish on Unimicron and NYPCB because we see their fundamentals finally starting to inflect meaningfully, driven by our bottom-up analysis that the ABF substrate market will enter undersupply from 2027 onwards.

Unimicron – What's Changed

Estimate revisions

Our 2025 EPS estimate declines 3%, but our 2026 and 2027 EPS estimates increase 6% and 60%, as we factor in more bullish assumptions for both ABF and BT business, to reflect our expectations of a reversion to up-cycle and the subsequent price hikes. The discrepancies in net income and EPS estimate changes are owing to share counts changes.

We also introduce our 2028 estimates.

Exhibit 49: Estimate revisions

Year to Dec. 31 (NT\$m)	2025E			2026E			2027E			2028E	
	New	Old	% diff	New	Old	% diff	New	Old	% diff	New	
P&L Summary											
Net sales	131,241	130,303	1%	159,901	141,431	13%	207,853	149,448	39%	260,217	
COGS	(113,537)	(112,195)		(133,952)	(118,335)		(165,274)	(121,197)		(193,991)	
Gross profit	17,704	18,107	-2%	25,949	23,096	12%	42,579	28,251	51%	66,226	
Operating expenses	(11,549)	(11,065)		(12,787)	(11,231)		(13,524)	(11,098)		(13,999)	
Operating income	6,155	7,042	-13%	13,162	11,865	11%	29,056	17,152	69%	52,227	
Non-operating income	781	(82)		1,852	2,127		1,852	2,127		1,852	
Pre-tax income	6,936	6,960	0%	15,014	13,992	7%	30,907	19,279	60%	54,078	
Net income	4,942	5,037	-2%	11,652	10,833	8%	24,198	15,004	61%	42,446	
Diluted EPS (NT\$)	3.22	3.31	-3%	7.58	7.12	6%	15.74	9.86	60%	27.62	
Margins											
			ppt			ppt				ppt	
Gross margin	13.5%	13.9%	-0.4	16.2%	16.3%	-0.1	20.5%	18.9%	1.6	25.5%	
Operating margin	4.7%	5.4%	-0.7	8.2%	8.4%	-0.2	14.0%	11.5%	2.5	20.1%	
Pretax margin	5.3%	5.3%	-0.1	9.4%	9.9%	-0.5	14.9%	12.9%	2.0	20.8%	
Net margin	3.8%	3.9%	-0.1	7.3%	7.7%	-0.4	11.6%	10.0%	1.6	16.3%	

Source: Morgan Stanley Research (E) estimates.

Price target and valuation methodology

Our 12-month price target is NT\$500 (up from NT\$120.75)

We change our valuation methodology from P/B back to a multi-stage residual income

(RI) valuation model: We believe the company's earnings will return to a growth track from 2026 onwards, driven by the high-end ABF substrate market's reversion to undersupply and Unimicron's dominant supply share for AI ASICs, which puts it in a favorable position to benefit from rising AI investments. We believe the overall ABF substrate market bottomed in 2H24, and growth will accelerate from 2H26 onwards with the ramp of multiple next-generation, large body size ABF substrates for new server GPUs, CPUs, ASICs and networking chips. New technologies, such as embedded passives and Intel's EMIB-T, will also increase the manufacturing complexity of the package substrates, which adds to the undercapacity as usable output decreases.

Similar to our analysis of other tech hardware companies within our coverage, we use residual income (RI) valuation model to value Unimicron – we think it derives the most accurate value of the firm given that it takes into account cost of equity. Key assumptions:

- Cost of equity of 9.2%
 - Risk-free rate of 1% (10-year Taiwan government note yield)
 - Equity risk premium of 8.7%,
 - Beta of 1.0

- Medium-term growth rate of 10% – fair given our analysis of the substrate market over the next couple of years
- Terminal growth rate of 3% (similar to the rest of the companies under our tech hardware coverage)

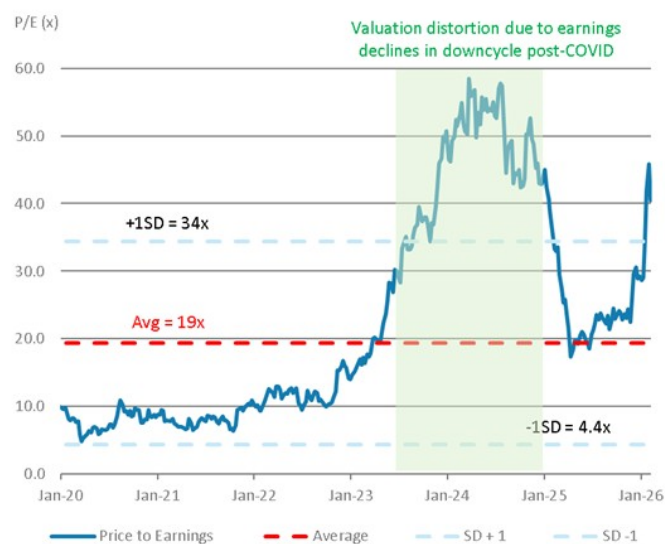
Our bull and bear case values increase by similar magnitudes, to NT\$850 and NT\$166, respectively.

Exhibit 50: Unimicron: residual income (RI) model

	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E
Total Equity	130,689	162,078	179,820	199,337	220,805	244,420	270,397	298,971	330,403	364,978	396,641
Net Profit	24,198	42,446	46,691	51,360	56,496	62,145	68,360	75,196	82,715	90,987	93,716
Return on Equity	20.0%	29.0%	27.3%	27.1%	26.9%	26.7%	26.6%	26.4%	26.3%	26.2%	24.6%
Beta	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Equity Risk Premium (Rm-Rf)	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%
Risk Free Rate (Rf)	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%
Cost of Equity	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%
Residual Income	12,053	25,901	29,393	32,213	35,315	38,726	42,479	46,606	51,146	56,140	56,324
Spread	10.8%	19.8%	18.1%	17.9%	17.7%	17.5%	17.4%	17.2%	17.1%	17.0%	15.4%
Beginning Equity Capital	130,689										
PV of Forecast Period	247,463										
PV of Continuing Value	390,291										
Equity Value	768,443										
No. of Shares	1,537										
Projected Price	500.0										

Source: Morgan Stanley Research estimates.

Exhibit 51: Unimicron: 2020-present P/E chart



Source: TEJ, Morgan Stanley Research estimates.

Exhibit 52: Unimicron: past 10-year P/B chart



Source: TEJ, Morgan Stanley Research estimates.

Financials

Exhibit 53: Quarterly Estimates

NT\$ mn	1Q25	2Q25	3Q25	4Q25E	1Q26E	2Q26E	3Q26E	4Q26E	2024	2025E	2026E	2027E
Sales	30,090	32,466	33,994	34,691	35,068	38,306	41,649	44,878	115,373	131,241	159,901	207,853
Cost of goods sold	-26,063	-28,221	-29,446	-29,807	-29,902	-32,363	-34,645	-37,043	-99,058	-113,537	-133,952	-165,274
Gross profit	4,026	4,246	4,549	4,884	5,167	5,943	7,004	7,835	16,315	17,704	25,949	42,579
Operating expenses	-2,759	-2,736	-3,016	-3,037	-3,042	-3,171	-3,256	-3,318	-11,198	-11,549	-12,787	-13,524
Operating profit	1,267	1,509	1,533	1,847	2,125	2,772	3,748	4,517	5,117	6,155	13,162	29,056
Interest income (loss)	103	62	-7	-7	-7	-7	-7	-7	494	151	-28	-28
Investment income (loss)	32	-13	-1	20	20	20	20	20	28	38	80	80
Gains (Losses) from investment disposal	0	0	0	0	0	0	0	0	0	0	0	0
Gains from FAs disposal	3	337	-29	0	0	0	0	0	-446	310	0	0
Exchange gain	452	-1,281	487	0	0	0	0	0	1,015	-342	0	0
Others	-483	-257	964	400	450	450	450	450	1,113	624	1,800	1,800
Pre-tax profit	1,373	357	2,947	2,260	2,588	3,235	4,211	4,980	7,321	6,936	15,014	30,907
Net Income for Minority	9	231	256	50	50	50	50	50	474	546	200	200
Income tax	-450	-97	-496	-407	-530	-809	-926	-896	-1,765	-1,449	-3,162	-6,509
Net profit	915	30	2,194	1,803	2,007	2,376	3,235	4,034	5,082	4,942	11,652	24,198
EPS (NT\$)	0.59	0.02	1.43	1.17	1.30	1.54	2.10	2.62	3.31	3.22	7.58	15.74
Margins (%)												
Gross margin	13.4%	13.1%	13.4%	14.1%	14.7%	15.5%	16.8%	17.5%	14.1%	13.5%	16.2%	20.5%
Operating margin	4.2%	4.6%	4.5%	5.3%	6.1%	7.2%	9.0%	10.1%	4.4%	4.7%	8.2%	14.0%
Pre-tax margin	4.6%	1.1%	8.7%	6.5%	7.4%	8.4%	10.1%	11.1%	6.3%	5.3%	9.4%	14.9%
Net margin	3.0%	0.1%	6.5%	5.2%	5.7%	6.2%	7.8%	9.0%	4.4%	3.8%	7.3%	11.6%
QoQ Growth												
Sales	2%	8%	5%	2%	1%	9%	9%	8%				
Gross profit	18%	5%	7%	7%	6%	15%	18%	12%				
Operating profit	86%	19%	2%	20%	15%	30%	35%	21%				
Pre-tax profit	331%	-74%	726%	-23%	15%	25%	30%	18%				
Net profit	1531%	-97%	7310%	-18%	11%	18%	36%	25%				
YoY Growth												
Sales	14%	16%	7%	18%	17%	18%	23%	29%	11%	14%	22%	30%
Gross profit	-6%	15%	-8%	44%	28%	40%	54%	60%	-20%	9%	47%	64%
Operating profit	-20%	68%	-22%	172%	68%	84%	145%	145%	-43%	20%	114%	121%
Pre-tax profit	-59%	-83%	95%	608%	88%	807%	43%	120%	-53%	-5%	116%	106%
Net profit	-62%	-98%	120%	3116%	119%	7926%	47%	124%	-58%	-3%	136%	108%

Source: Company data, Morgan Stanley Research (E) estimates.

Exhibit 54: Consolidated Financial Summary**Income Statement**

NT\$ mn; FY End Dec	2024	2025E	2026E	2027E	2028E
Net sales	115,373	131,241	159,901	207,853	260,217
Cost of goods sold	-99,058	-113,537	-133,952	-165,274	-193,991
Gross profit	16,315	17,704	25,949	42,579	66,226
Operating expenses	-11,198	-11,549	-12,787	-13,524	-13,999
Sales & Marketing	-1,637	-1,635	-1,858	-2,258	-2,607
G&A	-4,518	-4,903	-5,253	-5,198	-5,142
R&D	-5,043	-5,010	-5,676	-6,068	-6,250
Operating profit	5,117	6,155	13,162	29,056	52,227
Interest income	494	151	-28	-28	-28
Investment income	28	38	80	80	80
Gains from investment disposal	0	0	0	0	0
Gains from fixed assets disposal	-446	310	0	0	0
Exchange gain	1,015	-342	0	0	0
Other non-operating income	1,113	624	1,800	1,800	1,800
Net non-operating profit	2,204	781	1,852	1,852	1,852
Pre-tax profit	7,321	6,936	15,014	30,907	54,078
Net Income for Minority	-474	-546	-200	-200	-200
Income tax	-1,765	-1,449	-3,162	-6,509	-11,432
Net profit	5,082	4,942	11,652	24,198	42,446
Reported EPS (NT\$)	3.31	3.22	7.58	15.74	27.62

Balance Sheet

NT\$ mn; FY End Dec	2024	2025E	2026E	2027E	2028E
Cash	43,753	50,797	64,028	103,790	154,926
Mkt Securities	99	99	99	99	99
AR/NR	23,617	26,865	32,732	42,548	53,266
Inventory	14,863	17,035	20,098	24,798	29,106
Others	4,537	5,161	6,288	8,173	10,232
Current Assets	86,869	99,957	123,245	179,408	247,631
Long-term Investments	911	911	911	911	911
Fixed Assets	120,536	122,367	127,888	122,358	117,506
Other Assets	22,531	22,531	22,531	22,531	22,531
Total Assets	230,847	245,766	274,576	325,208	388,579
S-t Borrowings	7,188	7,188	7,188	7,188	7,188
AP/NP	15,117	17,326	20,442	25,222	29,604
Other S-T Liabilities	36,077	41,350	48,785	60,192	70,651
Other LT Liabilities	39,973	45,471	55,401	72,015	90,157
L-t Debt	32,905	31,904	30,903	29,902	28,901
Total Liabilities	131,260	143,239	162,719	194,519	226,501
Common Shares	15,300	15,300	15,300	15,300	15,300
Other Shareholders' Equity	84,287	87,227	96,557	115,389	146,778
Total Equity	99,587	102,527	111,857	130,689	162,078
Total Liab./Shrholder's Equity	230,847	245,766	274,576	325,208	388,579

Cash Flow Statement

NT\$ mn; FY End Dec	2024	2025E	2026E	2027E	2028E
Cashflow from Operations	10,294	24,663	31,400	43,727	59,086
Net Profits	5,082	4,942	11,652	24,198	42,446
Depreciation	17,231	18,321	19,334	19,823	18,965
Investment losses (income)	0	-38	-80	-80	-80
Investment Disposal Loss	0	0	0	0	0
Change in Working Capital	-5,856	2,062	1,621	1,672	-186
Other adjustments	-6,164	-624	-1,127	-1,886	-2,059
Cashflow from Investing	-29,230	-20,153	-24,855	-14,292	-14,114
(Purchases) Sale of fixed asset	-25,735	-20,153	-24,855	-14,292	-14,114
(Purchases) Sale of LT investment	0	0	0	0	0
Other adjustments	-3,495	0	0	0	0
Cashflow from financing	8,336	2,191	6,687	10,327	6,164
Increase in L-T debt	8,388	-1,001	-1,001	-1,001	-1,001
Increase in S-T debt	4,527	0	0	0	0
Issuance of stock	0	0	0	0	0
Other adjustments	-3	5,498	9,930	16,614	18,142
Exchange Rate Adjustment	1,494	342	0	0	0
Net change in cash	-9,106	7,043	13,231	39,762	51,136

Financial Ratios

	2024	2025E	2026E	2027E	2028E
Margins					
Gross margin	14.1%	13.5%	16.2%	20.5%	25.5%
Operating expenses/sales	-9.7%	-8.8%	-8.0%	-6.5%	-5.4%
Operating margin	4.4%	4.7%	8.2%	14.0%	20.1%
Tax rate	24.1%	20.9%	21.1%	21.1%	21.1%
Pre-tax margin	6.3%	5.3%	9.4%	14.9%	20.8%
Net Margin	4.4%	3.8%	7.3%	11.6%	16.3%
YoY Growth					
Turnover	10.9%	13.8%	21.8%	30.0%	25.2%
Gross Profit	-19.6%	8.5%	46.6%	64.1%	55.5%
Operating Profits	-42.6%	20.3%	113.8%	120.8%	79.7%
Pretax Profits	-53.0%	-5.2%	116.4%	105.9%	75.0%
Net Profits	-57.6%	-2.8%	135.8%	107.7%	75.4%
Adjusted Cash Dividend (NT\$)	1.50	1.46	3.44	7.14	12.53
Net Debt/Equity	-3.8%	-11.5%	-23.3%	-51.1%	-73.4%
Liabilities/Equity	131.8%	139.7%	145.5%	148.8%	139.7%
Liabilities/Assets	56.9%	58.3%	59.3%	59.8%	58.3%
ROAE	5.2%	4.9%	10.9%	20.0%	29.0%
ROAA	2.3%	2.1%	4.5%	8.1%	11.9%
A/R Turnover (days)	66.7	70.2	68.0	66.1	67.2
A/P Turnover (days)	48.6	52.1	51.5	50.4	51.6
Inventory Turnover (days)	47.5	51.3	50.6	49.6	50.7
Cash conversion (days)	65.6	69.3	67.2	65.2	66.3

Source: Company data, Morgan Stanley Research (E) estimates.

NYPCCB – What's Changed

Estimate revisions

Our 2025 EPS estimate declines 10%, but our 2026 and 2027 EPS estimates increase 11% and 71%, as we factor in more bullish assumptions for both ABF and BT business, to reflect our expectations of a reversion to up-cycle and the subsequent price hikes. We also introduce our 2028 estimates.

Exhibit 55: Estimate revisions

NT\$ mn	2025E			2026E			2027E			2028E	
	New	Old	Variance	New	Old	Variance	New	Old	Variance	New	
Net sales	40,173	40,251	0%	50,961	48,401	5%	57,658	49,338	17%	70,659	
COGS	-36,673	-36,596	0%	-44,101	-42,320	4%	-46,537	-42,596	9%	-50,928	
Gross profit	3,500	3,655	-4%	6,860	6,081	13%	11,121	6,742	65%	19,731	
Operating expenses	-1,663	-1,618	3%	-1,820	-1,699	7%	-1,918	-1,765	9%	-2,066	
Operating income	1,837	2,037	-10%	5,040	4,383	15%	9,203	4,977	85%	17,664	
Non-operating income	91	32	182%	687	793	-13%	687	793	-13%	687	
Pre-tax income	1,928	2,069	-7%	5,727	5,176	11%	9,890	5,770	71%	18,351	
Income tax	-321	-293	9%	-915	-828	11%	-1,575	-921	71%	-2,927	
Net income	1,607	1,776	-10%	4,813	4,348	11%	8,315	4,849	71%	15,424	
EPS (NT\$)	2.48	2.74	-10%	7.42	6.70	11%	12.82	7.48	71%	23.78	
Margins (%)			ppt			ppt			ppt		
Gross margin	8.7%	9.1%	-0.4	13.5%	12.6%	0.9	19.3%	13.7%	5.6	27.9%	
Operating margin	4.6%	5.1%	-0.5	9.9%	9.1%	0.8	16.0%	10.1%	5.9	25.0%	
Pre-tax margin	4.8%	5.1%	-0.3	11.2%	10.7%	0.5	17.2%	11.7%	5.5	26.0%	
Net margin	4.0%	4.4%	-0.4	9.4%	9.0%	0.5	14.4%	9.8%	4.6	21.8%	

Source: Morgan Stanley Research (E) estimates.

Price target and valuation methodology

We change our valuation methodology from P/B back to a multi-stage residual income

(RI) valuation model: We believe the company's earnings will return to a growth track from 2026 onwards, driven by the high-end ABF substrate market's reversion to undersupply, NYPCCB's dominant supply share for networking chips, and its proactive pricing strategy, which puts it in a favorable position to benefit from rising AI investments. We believe the overall ABF substrate market bottomed in 2H24, and growth will accelerate from 2H26 onwards with the ramp of multiple next-generation, large body size ABF substrates for new server GPUs, CPUs, ASICs and networking chips. New technologies, such as embedded passives and Intel's EMIB-T, will also increase the manufacturing complexity of the package substrates, which adds to the undercapacity as usable output decreases.

Similar to our analysis of other tech hardware companies within our coverage, we use a residual income (RI) valuation model to value NYPCCB – we think it derives the most accurate value of the firm given that it takes into account cost of equity. Key assumptions:

- Cost of equity of 9.3%
 - Risk-free rate of 1% (10-year Taiwan government note yield)
 - Equity risk premium of 8.7%
 - Beta of 1.0
- Medium-term growth rate of 12% (fair given our analysis of the substrate market over the next couple of years).
- Terminal growth rate of 3% (similar to the rest of the companies under our tech hardware coverage)

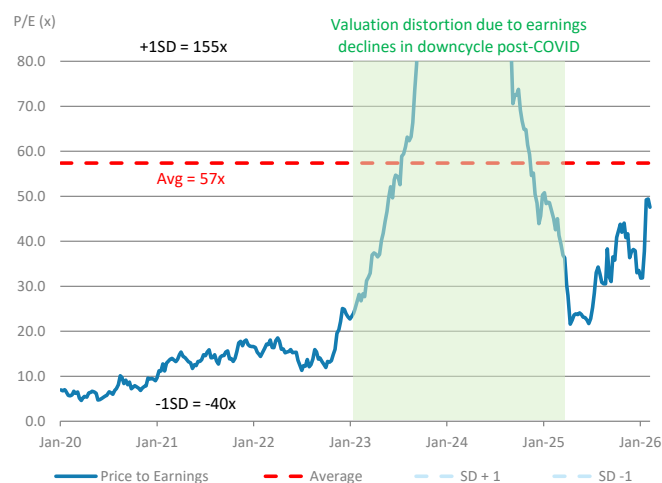
Our bull and bear case values increase by similar magnitudes, to NT\$920 and NT\$262, respectively.

Exhibit 56: NYPCB: residual income (RI) model

	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E
Total Equity	55,979	66,404	71,383	76,959	83,205	90,200	98,035	106,809	116,637	127,644	136,535
Net Profit	8,315	15,424	17,275	19,348	21,670	24,271	27,183	30,445	34,099	38,190	39,336
Return on Equity	15.6%	25.2%	25.1%	26.1%	27.1%	28.0%	28.9%	29.7%	30.5%	31.3%	29.8%
Beta	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Equity Risk Premium (Rm-Rf)	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%
Risk Free Rate (Rf)	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%
Cost of Equity	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%
Residual Income	3,208	8,925	10,499	12,008	13,696	15,583	17,696	20,059	22,704	25,664	26,187
Spread	6.3%	15.9%	15.8%	16.8%	17.8%	18.7%	19.6%	20.5%	21.3%	22.0%	20.5%
Beginning Equity Capital	55,979										
PV of Forecast Period	100,507										
PV of Continuing Value	177,544										
Equity Value	334,029										
No. of Shares	649										
Projected Price	515.0										

Source: Morgan Stanley Research estimates.

Exhibit 57: NYPCB: 2020-present P/E chart



Source: TEJ, Morgan Stanley Research estimates.

Exhibit 58: NYPCB: past 10-year P/B chart



Source: TEJ, Morgan Stanley Research estimates.

Financials

Exhibit 59: Quarterly Estimates

	1Q25	2Q25	3Q25	4Q25E	1Q26E	2Q26E	3Q26E	4Q26E	2024	2025E	2026E	2027E
Net sales	8,458	9,583	10,967	11,165	11,553	12,609	13,270	13,529	32,283	40,173	50,961	57,658
COGS	(8,028)	(8,814)	(9,918)	(9,913)	(10,195)	(10,988)	(11,405)	(11,512)	(31,926)	(36,673)	(44,101)	(46,537)
Gross profit	430	769	1,050	1,251	1,358	1,621	1,864	2,017	358	3,500	6,860	11,121
Operating expenses	(397)	(403)	(429)	(434)	(440)	(453)	(463)	(464)	(1,624)	(1,663)	(1,820)	(1,918)
SG&A	(397)	(403)	(429)	(434)	(440)	(453)	(463)	(464)	(1,624)	(1,663)	(1,820)	(1,918)
R&D	0	0	0	0	0	0	0	0	0	0	0	0
Operating income	33	366	620	818	918	1,168	1,402	1,553	(1,267)	1,837	5,040	9,203
Non-operating income	223	(593)	287	173	174	161	178	173	1,430	91	687	687
Interest income	45	47	19	19	19	19	19	19	238	130	76	76
Investment income	5	(8)	9	4	5	(8)	9	4	27	11	11	11
Disposal of investment	0	0	0	0	0	0	0	0	0	0	0	0
Exchange gain	124	(699)	167	0	0	0	0	0	557	(407)	0	0
Other	49	67	91	150	150	150	150	150	608	357	600	600
Pre-tax income	256	(227)	907	991	1,092	1,329	1,580	1,727	163	1,928	5,727	9,890
Income tax	(49)	39	(182)	(129)	(164)	(239)	(253)	(259)	41	(321)	(915)	(1,575)
Minority interest	0	0	0	0	0	0	0	0	0	0	0	0
Net income	207	(187)	725	862	928	1,090	1,327	1,468	204	1,607	4,813	8,315
Adj.wtd.avg.shrs (mn)	646	646	646	646	646	646	646	646	649	649	649	649
EPS	0.32	(0.29)	1.12	1.33	1.44	1.69	2.05	2.27	0.31	2.48	7.42	12.82
Fully Diluted shares (mn)	646	646	646	646	646	646	646	646	649	649	649	649
Diluted EPS (NT\$)	0.32	(0.29)	1.12	1.33	1.44	1.69	2.05	2.27	0.31	2.48	7.42	12.82
Margins (%)												
Gross Margin	5.1	8.0	9.6	11.2	11.8	12.9	14.0	14.9	1.1	8.7	13.5	19.3
Operating Margin	0.4	3.8	5.7	7.3	7.9	9.3	10.6	11.5	(3.9)	4.6	9.9	16.0
Pretax Margin	3.0	(2.4)	8.3	8.9	9.4	10.5	11.9	12.8	0.5	4.8	11.2	17.2
Net Margin	2.5	(2.0)	6.6	7.7	8.0	8.6	10.0	10.8	0.6	4.0	9.4	14.4
QoQ Growth (%)												
Sales	7	13	14	2	3	9	5	2				
Gross Profit	2,441	79	37	19	9	19	15	8				
Operating Profit	NM	10	69	32	12	27	20	11				
Pretax Profit	63	-188	NM	9	10	22	19	9				
Net Profit	16	-190	NM	19	8	17	22	11				
YoY Growth (%)												
Sales	19	18	19	42	37	32	21	21	(24)	24	27	13
Gross Profit	NM	319	95	7,289	216	111	78	61	(96)	879	96	62
Operating Profit	NM	NM	396	NM	2,673	219	126	90	(120)	NM	174	83
Pretax Profit	NM	-226	1,247	529	326	NM	74	74	(98)	1,082	197	73
Net Profit	NM	-258	1,131	383	347	NM	83	70	(96)	689	199	73

Source: Company data, Morgan Stanley Research (E) estimates.

Exhibit 60: Consolidated Financial Summary

Consolidated Income Statement

NT\$m (Year End Dec 31)	2024	2025E	2026E	2027E	2028E
Net sales	32,283	40,173	50,961	57,658	70,659
COGS	(31,926)	(36,673)	(44,101)	(46,537)	(50,928)
Gross profit	358	3,500	6,860	11,121	19,731
Operating expenses	(1,624)	(1,663)	(1,820)	(1,918)	(2,066)
Operating income	(1,267)	1,837	5,040	9,203	17,664
Non-operating income	1,430	91	687	687	687
Interest income	238	130	76	76	76
Investment income	27	11	11	11	11
Disposal of investment	-	-	-	-	-
Exchange gain	557	(407)	-	-	-
Other	608	357	600	600	600
Pre-tax income	163	1,928	5,727	9,890	18,351
Income tax	41	(321)	(915)	(1,575)	(2,927)
Minority interests	-	-	-	-	-
Net income	204	1,607	4,813	8,315	15,424
Adj. wtd. Avg. shrs (m)	649	649	649	649	649
Reported EPS (NT\$)	0.31	2.48	7.42	12.82	23.78
Diluted shrs (m)	649	649	649	649	649
Diluted EPS (NT\$)	0.31	2.48	7.42	12.82	23.78

Consolidated Balance Sheet

NT\$m (Year End Dec 31)	2024	2025E	2026E	2027E	2028E
Cash	8,481	14,331	23,048	32,033	46,298
Mkt securities	106	106	106	106	106
AR/NR	6,996	8,706	11,044	12,495	15,313
Inventory	4,101	4,711	5,665	5,978	6,542
Others	991	1,233	1,564	1,770	2,169
Current Assets	20,675	29,087	41,427	52,382	70,428
Long-term investments	460	460	460	460	460
Fixed assets	39,991	36,489	33,154	30,369	28,024
Other assets	1,813	1,813	1,813	1,813	1,813
Total Assets	62,938	67,849	76,854	85,024	100,725
S/T borrowings	-	-	-	-	-
AP/NP	1,518	1,743	2,096	2,212	2,421
Other ST liabilities	5,147	5,912	7,109	7,502	8,210
Total Current Liabilities	6,664	7,655	9,206	9,714	10,631
L/T debt	-	-	-	-	-
Other LT liabilities	10,824	13,469	17,086	19,331	23,690
Total Liabilities	17,488	21,124	26,292	29,045	34,321
Common shares	6,462	6,381	6,381	6,381	6,381
Retained Earnings	12,389	13,744	17,582	22,998	33,424
Other SH' Equity	26,600	26,600	26,600	26,600	26,600
Total Shareholders' Equity	45,450	46,724	50,562	55,979	66,404
Total Liab./SH's Equity	62,938	67,849	76,854	85,024	100,725

Consolidated Cash Flow Statement

NT\$m (Year End Dec 31)	2024	2025E	2026E	2027E	2028E
Operating Cashflow	2,160	6,824	8,933	12,479	17,713
Net Profits	204	1,607	4,813	8,315	15,424
Depreciation & Amort.	6,460	6,798	6,203	5,636	5,163
Investment losses/(income)	(27)	(11)	(11)	(11)	(11)
Working capital change	1,241	(1,329)	(1,741)	(1,256)	(2,465)
Other adjustments	(5,718)	(242)	(331)	(206)	(399)
Investing Cashflow	(2,602)	(3,296)	(2,869)	(2,851)	(2,818)
Capex	(2,379)	(3,400)	(3,000)	(3,000)	(3,000)
Change of L/T investment	0	0	0	0	0
Change of S/T investment	0	0	0	0	0
Other adjustments	(306)	0	0	0	0
Financing Cashflow	(3,843)	1,916	2,653	(642)	(630)
Increase in L/T debt	0	0	0	0	0
Increase in S/T debt	0	0	0	0	0
Issuance of stock	0	(81)	0	0	0
Cash dividends	(3,554)	(649)	(964)	(2,888)	(4,989)
Other adjustments	(289)	2,645	3,617	2,245	4,359
FX adjustment	436	407	0	0	0
Net change in cash	(3,849)	5,850	8,717	8,985	14,265

Consolidated Financial Ratios

	2024	2025E	2026E	2027E	2028E
Margins (%)					
Gross margin	1.1	8.7	13.5	19.3	27.9
Operating margin	(3.9)	4.6	9.9	16.0	25.0
Pretax margin	0.5	4.8	11.2	17.2	26.0
Net margin	0.6	4.0	9.4	14.4	21.8
YoY growth (%)					
Sales	(23.6)	24.4	26.9	13.1	22.5
Operating profits	(120.0)	NM	174.4	82.6	91.9
Pretax profits	(97.7)	1,082.2	197.1	72.7	85.6
Net profits	(96.5)	688.9	199.5	72.8	85.5
Others					
Cash dividend payout (%)	318%	60%	60%	60%	60%
Cash div (NT\$)	1.00	1.49	4.45	7.69	14.27
Yield (%)	0%	0%	1%	2%	4%
Net Debt/Equity (%)	-19%	-31%	-46%	-57%	-70%
Liabilities/Equity (%)	38%	45%	52%	52%	52%
Liabilities/Assets (%)	28%	31%	34%	34%	34%
ROAE (%)	0%	3%	10%	16%	25%
ROAA (%)	0%	2%	7%	10%	17%
AR/NR Turnover (days)	79	71	71	75	72
Inventory Turnover (days)	46	44	43	46	45
AP/NP Turnover (days)	26	16	16	17	17
Cash conversion cycle	98	99	98	103	100

Source: Company data, Morgan Stanley Research (E) estimates.

SEMCO – What's Changed

Estimate revisions

We update our estimates to reflect 4Q25 results and 2027 guidance, and adopt a more bullish view on the ABF substrates business. The company is fully booked throughout 2027 for ABF and is currently evaluating capacity expansion plans that are expected to come online from 2028 onwards. Contract renewals are being negotiated on more preferential terms, further supporting longer-term earnings visibility.

We maintain our long-term forecasts broadly unchanged, supported by an improving product mix driven by higher-value products in ABF. As a result, our EPS estimates rise +7% for 2026, +4% for 2027, and +13% for 2028.

Exhibit 61: Estimate revisions

(W bn)	FY26E			FY27E			FY28E		
	Previous	Revised	Change	Previous	Revised	Change	Previous	Revised	Change
Sales	13,450	13,200	(2%)	16,538	16,425	(1%)	19,864	19,159	(4%)
Optics	3,884	3,685	-5%	3,955	3,571	-10%	4,270	3,583	-16%
Substrate	3,177	3,221	1%	4,390	4,434	1%	5,695	5,680	0%
Components (MLCC)	6,389	6,294	-1%	8,193	8,419	3%	9,899	9,895	0%
Operating Profit	1,547	1,728	12%	2,161	2,387	10%	1,612	2,871	78%
Optics	117	147	26%	138	166	20%	213	179	-16%
Substrate	456	466	2%	627	672	7%	716	855	20%
Components (MLCC)	974	1,115	14%	1,395	1,550	11%	1,683	1,836	9%
OP Margin	11.5%	13.1%	1.6pp	13.1%	14.5%	1.5pp	13.2%	15.0%	1.8pp
Optics	3.0%	4.0%	1.0pp	3.5%	4.6%	1.1pp	5.0%	5.0%	0.0pp
Substrate	14.3%	14.5%	0.1pp	14.3%	15.1%	0.9pp	12.6%	15.1%	2.5pp
Components (MLCC)	15.2%	17.7%	2.5pp	17.0%	18.4%	1.4pp	17.0%	18.6%	1.6pp
Net Profit	1,307	1,462	12%	1,778	1,948	10%	1,992	2,294	15%
EPS for consensus	17,335	18,474	7%	23,630	24,630	4%	25,110	28,309	13%

Source: Company Reports, Morgan Stanley Research estimates

Price target and valuation methodology

Based on our residual income model, our new price target is W430K, 14% upside from the current level. It implies 3.1x 2026e P/B. With our prior bull case now playing out, we shift it to our base case and raise both base and bull case scenario values, reflecting continued AI hardware momentum and stronger margin assumptions for AI-focused MLCC and FC-BGA products.

We raise our bull case value to W490K (from W400K), reflecting higher peak earnings assumptions and multiple expansion under our supercycle framework. The revised bull case value implies 3.5x 2026E P/B, above the 2018 and 2020 cycle peaks of 2.4x, yet still below the historical high of 3.8x. Our bear case value remains unchanged, reflecting our view of limited downside to our trough earnings and multiple assumptions.

Exhibit 62: Residual Income Model

SEMCO (009150.KS)

Units are denominated in KRW'bn unless stated otherwise

RIM												
Fiscal period	FY25A	FY26E	FY27E	FY28E	FY29E	FY30E	FY31E	FY32E	FY33E	FY34E	FY35E	Terminal
Forecast year	0	1	2	3	4	5	6	7	8	9	10	value
Total Shareholder Equity	9,471	10,755	12,526	14,642	16,941	19,703	23,067	27,289	32,222	37,792	43,755	45,942
Residual Income		450	784	935	904	1,114	1,410	1,889	2,142	2,254	2,070	1,866
Core Net Profit		1,462	1,948	2,294	2,483	2,946	3,548	4,407	5,118	5,755	6,147	6,455
ROE		14.45%	16.74%	16.89%	15.72%	16.08%	16.59%	17.50%	17.20%	16.44%	15.08%	14.4%
Discount period	0.00	1.00	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00	10.00	
Discount factor	1.00	0.91	0.83	0.75	0.68	0.62	0.56	0.51	0.47	0.42	0.39	0.39
PV of Equity Capital	0.0	409.3	648.3	702.7	617.4	691.9	795.9	969.3	999.3	955.9	798.0	719.6
Beginning Equity Capital	9,016											
PV of Equity Capital	8,308											
PV of Continuing Value	15,111											
Total Equity Value	32,434											
FD shares outstanding ('000)	75,547											
Per share value	430,000											
Current Price	375,000											
Upside Potential	15%											

Source: Company Reports, Morgan Stanley Research estimates

Financials

Exhibit 63: Financial Summary

Income Statement					
(W bn)	FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	10,294	11,314	13,200	16,425	19,159
Gross Profit	1,959	2,277	3,456	4,775	5,741
SG&A	-1,224	-1,364	-1,728	-2,387	-2,871
EBITDA	5,620	2,527	2,956	3,645	4,238
Operating profit/loss	735	913	1,728	2,387	2,871
Financial Income	-1	12	147	176	148
Pretax profit	797	896	1,875	2,564	3,018
Net Profit	703	726	1,462	1,948	2,294
Margins					
Gross Margin	19.0%	20.1%	26.2%	29.1%	30.0%
EBITDA Margin	54.6%	22.3%	22.4%	22.2%	22.1%
Operating Margin	7.1%	8.1%	13.1%	14.5%	15.0%
Net Income	6.8%	6.4%	11.1%	11.9%	12.0%
Growth Rates					
Sales		9.9%	16.7%	24.4%	16.6%
EBITDA		-55.0%	17.0%	23.3%	16.2%
Operating Income		24.3%	89.2%	38.2%	20.2%
Net Income		3.3%	101.2%	33.3%	17.7%
Balance Sheet					
(W bn)	FY24A	FY25A	FY26E	FY27E	FY28E
Cash & Cash Equivalents	2,013	4,377	4,601	4,863	5,348
Trade and other receivables	1,492	1,577	2,075	2,509	3,013
Inventory	2,251	2,171	2,560	2,916	3,394
Other current assets	136	141	141	141	141
Current Assets	5,892	8,265	9,377	10,429	11,895
Tangible Assets	5,933	5,936	6,108	6,425	7,038
Intangible Assets	146	144	149	156	171
Other non-current assets	435	370	370	370	370
Total Assets	12,792	15,307	17,008	19,001	21,819
Trade and other payables	1,212	2,749	3,238	3,535	4,315
Short term Borrowings	1,580	1,931	1,860	1,784	1,706
Other current liabilities	53	82	82	82	82
Current Liabilities	3,057	4,981	5,398	5,620	6,322
Borrowings	0	56	56	56	56
Long-term other payables	84	127	127	127	127
Other Non-current liabilities	634	656	656	656	656
Total Liabilities	3,777	5,836	6,253	6,475	7,177
Capital Stock	388	388	388	388	388
Retained Earnings	6,490	6,871	8,090	9,773	11,734
Total Shareholders' Equity	9,016	9,471	10,755	12,526	14,642
Total Liabilities & SE	12,792	15,307	17,008	19,001	21,819
Net Debt	1,147	(403)	(770)	(1,183)	(1,823)
Cash Flow Statement					
(W bn)	FY24A	FY25A	FY26E	FY27E	FY28E
Net Income (Loss)	703	726	1,462	1,948	2,294
Add: Depreciation & Amortization	4,885	1,613	1,228	1,258	1,367
Less: Changes in Working Capital	319	2,094	(399)	(493)	(202)
Less: Net tax expense	(601)	(201)	(414)	(615)	(724)
Less: Net interest expense	383	84	(71)	(75)	(78)
Cash flows from operating activities	1,430	3,360	1,806	2,023	2,657
Acquisition of PP&E	(772)	(1,022)	(1,404)	(1,583)	(1,995)
Other Operating cashflows	(4,260)	(956)	0	0	0
Cash flows from Investing activities	(806)	(1,032)	(1,404)	(1,583)	(1,995)
Changes in LT Borrowings	8	0	0	0	0
Changes in Short term Borrowing	152	555	0	0	0
Dividend Payout	0	0	0	0	0
Other Investing Cashflows	(381)	(199)	0	0	0
Cash flows from Financing activities	(309)	41	(178)	(178)	(178)
Cash and cash equivalents - closing	2,013	4,377	4,601	4,863	5,348
Increase(decrease) in cash & cash equivalents	315	2,369	224	262	484
FX impact	29	(5)	0	0	0
Ratio Analysis					
(W bn)	FY24A	FY25A	FY26E	FY27E	FY28E
ROE (AOY)	8.0%	7.5%	13.8%	16.0%	15.7%
ROE (EOY)	7.5%	7.3%	13.0%	14.9%	14.6%
ROA	5.6%	4.9%	8.6%	10.3%	10.5%
ROIC (AOY)	5.6%	6.4%	10.8%	12.9%	13.5%
Asset Turnover	106.9	164.2	162.2	138.3	135.3
Days sales outstanding	0.0	0.0	0.0	0.0	0.0
Days inventory outstanding	53.0	50.9	57.4	55.8	57.6
Days payable outstanding	86.2	76.2	81.5	75.8	76.3
Current Ratio	1.9	1.7	1.7	1.9	1.9
Debt to Equity	41.9%	61.6%	58.1%	51.7%	49.0%
Net Debt to Equity	12.7%	-4.3%	-7.2%	-9.4%	-12.5%
EPS (MW)	8,990	9,196	18,474	24,630	28,309
BVPS (MW)	116,340	122,021	138,145	160,425	186,384
DPS	1,800	2,350	2,350	2,350	2,350
EBITDA	(132)	(164)	(414)	(615)	(724)
P/E	17	26	20	15	13
P/BV	1.3	2.0	2.7	2.3	2.0
EV/EBITDA	(96)	(109)	(67)	(44)	(37)
EV/Sales	1.2	1.6	2.1	1.7	1.4
Dividend Yield (%)	1	1	1	1	1
FCF	690	2,525	668	878	1,239
Share Price (KRW)	153,200	241,000	375,000	375,000	375,000
Shares Outstanding (000 AOY)	75,547	75,547	75,547	75,547	75,547

Source: Company Reports, FactSet, Morgan Stanley Research estimates

Risk Reward – Unimicron (3037.TW)

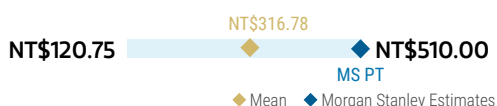
Price hikes and spec upgrade to support higher earnings estimates; OW

PRICE TARGET NT\$500.00

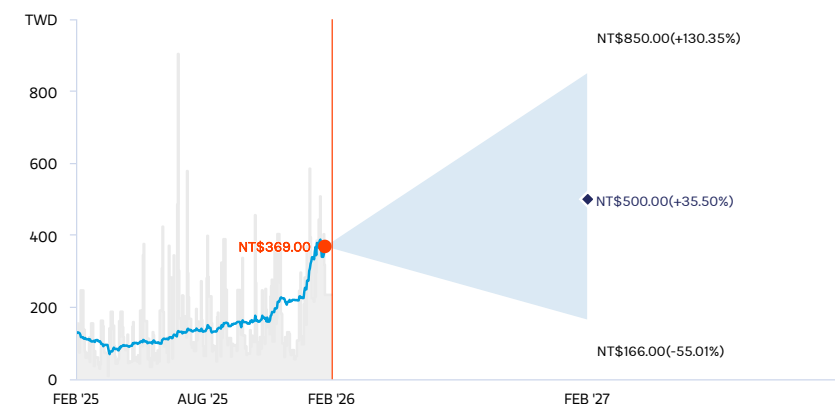
Base case, residual income (RI) valuation model, which we think derives the most accurate value of the firm given that it takes into account cost of equity. We use a cost of equity of 9.2% [risk-free rate of 1% (10-year Taiwan government note yield), equity risk premium of 8.7%, and a beta of 1.0], a medium-term growth rate of 10%, and a terminal growth rate of 3%..

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



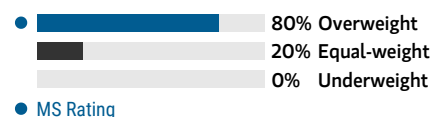
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect the ABF substrate market to revert to undersupply from 2027 onwards, with the undersupply gap reaching 5-10% by 2030.
- This is primarily driven by AI demand, with next-gen AI/networking chips adopting largerbody size ABF substrates.
- In the near to medium term, the short supply of T-glass will be a key watch point, but we believe this will mainly affect low-end ABF and BT substrates, as suppliers prioritize supply to the AI chip vendors.
- Our price target implies 32x 2027e P/E, above the 10-20x range in which it traded in the previous up-cycle. We deem this fair given AI chip substrate spec upgrade and suppliers' ability to raise prices.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*
Secular Growth: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

NT\$850.00

54x 2027e P/E

Demand for AI servers, general servers and PCs is much stronger than expected: Pricing increases more than expected, 30%+ in 2026-27, driving stronger margin expansion for Unimicron.

BASE CASE

NT\$500.00

32x 2027e P/E

Entering up-cycle driven by AI chip upgrade: Robust demand for AI GPUs and ASICs, supported by continued data center infrastructure investments by CSPs and neo cloud, leads to undersupply in the high-end ABF market from 2027 onwards. PC demand is sluggish, while general servers still show double-digit unit growth in 2026. ASP has bottomed and started to inflect upwards, driven by both raw material price hikes and substrate spec upgrades.

BEAR CASE

NT\$166.00

11x 2027e P/E

Demand for AI servers and general servers weakens, with PC demand weakening further: The result is greater ABF oversupply; thus, there are pricing declines of ~15-20% in 2026-27, leading to weaker margins for Unimicron over the next two years.

Risk Reward – Unimicron (3037.TW)

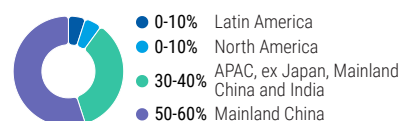
KEY EARNINGS INPUTS

Drivers	2024	2025e	2026e	2027e
IC substrate ASP YoY (%)	(13.4)	6.4	19.7	23.5
IC substrate GM (%)	16.9	14.6	18.1	23.3

INVESTMENT DRIVERS

- 5G smartphone demand
- 5G base station demand
- Automotive demand

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

4/5
MOST
3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Better-than-expected ABF substrate demand from PC and server customers
- Capex cuts; halt to its capacity expansion plan
- Continued yield issues of alternative technology that doesn't require substrate (e.g. CoWoP)

RISKS TO DOWNSIDE

- Sudden demand shortfall
- Technological change that would not require ABF substrates
- Intensifying competition
- Yield issues or production hiccups when ramping new capacity

OWNERSHIP POSITIONING

Inst. Owners, % Active 63.7%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT\$, mn) 141,431 159,901 179,618
160,289

Net income (NT\$, mn) 9,321 11,652 22,626
15,319

EPS (NT\$) 6.09 7.58 14.79
9.97

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Nan Ya PCB (8046.TW)

Price hikes and spec upgrade to support higher earnings estimates; OW

PRICE TARGET NT\$515.00

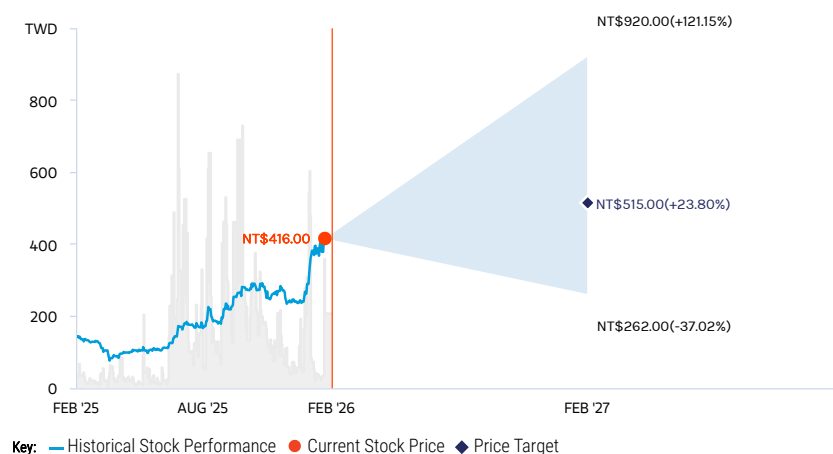
Base case, residual income (RI) valuation model, which we think derives the most accurate value of the firm given that it takes into account cost of equity. We use a cost of equity of 9.3% [risk-free rate of 1% (10-year Taiwan government note yield), equity risk premium of 8.7%, and a beta of 1.0], a medium-term growth rate of 12%, and a terminal growth rate of 3%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART

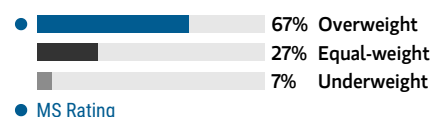


Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- Price hikes for BT and ABF substrates, driven by raw material cost increases, will likely drive margin improvement for NYPCB.
- We believe the ABF substrate market will become undersupplied again from 2027 onwards, driven by AI demand, with next-gen AI/networking chips adopting large-body size ABF substrates.
- NYPCB should benefit from ABF substrate structural tailwinds as well - rising data traffic, big data analysis, AI/ML, and 5G.
- Our price target implies 40x 2027e P/E, above the 8-25x range in which it traded in the previous up-cycle. We deem this fair given ABF substrate spec upgrade and suppliers' ability to raise prices.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*
Secular Growth: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

NT\$920.00

72x 2027e P/E

BT price increases further in 1H26, and ABF becomes undersupplied faster than expected (current expectation is from 2027 onwards), driving even stronger margin expansion for Nan Ya PCB.

BASE CASE

NT\$515.00

40x 2027e P/E

Entering up-cycle driven by AI chip upgrade: Robust demand for AI GPUs/ASICs and networking chips, supported by continued data center infrastructure investments by CSPs and neo cloud, leads to undersupply in the high-end ABF market from 2027 onwards. PC demand is sluggish, while general servers still show double-digit unit growth in 2026. ASP has bottomed and started to inflect upwards, driven by both raw material price hikes and substrate spec upgrades.

BEAR CASE

NT\$262.00

20x 2027e P/E

ABF demand weakens further, resulting in greater oversupply in the next few years. Thus, ABF pricing decreases 20% or more in 2026-27, leading to weaker margins for Nan Ya PCB.

Risk Reward – Nan Ya PCB (8046.TW)

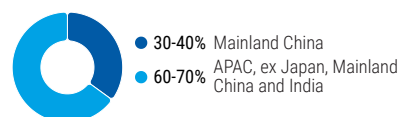
KEY EARNINGS INPUTS

Drivers	2024	2025e	2026e	2027e
ABF ASP y/y (%)	(27.6)	9.6	12.1	12.5
ABF GM (%)	4.0	14.8	19.5	26.7

INVESTMENT DRIVERS

- 5G smartphone demand
- 5G base station demand
- Automotive demand

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

4/5
MOST
3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Better-than-expected ABF and BT demand
- Faster-than-expected pickup in AI and 5G demand
- Stronger-than-expected ASP hike
- Continued yield issues of alternative technology that doesn't require substrate (e.g. CoWoP)

RISKS TO DOWNSIDE

- A sudden demand shortfall, creating headwinds for ABF substrate demand and pricing
- Technological change that would not require ABF substrates
- Intensifying competition

OWNERSHIP POSITIONING

Inst. Owners, % Active 63.3%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2025e

Sales / Revenue (NT\$, mn) 39,905 40,173 40,251
40,127

Net income (NT\$, mn) 1,419 1,607 2,102
1,724

EPS (NT\$) 2.19 2.48 3.25
2.67

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Samsung Electro-Mechanics (009150.KS)

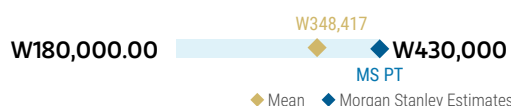
OW: Catalysts for Growth in AI and ASIC

PRICE TARGET W430,000

Base case, derived from our residual income-derived valuation methodology. We apply a 10.5% cost of equity (beta 1.0, equity risk premium of 6.5% and risk-free rate of 4%) and a terminal growth rate of 5%. Our base year is 2026 and our forecast period is 2026-35. We triangulate with relative P/B valuation. Our base case 2026e P/B is 3.1x, which is the company's historical average.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART

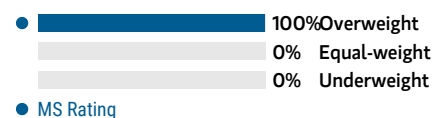


Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We prefer ASIC chip ABF substrates as a way to play custom silicon in the AI chip supply chain.
- SEMCO's rising exposure to and share gains with ASIC customers drive new secular growth and structurally higher returns given the substantial content increase amid rising complexity in ABF multilayer core.
- Given the ASIC/auto computing opportunity – and coupled with 1) earnings expectations bottoming based on the current outlook, and 2) the prospect of eventual cyclical recovery – we think the shares will look to discount accelerating EPS estimate revisions.
- NTM P/B is below its mid-cycle average, indicating continued valuation upside on top of earnings.
- Our price target implies 2026e P/B of 3.1x.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

W490,000

3.5x 2026e P/B

ABF opportunities from major US CSPs drive meaningful content growth, and we estimate future ASIC content and volume growth will continue to outpace. Customer diversification into non-mobile companies also lowers earnings volatility and improves profitability. Our bull case implies a 2026e PB above the 2018-2019 peak of 2.4x but still below historical peak of 3.8x. We believe this premium is supported by what we view as an unprecedented supercycle in the underlying industry dynamics.

BASE CASE

W430,000

3.1x 2026e P/B

We see a number of catalysts ahead for SEMCO, including a large ABF opportunity in ASIC chip wins and FCF inflection. Although the stock has moved higher, our long-term conviction in custom silicon opportunities via ABF substrates has grown substantially.

BEAR CASE

W180,000

1.3x 2026e P/B

MLCC localization trend intensifies with more competition from Chinese suppliers resulting in increased pricing pressure. Substrate business remains soft amid weak macro climate and tepid end demand. China's smartphone demand slows, coupled with growing geopolitical tension/macroeconomic uncertainty driven by the US imposition of tariffs. We reflect slower content growth for MLCC and substrates.

Risk Reward – Samsung Electro-Mechanics (009150.KS)

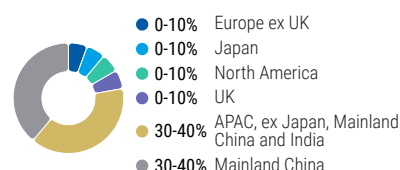
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Revenue Growth (%)	9.9	16.7	24.4	16.6
OPM (%)	8.1	13.1	14.5	15.0
CAPEX (%)	8.7	10.6	9.6	10.4

INVESTMENT DRIVERS

- AI-server, High-speed network, Auto 2.0, and smartphone content growth
- Customer diversification away from Samsung Electronics, targets major CSPs
- MLCC pricing/margin trend

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

2/5
MOST

3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Price hikes driven by MLCC shortage
- Better-than-expected smartphone demand
- Higher MLCC margin from better yield and product mix
- Strong tailwind from China's consumption-friendly policies

RISKS TO DOWNSIDE

- Significant pullback in Samsung's mobile flagship product cycle and SEMCO's lower market share
- Execution risk in penetrating Chinese smartphone customers
- Consumer demand headwinds in 2026

OWNERSHIP POSITIONING

Inst. Owners, % Active 58.5%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (W, bn) 11,942 13,200 14,358
12,878

EBITDA (W, bn) 1,893 2,322 2,956

EBIT (W, bn) 1,005 1,334 1,728

EPS (W) 9,806 13,893 18,474

◆ Mean ◆ Morgan Stanley Estimates
Source: Refinitiv, Morgan Stanley Research

Appendix

Reason to use ABF substrates vs. BT substrates

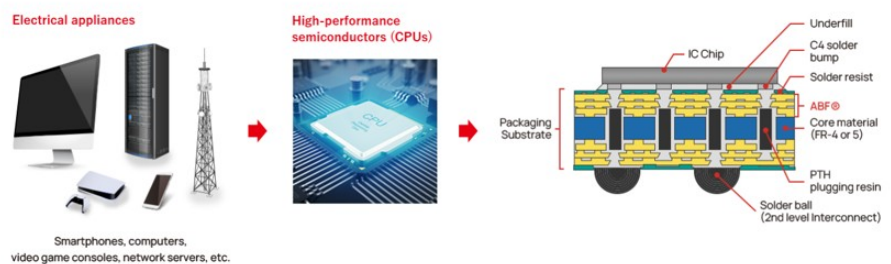
BT (Bismaleimide Triazine) substrates are typically used in consumer chips, mobile SoCs (System on a Chip), RF (radio frequency) chips, PMICs (power management IC), etc. BT substrates typically support coarser wiring, larger vias, lower routing density, and have lower cost. BT substrates are typically used when I/O (input/output) count is moderate, and cost is a concern.

ABF substrates, on the other hand, support higher layer count, very high routing density, microvias, and better electrical performance at higher speeds. As a result, ABF substrates are typically used in CPUs, GPUs, AI accelerators, networking chips, and FPGAs (Field-Programmable Gate Arrays). ABF substrates are also much more expensive, so they're used in higher-end chips that require finer traces, smaller vias, and more routing per layer. They are overkill for lower-end chips, where BT substrates are good enough.

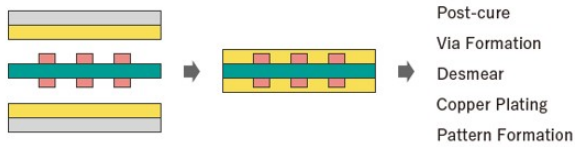
ABF substrates have emerged as a critical bottleneck for high-performance computing:

This reflects their ability to support the ultra-fine circuitry required by CPUs, GPUs, ASICs. Unlike traditional substrates, ABF substrates allow for highly precise line/space patterns (5/7um and below) that facilitate countless electrical connections between the silicon die and the rest of the system. Compared to BT substrates, ABF substrates also boast low dielectric loss tangent (Df), lower surface roughness, and low coefficient of thermal expansion (CTE), ensuring structural integrity, which is particularly important for large AI chips as it prevents signal degradation and the delicate interconnects from failing.

Exhibit 64: Use of ABF in IC packaging

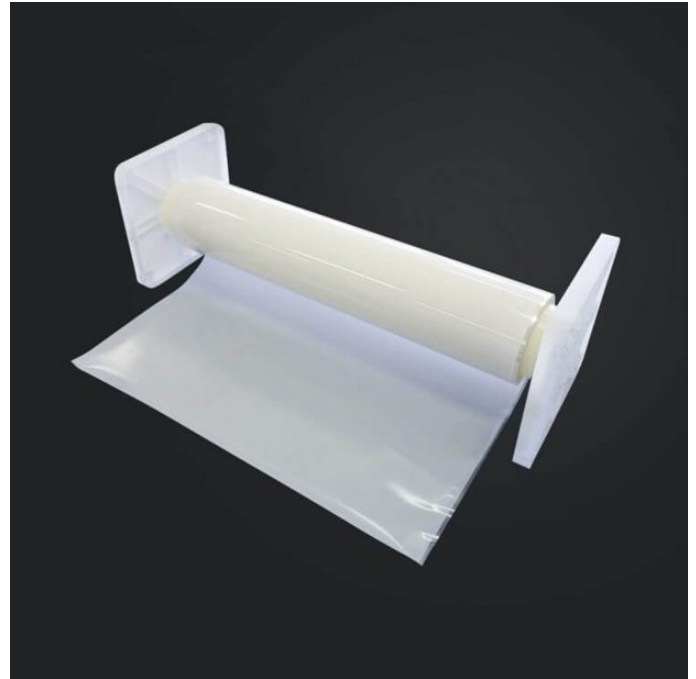


Source: Ajinomoto Fine-Techno Co.

Exhibit 65: How ABF is laminated onto the core layers**Instructions for use**

After lamination on the substrate, the wiring is formed through various processes.

Source: Ajinomoto Fine-Techno Co.

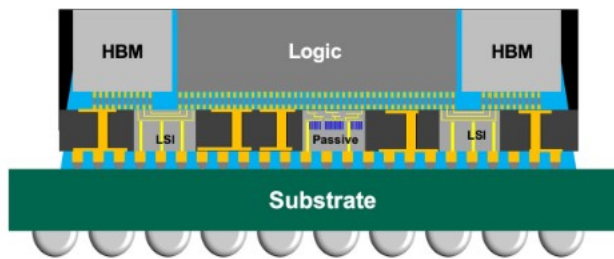
Exhibit 66: Ajinomoto Build-up Film (ABF)

Source: Ajinomoto.

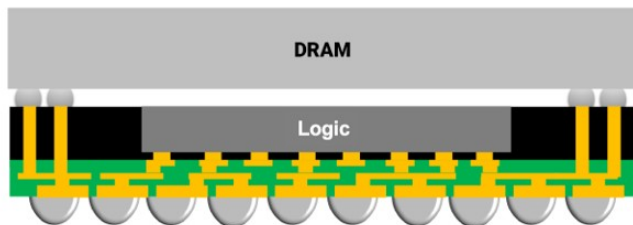
The usage of substrates in different packaging technologies

The relationship between various packaging technologies should be best understood as a progression of interconnect density. At the foundation sits the ABF substrate, which serves as the mechanical and electrical base. The gap is bridged between dense silicon and the substrate, redistribution layers (RDL) or interposer (silicon, organic or glass-based) to route signals. High-end integration methods like TSMC's Chip-on-Wafer-on-Substrate (CoWoS) utilize a large silicon interposer to link logic and memory at generous bandwidths, whereas Intel's Embedded Multi-die Interconnect Bridge (EMIB) embeds small silicon bridges directly into the substrates to achieve similar performance at lower costs.

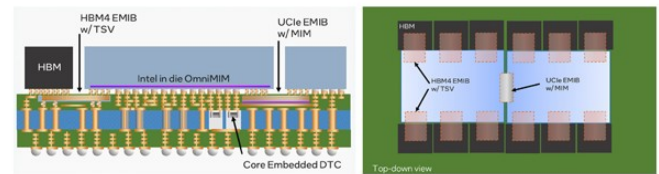
Integrated Fan-Out (InFO), on the other hand, is quite different from the aforementioned packaging technologies. InFO focuses on thinning the package for consumer electronic devices by removing traditional substrates entirely in favor of fine-pitched RDLs. This approach has been widely adopted by Apple's A series smartphone SoC (Systems on a Chip).

Exhibit 67: TSMC CoWoS-L cross section

Source: TSMC.

Exhibit 69: TSMC InFO_PoP cross section

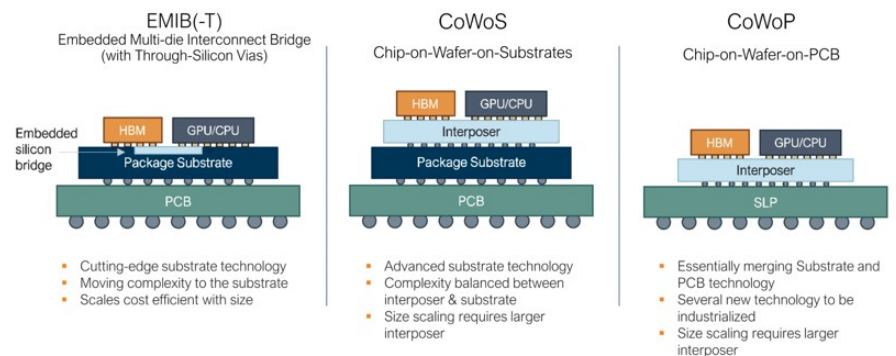
Source: TSMC.

Exhibit 68: Intel EMIB cross section and top-down view

Source: Intel.

Exhibit 70: The role of substrates and PCBs in advanced packaging

The role of Substrates and PCBs in Advanced Packaging



Source: AT&S.

Design choices in advanced packaging now dictate the fundamental architecture of the substrate: When an IC designer opts for 2.5D or 3D stacking, the substrate must be engineered to handle significantly higher power delivery demands while maintaining signal integrity across a much larger surface area. This leads to increases in substrate dimensions and layer counts. As chips grow larger to accommodate more components like high-bandwidth memory (HBM), the substrate must physically expand to match this footprint, with more and more designs exceeding 100mm x 100mm in area size.

Meanwhile, to manage the complex routing and thermal challenges of these advanced chips, layer counts have surged from ~8 layers in PC CPUs to 20+ layers. The increase in substrate dimensions and layer counts significantly drags down manufacturing yields, which lowers real output and further adds to the capacity pressure of high-end ABF substrates. Now substrates are no longer a commodity; rather, they are a performance-enabling component that must be co-designed with the silicon that sits atop.

What is glass core substrate and what are the pros and cons of adopting it?

Glass core substrate is a type of next-generation substrate that replaces the traditional core layer in an ABF substrate with a sheet of glass. It offers superior flatness and rigidity (low CTE) compared to today's organic substrates, which could help suppliers overcome the physical limits of organic materials for future generations of massive AI chips (based on our supply chain checks, 150mm x 150mm could be the area size limit for organic substrates).

Glass core substrate also boasts better higher interconnect density by using through-glass vias (TGV), which are of smaller diameters and finer via pitch compared to the laser-drilled holes on an organic substrate, which enables a significant increase in routing density and connections per package, which are essential for advanced packaging.

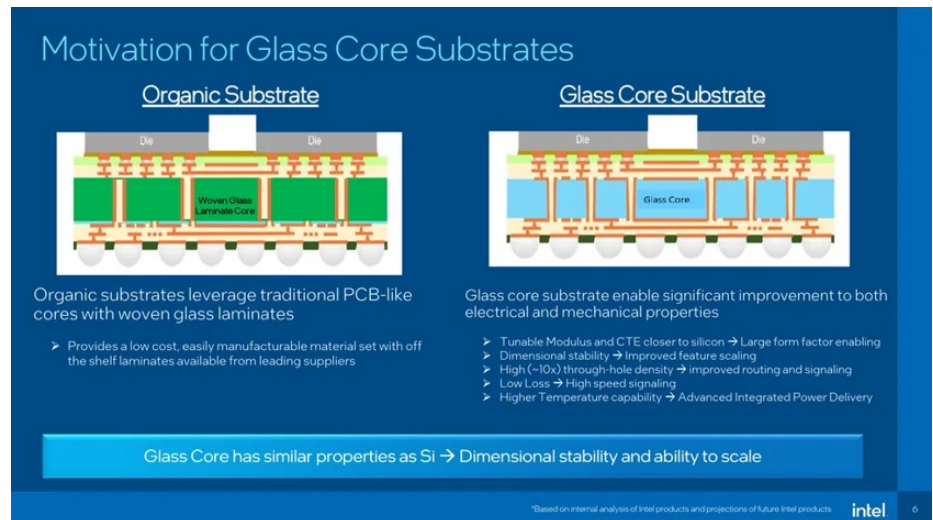
Another prospect for glass core substrate lies in the application in Co-Packaged Optics (CPO) – photonic components could be integrated into the package directly. Because glass is transparent, it can act as a 'waveguide'; AI chips might be able to communicate via light traveling through the glass core itself, instead of copper wiring, which reduces power consumption.

While the technical advantages of glass core substrate are appealing, there remain multiple challenges for the supply chain to overcome in order to achieve mass production at a reasonable costs:

- **Fragility:** Glass is brittle in nature, which makes it more prone to cracks and challenging to handle.
- **Fabrication of TGVs is costly and slow:** Traditional laser drilling method is slow for mass production of TGVs, so some suppliers have shifted to laser-induced deep etching, which uses laser to weaken the glass followed by a chemical etch to open the holes. Filling these TGVs with copper (a process called metallization) is also difficult with current technology.
- **Adhesion challenge:** Metal and glass don't naturally stick together. The copper layers could peel off while being repeatedly heated to high temperatures and cooling.
- **Inspection difficulties:** The transparency of glass breaks traditional automated optical inspection (AOI) machines, making inspection difficult.
- **Infant ecosystem:** Compared to organic substrates, the entire ecosystem of glass core substrate is still in its infancy, making standardization and cost-down difficult.

All in, this makes the cost of producing a substrate with a glass core *multiple times* more expensive today, compared to the current product offering. Over time, as technology improves and costs come down, we could see this glass core solution becoming more mainstream, but in the near term, we don't expect to see much adoption.

Exhibit 71: Organic Substrate vs. Glass Core Substrate



Source: Intel.

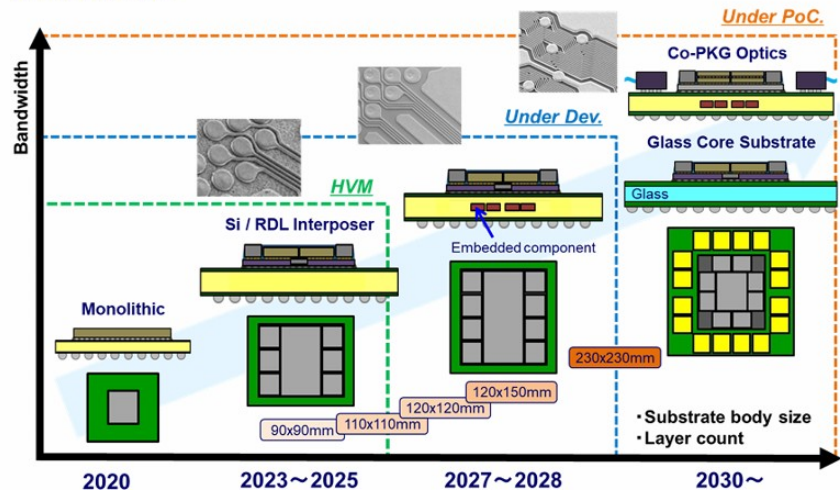
Embedded passives: is this a new trend and how much difficulty increase does this bring?

The embedding of passive components, such as capacitors and resistors, directly into the substrate is a technological trend aimed at improving electrical performance in high-frequency applications. By placing these components microns away from the silicon die rather than counted on the surface (millimeters away), engineers can significantly reduce "parasitic inductance," which is essential for stable power delivery in AI hardware.

Nonetheless, this trend adds to the manufacturing difficulty of substrates, as it significantly complicates the build-up process and introduces new failure points (in addition to the existing many) within the high-end substrate. This also raises the repair and maintenance costs of the vendors and potentially their customers, because if a single embedded passive component is defective, the entire package might need to be scrapped.

Exhibit 72: Technological Advancement and Development of IC Package Substrate

- Besides becoming larger and more multi-layered, IC package substrates are expected to advance in component embedding.
- Maintain the position as the top supplier by developing essential elemental-technologies across all areas.



Source: Ibidem.

Valuation Methodology and Risks

Ibiden (4062.T)

Derived from the base case. DCF assumptions: 1.5% risk-free rate, 1.28 beta, 5.0% risk premium, for 7.5% WACC. Terminal growth of 0% from F3/36.

Risks to Upside

- High-value-added ABF package products support earnings increase beyond our expectation from F3/27 2H .
- Sales exceed our base-case scenario as material shortages for FC packages are resolved

Risks to Downside

- Sales are weaker than base case because key material suppliers cannot supply enough material.
- We estimate OP sensitivity to FX changes at ¥800mn for ¥1/US\$ and ¥100mn for ¥1/€.

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
5. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

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(as of January 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1520	41%	425	48%	28%	685	42%
Equal-weight/Hold	1579	43%	375	42%	24%	713	44%
Not-Rated/Hold	4	0%	1	0%	25%	1	0%
Underweight/Sell	592	16%	89	10%	15%	232	14%
Total	3,695		890			1631	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Stock Price, Price Target and Rating History (See Rating Definitions)

Ibiden (4062.T) - As of 02/20/26 GMT in JPY
Industry : Electronic Components



Stock Rating History: 2/1/21 : O/I; 6/1/22 : U/I; 6/6/23 : E/I; 9/8/23 : E/A; 1/17/24 : U/A; 4/10/24 : U/I; 1/22/25 : E/I;
2/28/25 : O/I; 11/26/25 : E/I; 2/4/26 : U/I

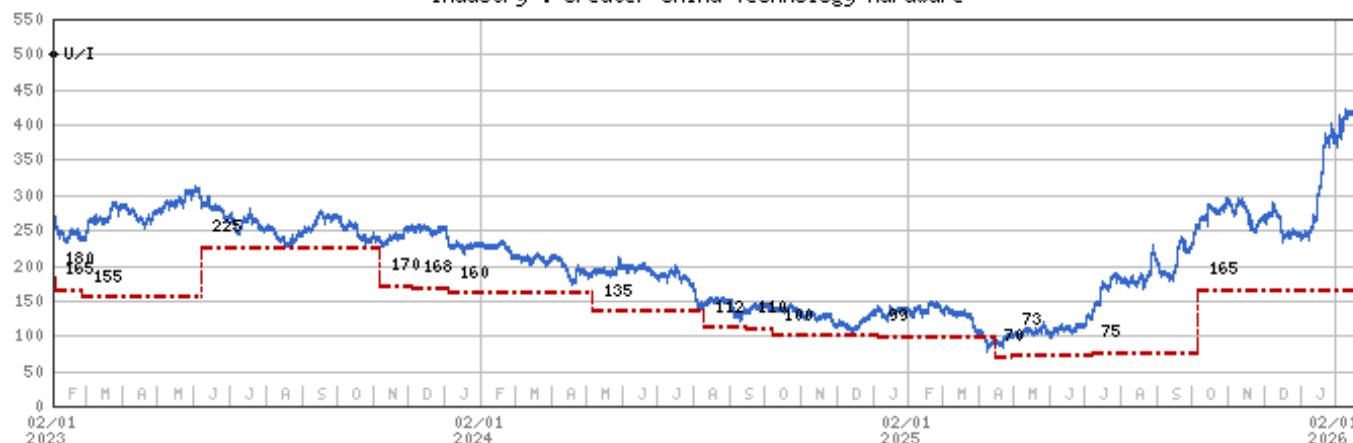
Price Target History: 1/14/21 : 2850; 2/4/21 : 3400; 4/12/21 : 3200; 6/23/21 : 3650; 8/24/21 : 3750; 9/27/21 : 4000; 10/29/21 : 4200;
1/17/22 : 4250; 2/4/22 : 4100; 4/11/22 : 3750; 6/1/22 : 1875; 6/30/22 : 1550; 10/27/22 : 1750; 1/17/23 : 2000; 2/3/23 : 1750;
3/22/23 : 1700; 6/6/23 : 3900; 9/8/23 : 4400; 10/26/23 : 4000; 1/17/24 : 3000; 2/1/24 : 2850; 5/2/24 : 2450; 9/20/24 : 1875;
1/22/25 : 2500; 2/4/25 : 2300; 2/28/25 : 2500; 6/13/25 : 3750; 8/13/25 : 4150; 9/17/25 : 4750; 11/26/25 : 6500

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)
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Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Nan Ya PCB (8046.TW) - As of 02/20/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 2/1/21 : NA/I; 10/22/21 : O/I; 6/1/22 : E/I; 12/21/22 : U/I

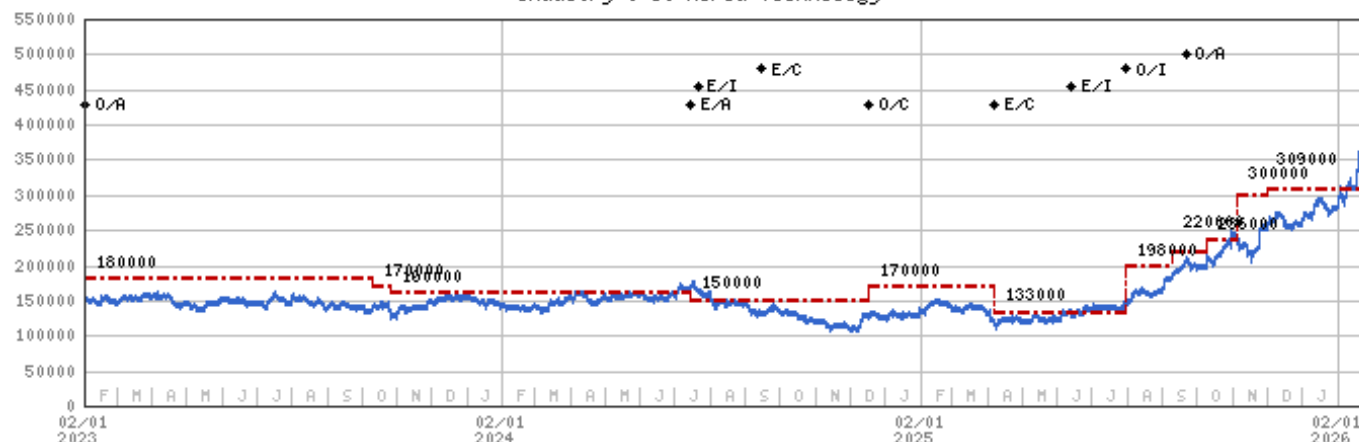
Price Target History: 10/22/21 : 580; 11/9/21 : 630; 2/9/22 : 650; 2/25/22 : 670; 6/1/22 : 420; 8/23/22 : 275; 10/27/22 : 235;
12/21/22 : 180; 2/2/23 : 165; 2/25/23 : 155; 6/7/23 : 225; 11/7/23 : 170; 12/4/23 : 168; 1/4/24 : 160; 5/7/24 : 135; 8/9/24 : 112;
9/15/24 : 110; 10/8/24 : 100; 1/6/25 : 99; 4/15/25 : 70; 4/29/25 : 73; 7/7/25 : 75; 10/6/25 : 165

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Samsung Electro-Mechanics (009150.KS) - As of 02/20/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 2/1/21 : O/A; 5/12/21 : E/A; 7/19/21 : E/I; 8/12/21 : E/C; 6/1/22 : U/C; 10/4/22 : U/A; 11/9/22 : O/A;
7/15/24 : E/A; 7/21/24 : E/I; 9/15/24 : E/C; 12/18/24 : O/C; 4/7/25 : E/C; 6/13/25 : E/I; 7/31/25 : O/I; 9/21/25 : O/A

Price Target History: 1/19/21 : 240000; 3/1/21 : 250000; 5/12/21 : 180000; 10/19/21 : 170000; 10/27/21 : 160000; 1/24/22 : 180000;
3/18/22 : 170000; 6/1/22 : 140000; 11/9/22 : 180000; 10/11/23 : 170000; 10/27/23 : 160000; 7/15/24 : 150000; 12/18/24 : 170000;
4/7/25 : 133000; 7/31/25 : 198000; 9/10/25 : 220000; 10/10/25 : 236000; 11/5/25 : 300000; 12/1/25 : 309000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)
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Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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Unimicron (3037.TW) - As of 02/20/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 2/1/21 : NA/I; 10/22/21 : O/I; 6/1/22 : E/I; 2/22/23 : U/I; 4/15/25 : E/I

Price Target History: 10/22/21 : 188.06; 10/27/21 : 237.55; 2/3/22 : 267.24; 2/24/22 : 366.22; 6/1/22 : 237.55; 8/23/22 : 178.16;
10/27/22 : 133.62; 12/21/22 : 123.72; 2/2/23 : 116.79; 2/22/23 : 98.98; 6/7/23 : 138.57; 11/7/23 : 125.7; 1/24/24 : 118.77;
4/15/25 : 81.66; 7/30/25 : 120.75

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Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

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INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (02/20/2026)
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Andy Meng, CFA

AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK\$36.90
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb70.02
BYD Electronics (0285.HK)	O (04/28/2023)	HK\$32.38
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb11.85
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb18.92
Eoptolink Technology Inc Ltd (300502.SZ)	U (09/11/2025)	Rmb366.11
Genius Electronic Optical Co. Ltd. (3406.TW)	E (04/23/2025)	NT\$429.50
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb6.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.38
Largan Precision (3008.TW)	E (10/17/2025)	NT\$2,290.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb10.68
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb9.80
Q Technology (Group) Company Ltd (1478.HK)	E (11/10/2023)	HK\$8.62
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb85.45
Shanghai Conant Optical Co Ltd (2276.HK)	E (09/11/2025)	HK\$63.75
Shenzhen Transsion Holdings Co Ltd (688036.SS)	O (10/24/2023)	Rmb58.63
Sunny Optical (2382.HK)	O (07/10/2025)	HK\$56.95
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb311.58
Wingtech Technology Co Ltd (600745.SS)	E (11/10/2023)	Rmb33.18
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK\$35.36
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb213.39
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	E (04/20/2023)	HK\$111.60
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb124.50
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb26.51
Zhongji Innolight Co Ltd (300308.SZ)	O (11/06/2023)	Rmb531.00
ZTE Corporation (0763.HK)	E (03/11/2024)	HK\$27.02
ZTE Corporation (000063.SZ)	U (07/02/2021)	Rmb37.16

Derrick Yang

Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT\$1,450.00
Advantech (2395.TW)	O (01/20/2021)	NT\$305.00
AirTAC International (1590.TW)	O (04/16/2025)	NT\$1,200.00
AU Optronics (2409.TW)	E (02/10/2026)	NT\$16.15
Bizlink (3665.TW)	O (03/10/2025)	NT\$1,365.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb4.18
BOE Varitronix Ltd (0710.HK)	O (06/20/2023)	HK\$4.75
Chenbro (8210.TW)	O (07/23/2025)	NT\$975.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT\$1,050.00
E Ink Holdings Inc. (8069.TWO)	E (01/07/2026)	NT\$189.50
Ennostar Inc (3714.TW)	U (09/23/2022)	NT\$35.70
Hiwin Technologies Corp. (2049.TW)	U (05/12/2025)	NT\$232.00
Innolux (3481.TW)	E (04/07/2025)	NT\$21.95
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT\$3,030.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb34.29
Radiant Opto-Electronics Corporation (6176.TW)	E (03/01/2024)	NT\$126.50
Sanan Optoelectronics (600703.SS)	U (08/21/2023)	Rmb15.91
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.56
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb9.65
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb141.13

Howard Kao

Acer Inc. (2353.TW)	U (04/23/2025)	NT\$27.10
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT\$523.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT\$30.60
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK\$5.59
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT\$229.50

Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT\$829.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb65.33
Lenovo (0992.HK)	E (11/16/2025)	HK\$9.22
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT\$1,645.00
Nan Ya PCB (8046.TW)	O (02/22/2026)	NT\$416.00
Pegatron Corporation (4938.TW)	U (11/16/2025)	NT\$70.50
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT\$286.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb64.25
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb235.00
Unimicron (3037.TW)	O (02/22/2026)	NT\$369.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT\$131.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT\$3,740.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT\$256.00
Zhen Ding (4958.TW)	E (08/02/2022)	NT\$178.00
Sharon Shih		
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT\$1,605.00
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT\$1,065.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT\$192.50
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT\$1,260.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT\$1,545.00
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb54.75
Foxconn Technology (2354.TW)	U (04/23/2025)	NT\$58.70
GoerTek Inc (002241.SZ)	U (04/23/2025)	Rmb26.16
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT\$227.00
LandMark Optoelectronics Corporation (3081.TWO)	O (01/19/2026)	NT\$1,085.00
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb15.39
Lite-On Technology (2301.TW)	E (01/15/2025)	NT\$179.50
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb50.80
Sunonwealth Electric Machine Industry Co (2421.TW)	E (02/23/2024)	NT\$144.50
Tong Hsing (6271.TW)	E (03/18/2019)	NT\$153.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT\$180.00

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* Historical prices are not split adjusted.